CAPSTONE PROJECT PROPOSAL

Online Management System for Crates N' Plates with Sales Analytics and Reservation Scheduling

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1.1 Background of the Study

The food service industry has remained one of the fastest-growing sectors of the global economy, continuously evolving in response to changing consumer behavior, technological innovations, and increasing market competition. Over the past decade, rapid advancements in information and communication technology (ICT) have transformed how restaurants, cafés, catering businesses, and other food service establishments conduct their daily operations. The growing accessibility of digital technologies has shifted business practices from traditional paper-based procedures to integrated and automated information systems that enhance efficiency, productivity, and customer satisfaction. As businesses strive to remain competitive in an increasingly digital environment, the adoption of information systems has become an essential component of operational management rather than merely a technological innovation.

The digital transformation of businesses has significantly influenced the hospitality and food service industries by changing how organizations manage customer transactions, reservations, inventory, employee coordination, financial reporting, and business decision-making. Traditional restaurant operations, which once depended heavily on handwritten records, manual computations, and paper-based documentation, have gradually been replaced by computerized systems capable of processing information in real time. These technological developments have enabled restaurant owners and managers to streamline business processes, reduce operational costs, improve service quality, and provide customers with faster and more reliable services.

The emergence of digital technologies has likewise altered customer expectations regarding restaurant services. Modern consumers increasingly expect businesses to provide efficient transaction processing, organized reservation systems, faster order fulfillment, and accurate service delivery. Customers no longer evaluate restaurants solely based on food quality but also consider service efficiency, convenience, responsiveness, and overall customer experience. As a result, restaurant businesses have been compelled to invest in digital solutions that improve operational performance while simultaneously enhancing customer satisfaction and organizational competitiveness.

Information systems have become fundamental tools that enable organizations to manage operational activities effectively by collecting, processing, storing, and distributing information needed for daily operations and strategic decision-making. Unlike manual systems, computerized information systems facilitate the rapid processing of large volumes of business data while minimizing the occurrence of human errors associated with manual record-keeping. They also provide managers with immediate access to accurate and up-to-date information that supports planning, monitoring, controlling, and evaluating business performance. Consequently, organizations that successfully integrate information systems into their operations are generally more capable of adapting to changing market conditions and responding efficiently to customer demands.

Within the restaurant industry, the implementation of integrated restaurant management systems has become increasingly important because restaurant operations involve multiple interconnected activities that require continuous coordination among employees. A single customer transaction often involves order processing, payment recording, food preparation, inventory deduction, receipt generation, customer service, and business reporting. When these activities are managed manually, delays and communication problems become more likely, particularly during periods of high customer demand. Integrated information systems eliminate these inefficiencies by connecting operational processes within a centralized platform where information can be updated and shared in real time among authorized users.

The increasing complexity of restaurant operations has also encouraged businesses to adopt Point-of-Sale (POS) systems as one of the primary components of restaurant management. Modern POS systems extend beyond simple payment processing by integrating customer orders, menu management, sales recording, inventory updates, employee monitoring, and financial reporting into a single application. Through automation, these systems significantly reduce transaction processing time, improve billing accuracy, eliminate repetitive manual computations, and maintain comprehensive records of completed transactions. Furthermore, the availability of digital transaction histories enables management to monitor business performance more effectively while reducing the administrative workload associated with manual documentation.

Beyond transaction processing, restaurants increasingly rely on integrated information systems to improve inventory management. Inventory represents one of the most valuable assets of food service establishments because the availability of ingredients and supplies directly affects the restaurant's ability to provide consistent customer service. Effective inventory management ensures that sufficient stocks are available to meet customer demand while minimizing food spoilage, excessive inventory costs, and unnecessary wastage. However, manual inventory monitoring often depends on handwritten logbooks, physical stock counting, and periodic inventory reconciliation, all of which require considerable time and are susceptible to human error. These limitations frequently result in inaccurate inventory records, delayed replenishment, inventory shortages, and overstocking, ultimately affecting operational efficiency and profitability.

Another operational area that has experienced significant technological advancement is reservation management. As customer demand continues to increase, restaurants are expected to provide convenient reservation services that allow customers to schedule dining experiences, private events, and catering services efficiently. Traditional reservation methods typically rely on telephone calls, handwritten reservation books, or manual scheduling procedures, making it difficult for employees to monitor reservation availability and prevent scheduling conflicts. Computerized reservation systems address these issues by maintaining centralized reservation databases that enable authorized personnel to confirm bookings, monitor reservation schedules, prevent duplicate reservations, and provide customers with timely updates regarding their bookings. These systems not only improve organizational efficiency but also enhance customer confidence by providing a more reliable reservation process.

The increasing availability of operational data has likewise highlighted the importance of business intelligence and sales analytics within restaurant management. Every customer transaction generates valuable information regarding purchasing behavior, product demand, sales performance, inventory consumption, and customer preferences. When properly analyzed, these data provide meaningful insights that enable restaurant owners to evaluate business performance, identify best-selling products, forecast customer demand, optimize menu offerings, and develop more effective marketing strategies. However, organizations that rely solely on manual record-keeping often fail to maximize the value of these operational data because the information remains fragmented, difficult to analyze, and time-consuming to consolidate. Consequently, many managerial decisions continue to depend on intuition rather than evidence-based analysis.

Sales analytics has therefore emerged as a critical component of modern restaurant management because it transforms raw operational data into meaningful information that supports strategic decision-making. Through graphical dashboards, statistical reports, trend analysis, and performance indicators, sales analytics enables managers to evaluate business performance from multiple perspectives. Restaurant owners can identify seasonal sales patterns, monitor revenue growth, compare product performance, analyze customer purchasing habits, and assess the effectiveness of promotional activities. These analytical capabilities provide organizations with a competitive advantage by allowing management to make informed decisions based on accurate and timely business information rather than relying exclusively on manual observations or historical experience.

The integration of these technological innovations demonstrates that digital transformation is no longer limited to large corporations but has become increasingly relevant to small and medium-sized enterprises (SMEs), including independently owned restaurants. SMEs represent a significant portion of the food service industry and contribute substantially to local economic development through employment generation, entrepreneurship, and community engagement. Nevertheless, many small restaurants continue to experience operational challenges due to limited technological adoption, financial constraints, and dependence on traditional management practices. As business competition intensifies and customer expectations continue to evolve, these establishments face increasing pressure to modernize their operational processes through the adoption of integrated information systems capable of improving efficiency, accuracy, and overall business performance.

The increasing adoption of restaurant management systems reflects the growing recognition that technology is an essential driver of operational efficiency within the food service industry. A Restaurant Management System (RMS) is an integrated information system designed to automate and coordinate the major operational activities of a restaurant, including order processing, billing, inventory management, reservation scheduling, employee management, customer information management, and business reporting. Unlike stand-alone applications that perform only a single function, an integrated restaurant management system allows information generated from one operational process to be shared automatically with other related processes. This integration eliminates redundant tasks, minimizes inconsistencies in business records, and enables management to obtain accurate operational information in real time.

One of the most essential components of a Restaurant Management System is the Point-of-Sale (POS) module. The Point-of-Sale system serves as the primary interface between customers and restaurant personnel during sales transactions. Traditionally, restaurants relied on handwritten order slips and manual computation of customer bills, which frequently resulted in computational errors, incomplete records, delayed service, and communication problems between the cashier and kitchen personnel. With the implementation of computerized POS systems, restaurants are able to automate customer order processing, payment computation, receipt generation, and transaction recording while simultaneously updating inventory records and generating sales reports. The integration of these functions not only improves transaction accuracy but also shortens customer waiting time and enhances the overall dining experience.

Beyond improving transaction processing, computerized Point-of-Sale systems also provide businesses with valuable operational data that can be utilized for business analysis. Every completed transaction contains information regarding purchased products, customer preferences, sales volume, transaction time, and payment details. When these data are systematically stored and analyzed, they become important resources that enable management to evaluate operational performance, monitor customer demand, and identify business opportunities. Consequently, modern POS systems are no longer viewed merely as payment processing tools but as essential information systems that support operational management and strategic decision-making.

Inventory management represents another critical aspect of restaurant operations that significantly influences business performance. Restaurants continuously consume raw ingredients, beverages, packaging materials, and other supplies to provide products and services to customers. Because many food products are perishable, maintaining an appropriate inventory level is essential to avoid food spoilage, unnecessary expenses, and interruptions in restaurant operations. Effective inventory management enables organizations to balance customer demand with available resources while minimizing operational waste and maximizing profitability.

Manual inventory monitoring, however, presents numerous operational challenges. Restaurant employees are typically required to record stock movement manually, update inventory logbooks, conduct physical inventory counts, and reconcile discrepancies between recorded inventory and actual stock levels. These procedures consume considerable time and effort while increasing the possibility of inaccurate inventory records, delayed updates, misplaced documentation, and human error. Furthermore, manual inventory monitoring limits management's ability to identify fast-moving products, determine reorder levels, monitor stock consumption, and forecast inventory requirements accurately. Such limitations often result in stock shortages, overstocking, unnecessary purchasing costs, and reduced operational efficiency.

The implementation of computerized inventory management systems addresses these challenges by automating inventory recording and monitoring processes. Through system integration, inventory quantities are automatically updated whenever customer transactions are completed, ensuring that inventory records remain accurate and current. Automated inventory systems also provide notifications for low-stock items, generate inventory reports, monitor inventory history, and assist management in making timely purchasing decisions. These capabilities improve inventory accuracy while reducing administrative workload and strengthening internal control over business resources.

Reservation scheduling has likewise become an increasingly important operational component within the restaurant industry. Restaurants offering dine-in services, function room reservations, catering packages, and special event accommodations require efficient scheduling systems to organize customer bookings and allocate available resources effectively. Traditional reservation procedures commonly involve handwritten reservation books, telephone confirmations, or manually maintained schedules. Although these methods are relatively simple, they become increasingly difficult to manage as customer demand grows and reservation volume increases.

Manual reservation systems often expose businesses to scheduling conflicts, duplicate bookings, overlooked reservations, delayed confirmations, and communication errors among employees. Because reservation information is maintained separately from other operational records, employees frequently spend additional time verifying booking availability and coordinating customer schedules. These inefficiencies not only reduce employee productivity but also negatively affect customer satisfaction by creating uncertainty regarding reservation confirmation and availability.

Computerized reservation systems overcome these operational limitations by maintaining a centralized reservation database accessible to authorized personnel. Through automated scheduling, reservation requests can be processed more efficiently while minimizing scheduling conflicts and duplicate bookings. Reservation systems also improve communication between restaurant personnel and customers by maintaining organized booking records and enabling timely reservation confirmation, modification, and cancellation. As a result, restaurants are better able to maximize seating capacity, improve customer service, and utilize available resources more effectively.

In recent years, business intelligence and sales analytics have become increasingly significant in supporting managerial decision-making within the hospitality industry.

Every operational activity performed by a restaurant generates valuable business data that can be transformed into meaningful information through analytical processes. Sales transactions reveal customer purchasing behavior, inventory records indicate consumption patterns, reservation histories demonstrate customer demand, and financial reports provide insights into business performance. Collectively, these operational data represent strategic organizational resources that support evidence-based decision-making.

Sales analytics refers to the systematic process of collecting, organizing, analyzing, and interpreting sales-related information to identify trends, evaluate product performance, monitor customer purchasing behavior, and measure organizational performance. Unlike traditional sales reports that merely summarize completed transactions, sales analytics enables management to understand the underlying factors influencing business performance. Analytical dashboards provide graphical representations of revenue growth, best-selling products, customer demand, sales trends, peak operating hours, and seasonal purchasing patterns. These insights enable restaurant owners to make informed decisions regarding pricing strategies, promotional campaigns, menu development, inventory planning, and resource allocation.

For small and medium-sized enterprises (SMEs), the application of sales analytics is particularly beneficial because it allows business owners to maximize limited organizational resources through data-driven planning. Instead of relying solely on intuition or personal experience, managers are able to evaluate objective business information before making operational and strategic decisions. Consequently, sales analytics contributes not only to improved financial performance but also to better customer satisfaction, more efficient resource utilization, and enhanced organizational competitiveness.

Despite the increasing availability of these technological solutions, many small restaurant businesses continue to rely on fragmented operational procedures that combine manual record-keeping with isolated digital applications. In many establishments, sales transactions, reservation records, inventory monitoring, and business reports are managed independently rather than through an integrated information system. Such fragmentation often results in duplicate data entry, inconsistent records, delayed information retrieval, and limited communication among employees. Moreover, because operational information is scattered across multiple records and documents, management experiences difficulties in obtaining a comprehensive view of business performance needed to support effective decision-making.

These operational challenges are particularly evident among independently owned restaurants and local food service establishments in developing economies, where financial limitations, technological constraints, and organizational practices often delay the adoption of integrated information systems. While many businesses recognize the potential benefits of digital transformation, the transition from traditional operational procedures to computerized management systems remains gradual. Consequently, manual processes continue to affect operational efficiency, data accuracy, customer service, and long-term business sustainability.

The Philippine food service industry reflects these developments as restaurants, cafés, and catering businesses increasingly adopt digital technologies to improve operational efficiency and remain competitive. However, despite the growing availability of computerized restaurant management solutions, numerous local establishments continue to encounter operational challenges associated with manual business processes, fragmented information management, and limited utilization of business data. These realities highlight the continuing need for practical, affordable, and integrated restaurant management systems specifically designed to address the operational requirements of local food service enterprises.

1.2 Statement of the Problem

In the Philippines, the food service industry has continued to expand due to increasing urbanization, changing consumer lifestyles, rising disposable income, and the growing preference for convenience-oriented dining experiences. Restaurants, cafés, catering businesses, and other food service establishments have become important contributors to the country's economic development by generating employment opportunities, supporting local suppliers, and stimulating tourism. The continuous growth of this industry has also intensified competition among food establishments, encouraging business owners to improve operational efficiency, service quality, and customer satisfaction through the adoption of innovative technologies.

The rapid advancement of digital technology has significantly influenced how local restaurants manage their daily operations. Many establishments have begun implementing computerized Point-of-Sale systems, online reservation platforms, inventory management applications, and digital payment solutions to improve service delivery and organizational performance. These technological innovations enable restaurant owners to automate repetitive tasks, reduce operational costs, improve communication among employees, and provide customers with more efficient and convenient services. Furthermore, digital transformation has enabled restaurant managers to obtain timely operational information that supports strategic planning and evidence-based decision-making.

Despite these developments, many small and medium-sized restaurants in the Philippines continue to rely on traditional management practices. Paper-based record keeping, handwritten sales reports, manual reservation logs, and physical inventory monitoring remain common among independently owned restaurants because of financial constraints, limited technological resources, and insufficient awareness regarding the benefits of integrated information systems. While these manual procedures may initially appear manageable, they often become inefficient as business operations expand and customer demand increases. Consequently, restaurant owners encounter increasing difficulties in managing operational information accurately and efficiently.

The continued dependence on manual operational procedures exposes restaurants to numerous challenges that affect both organizational performance and customer satisfaction. Manual sales recording increases the likelihood of computational mistakes, incomplete documentation, duplicate transactions, and delayed report preparation. Likewise, manually maintained reservation schedules often lead to overlapping bookings, reservation conflicts, delayed customer confirmations, and communication errors among employees. Inventory management also becomes increasingly difficult because stock movement must be monitored through handwritten records and periodic physical inventory counts, making inventory discrepancies more likely to occur. These operational limitations reduce organizational efficiency and hinder management's ability to make informed business decisions.

Another significant concern among small restaurant businesses is the limited utilization of operational data. Although restaurants generate substantial amounts of information through customer transactions, inventory movement, reservation records, and financial activities, much of these data remain underutilized when maintained through manual documentation. Without integrated analytical tools, restaurant owners often rely primarily on personal observation and experience when making decisions regarding inventory replenishment, pricing strategies, promotional activities, menu development, and resource allocation. While managerial experience remains valuable, relying solely on intuition may reduce the organization's ability to respond effectively to changing customer preferences and market conditions.

Business intelligence and sales analytics have therefore become increasingly important components of modern restaurant management. By transforming operational data into meaningful information, analytical systems enable managers to monitor business performance continuously and identify opportunities for organizational improvement. Sales analytics allows management to evaluate revenue trends, identify best-selling products, determine customer purchasing patterns, analyze seasonal demand, and assess the effectiveness of marketing campaigns. These analytical capabilities provide valuable support for evidence-based decision-making, allowing restaurant owners to allocate resources more effectively while improving operational efficiency and profitability.

Within this context, **Crates N' Plates**, located at **108 Delgado Street, Barangay Mabulo-Delgado, Iloilo City**, represents a growing local food service establishment that experienced the operational challenges commonly encountered by many independently owned restaurants. Established on **February 28, 2020**, under the ownership of **Engr. Aireen Joyce Saplada**, the business initially operated as a diner before expanding into a café in 2022. As customer demand continued to increase, the establishment diversified its services by offering dine-in meals, café products, restobar services, catering packages, private function room reservations, and event catering for various occasions held both inside and outside the restaurant premises. This continuous business expansion significantly increased the volume and complexity of the restaurant's operational activities.

As Crates N' Plates expanded its operations, the number of daily transactions, customer reservations, inventory movements, catering schedules, and administrative responsibilities also increased considerably. The restaurant was required to process customer orders efficiently while simultaneously coordinating food preparation, monitoring available inventory, managing reservations, preparing financial reports, and maintaining accurate business records. These interconnected operational activities required timely communication and accurate information sharing among managers, cashiers, kitchen personnel, and other employees. However, because most operational procedures remained manual, coordinating these activities became increasingly difficult as customer volume continued to grow.

The restaurant's order management process relied primarily on handwritten order slips prepared by cashiers before being physically forwarded to the kitchen for food preparation. Beverage orders were processed separately, requiring additional coordination between service personnel and kitchen staff. During busy operating hours, this manual workflow frequently increased the possibility of communication delays, misplaced order slips, duplicated orders, incomplete transactions, and extended customer waiting times. Since operational information was recorded manually, employees were also required to spend additional time verifying completed transactions and correcting documentation errors before reports could be prepared.

Inventory management likewise depended on handwritten inventory logbooks maintained separately for kitchen ingredients and beverage supplies. Employees conducted periodic inventory checking by comparing available stock with manually recorded transactions to determine inventory consumption and replenishment requirements. Although this procedure enabled management to maintain basic inventory control, it required considerable time, increased employee workload, and remained highly susceptible to recording inaccuracies. Delayed inventory updates occasionally made it difficult to determine actual stock availability, identify low-stock items promptly, and monitor inventory movement accurately. These limitations increased the risk of inventory shortages, overstocking, unnecessary purchasing expenses, and food wastage that could negatively affect restaurant profitability.

Reservation scheduling represented another operational challenge encountered by the establishment. Customer reservations for dine-in services, private function rooms, and catering events were maintained through handwritten reservation records and manual scheduling procedures. Before confirming reservations, employees manually checked existing booking records to determine schedule availability. As reservation requests increased, this process became increasingly difficult to manage because employees had to coordinate multiple bookings while simultaneously performing other operational responsibilities. The absence of a centralized reservation system increased the possibility of scheduling conflicts, duplicate reservations, overlooked bookings, delayed confirmations, and communication errors, all of which could negatively affect customer satisfaction and the professional image of the business.

The preparation of sales reports also remained dependent on manual documentation. Sales records were consolidated from receipts and handwritten transaction logs before daily, weekly, and monthly reports could be prepared. This reporting process consumed valuable time that could otherwise have been allocated to customer service and operational management. Because reports were prepared manually, management frequently experienced delays in obtaining accurate financial information needed to monitor business performance, evaluate sales trends, and support timely decision-making. Furthermore, the absence of automated reporting limited management's ability to access real-time operational information necessary for responding immediately to emerging business concerns.

These operational observations demonstrated that while Crates N' Plates had successfully expanded its business operations and customer base, its existing management practices had not evolved at the same pace as its organizational growth. The increasing complexity of restaurant operations highlighted the limitations of manual procedures and emphasized the need for a more efficient, integrated, and technology-driven approach to restaurant management. Consequently, implementing a computerized management system became not merely a technological enhancement but an operational necessity capable of supporting the restaurant's continued growth and long-term sustainability.

The growing dependence on information technology within the food service industry has resulted in the development of numerous restaurant management systems designed to improve operational efficiency and customer service. Various commercial and academic systems have introduced features such as Point-of-Sale (POS) transaction processing, inventory monitoring, online ordering, reservation management, employee scheduling, and financial reporting. These technological solutions have demonstrated the significant role of information systems in reducing manual workload, minimizing operational errors, and improving organizational productivity. However, despite the increasing availability of these systems, many existing solutions continue to focus on only one or a few operational functions rather than providing a comprehensive platform capable of integrating all major restaurant operations.

A review of previous studies revealed that many restaurant management systems concentrated primarily on automating customer ordering, billing, or inventory management. Other systems emphasized online reservations or customer relationship management without integrating sales analytics or comprehensive reporting capabilities. Although these systems successfully addressed particular operational concerns, they often failed to provide restaurant owners with a centralized source of information capable of supporting both day-to-day operations and long-term strategic planning. Consequently, management still needed to rely on multiple applications or manually consolidate information obtained from different operational processes before making important business decisions.

Another limitation observed in many existing restaurant management systems is the insufficient utilization of business intelligence and analytical tools. While transactional data are routinely collected during restaurant operations, many systems simply store these data without transforming them into meaningful information that supports managerial decision-making. Restaurant owners often receive basic sales reports but are not provided with detailed analyses of customer purchasing behavior, product performance, sales trends, inventory consumption, seasonal demand, or operational performance indicators. Without these analytical capabilities, management opportunities to optimize menu offerings, improve promotional strategies, forecast customer demand, and allocate organizational resources more effectively remain limited.

Similarly, many available systems have been developed to address the requirements of large restaurant chains or franchise businesses. These commercial solutions frequently contain extensive functionalities that exceed the operational requirements and financial capabilities of independently owned restaurants and small-to-medium enterprises (SMEs). As a result, many local restaurant owners continue to depend on manual operational procedures because commercially available systems may be expensive, difficult to customize, or incompatible with their existing workflows. This situation highlights the need for practical, user-friendly, and cost-effective information systems specifically designed to address the operational requirements of local restaurants while remaining adaptable to their existing business processes.

The researchers also recognized that digital transformation among local food establishments extends beyond merely replacing manual records with computerized systems. Effective digital transformation requires the integration of operational activities into a centralized information system capable of supporting communication, coordination, reporting, monitoring, and managerial decision-making. Information generated from sales transactions should automatically update inventory records; reservation schedules should be accessible in real time; reports should be generated instantly; and analytical dashboards should provide meaningful business insights that assist management in evaluating organizational performance. The integration of these operational functions enables businesses to maximize the value of their operational data while improving efficiency, accuracy, and service quality.

For Crates N' Plates, these operational requirements became increasingly evident as the restaurant continued to expand its products and services. The growth of the business resulted in increased transaction volumes, more frequent reservations, larger inventory requirements, and greater administrative responsibilities. While the restaurant successfully attracted a growing customer base, its continued reliance on manual operational procedures limited its ability to manage expanding business activities efficiently. Employees devoted considerable time to routine administrative tasks, management experienced delays in obtaining operational reports, and valuable business information remained fragmented across multiple manual records. These operational challenges highlighted the need for a more integrated management approach capable of supporting the restaurant's continuous development.

The researchers therefore identified a need to develop an information system that not only automated restaurant operations but also generated meaningful business intelligence capable of supporting managerial decision-making. Unlike conventional restaurant management applications that primarily focus on transaction processing, the proposed system sought to integrate Point-of-Sale (POS) management, reservation scheduling, inventory monitoring, sales analytics, report generation, and Role-Based Access Control (RBAC) into a single web-based platform. This integration enables operational information to flow automatically among different modules, eliminating duplicate data entry, improving record accuracy, and providing management with real-time access to essential business information.

A distinguishing feature of the proposed study is the incorporation of a **Sales Analytics Dashboard** designed specifically to assist management in monitoring organizational performance. Rather than merely presenting lists of completed transactions, the analytics module generates visual reports that summarize daily, weekly, monthly, and yearly sales performance, identify best-selling products, monitor revenue trends, evaluate inventory movement, and analyze customer purchasing patterns. These analytical capabilities provide management with valuable information that supports evidence-based decision-making regarding menu development, inventory replenishment, pricing strategies, promotional campaigns, and resource allocation. Consequently, the developed system extends beyond operational automation by functioning as a decision support tool capable of improving both operational and strategic management.

Another significant contribution of the proposed system is the implementation of an integrated **Reservation Scheduling Module** that manages dine-in reservations, private function room bookings, and catering reservations through a centralized scheduling platform. Unlike manual reservation procedures that require continuous employee verification and coordination, the proposed system provides organized reservation records, monitors booking availability, minimizes scheduling conflicts, and improves communication among restaurant personnel. This functionality contributes to more efficient reservation management while enhancing customer confidence and service reliability.

Furthermore, the implementation of **Role-Based Access Control (RBAC)** strengthens information security by assigning appropriate system permissions according to user responsibilities. Administrators, managers, cashiers, kitchen personnel, and other authorized users are granted access only to the modules necessary for performing their respective duties. This security mechanism protects confidential business information, reduces the risk of unauthorized system access, and improves accountability by maintaining activity records for authorized users. The inclusion of RBAC further distinguishes the proposed system from many manually operated restaurant management practices that provide limited control over access to sensitive operational information.

The researchers likewise considered the long-term sustainability of the proposed system by developing it as a web-based application. Through centralized data management, authorized users may access operational information using desktop computers, laptops, tablets, or other internet-connected devices. This flexibility improves accessibility while ensuring that operational records remain consistent, organized, and readily available whenever management requires timely business information. The centralized database likewise supports future system enhancement by providing a scalable platform capable of accommodating additional functionalities as the restaurant continues to grow.

Overall, the development of the **Online Management System for Crates N' Plates with Sales Analytics and Reservation Scheduling** was undertaken to address the operational limitations associated with manual restaurant management while supporting the organization's continuing growth and modernization. By integrating sales transaction processing, reservation scheduling, inventory monitoring, sales analytics, automated reporting, and secure user management into a unified information system, the proposed application was expected to improve operational efficiency, strengthen data accuracy, enhance customer service, support evidence-based managerial decision-making, and contribute to the long-term competitiveness and sustainability of Crates N' Plates.

Ultimately, this study sought to demonstrate that the successful integration of information technology into restaurant operations extends beyond the automation of routine business activities. The researchers recognized that information systems serve not only as operational tools but also as strategic organizational resources capable of transforming business data into valuable information for planning, monitoring, controlling, and decision-making. Through the implementation of a comprehensive Online Management System with Sales Analytics and Reservation Scheduling, the study aimed to provide Crates N' Plates with an efficient, reliable, and scalable digital solution that addressed its existing operational challenges while supporting continuous organizational improvement. The findings of this study were likewise expected to contribute to the growing body of knowledge on restaurant information systems by demonstrating how integrated technologies can enhance operational performance among small and medium-sized food service enterprises.

Every organization depends on effective operational processes to ensure the efficient delivery of products and services to its customers. Within the food service industry, restaurant operations involve numerous interconnected activities, including customer order processing, reservation management, inventory monitoring, sales recording, financial reporting, and customer service. Each of these operational functions contributes significantly to the overall performance of the business. Consequently, inefficiencies affecting one operational area frequently influence other business processes, ultimately reducing organizational productivity and customer satisfaction.

As restaurants continue to expand their operations, the volume of customer transactions, inventory movement, reservations, and administrative responsibilities likewise increases. Such organizational growth requires businesses to implement efficient management strategies capable of handling operational complexity while maintaining service quality. Information technology has therefore become an essential organizational resource that enables restaurant owners to automate routine procedures, improve communication among employees, maintain accurate operational records, and support informed managerial decision-making. Businesses that continue to depend primarily on manual operational procedures often experience increasing difficulties in maintaining efficiency as customer demand and business activities expand.

Crates N' Plates has experienced continuous business growth since its establishment by expanding its products and services to meet the changing needs of its customers. In addition to offering dine-in meals and café products, the establishment also provides restobar services, private function room reservations, and catering services for various events conducted both within and outside the restaurant premises. This continuous expansion has increased the complexity of the restaurant's daily operations, requiring more effective methods for managing customer transactions, reservation schedules, inventory records, and business reports.

Despite its continued growth, many operational procedures at Crates N' Plates remained dependent on manual documentation and traditional record-keeping practices. Customer transactions, reservation schedules, inventory records, and operational reports were managed separately using handwritten records and manual documentation. Although these procedures enabled the restaurant to perform its daily operations, they became increasingly inefficient as customer demand and transaction volume continued to increase. The absence of an integrated information system limited the restaurant's ability to coordinate operational activities efficiently while providing management with accurate and timely business information.

One of the major operational concerns observed by the researchers involved the manual processing of sales transactions. Customer orders were manually recorded before payment processing, requiring employees to prepare handwritten records and perform manual computations during each transaction. Although employees exercised care in recording sales information, manual procedures naturally increased the possibility of computational errors, duplicate entries, incomplete transaction records, and misplaced documentation. During periods of high customer demand, employees were required to process numerous transactions simultaneously, making it increasingly difficult to maintain complete accuracy while providing timely customer service.

The manual preparation of sales reports likewise presented significant administrative challenges. Restaurant personnel were required to consolidate receipts, review handwritten sales records, and manually prepare daily, weekly, and monthly reports before management could evaluate business performance. This reporting procedure required considerable time and effort while increasing the possibility of delayed reports and inaccurate financial summaries. Because business information was not immediately available, management experienced difficulties in monitoring daily revenue, evaluating operational performance, and making timely decisions regarding restaurant operations. Delays in report preparation also limited management's ability to respond immediately to operational concerns that required prompt attention.

Reservation management represented another area that required improvement. Crates N' Plates accommodates various customer reservations, including dine-in bookings, private function room reservations, and catering services for different occasions. These reservations require accurate scheduling and continuous coordination among restaurant personnel to ensure that customer requests are accommodated appropriately. However, because reservation information was maintained manually, employees were required to verify booking schedules individually before confirming customer reservations.

As reservation requests continued to increase, maintaining manual reservation records became progressively more challenging. Employees needed to coordinate multiple bookings while simultaneously managing other restaurant responsibilities, increasing the possibility of scheduling conflicts, duplicate reservations, delayed confirmations, overlooked bookings, and communication errors. These operational issues not only affected employee productivity but also had the potential to reduce customer satisfaction because reservation-related errors could inconvenience customers and negatively influence their perception of the restaurant's service quality.

Inventory monitoring also remained largely dependent on manual operational procedures. Separate inventory records were maintained for kitchen ingredients and beverage supplies through handwritten logbooks that required periodic updating and verification. Employees conducted physical inventory checking to determine available stock levels and manually compared inventory records with completed sales transactions. While this process provided a basic method of inventory control, it demanded significant employee effort and remained susceptible to recording inaccuracies and delayed updates.

Manual inventory monitoring created additional operational concerns because management experienced difficulties in accurately determining inventory movement, monitoring product consumption, identifying low-stock items, and establishing appropriate reorder schedules. Inaccurate inventory information increased the possibility of stock shortages, unnecessary overstocking, inventory discrepancies, food spoilage, and increased purchasing costs. These issues not only affected operational efficiency but also reduced the restaurant's ability to provide uninterrupted customer service, particularly when essential ingredients or supplies became unavailable during business operations.

Another significant concern identified during the conduct of the study was the absence of a centralized information system capable of integrating the restaurant's major operational processes. Sales transactions, reservation records, inventory information, and operational reports were maintained independently using separate manual records. Because these operational data were stored in different documents, retrieving accurate and up-to-date information often required employees to examine multiple logbooks, receipts, and handwritten reports before obtaining the information requested by management. This fragmented method of information management consumed valuable time and increased the possibility of inconsistencies among business records.

The absence of an integrated operational platform also limited communication among employees. Since operational information was manually transferred from one department to another, delays frequently occurred in transmitting customer orders, updating reservation schedules, recording inventory movement, and preparing business reports. During peak operating hours, employees devoted most of their attention to serving customers, leaving administrative tasks to be completed later. As a result, operational records were sometimes updated after transactions had already been completed, increasing the possibility of discrepancies between actual business activities and recorded information. These delays affected not only operational efficiency but also management's ability to monitor business performance accurately.

Another limitation observed by the researchers involved the restaurant's inability to utilize operational data effectively. Every completed transaction generated valuable information regarding customer purchasing behavior, sales performance, inventory consumption, and product demand. However, because these records were maintained manually, the restaurant was unable to transform them into meaningful information that could support managerial planning and decision-making. Business data remained scattered across receipts, reservation records, and inventory logbooks, making comprehensive analysis difficult and time-consuming.

The absence of a sales analytics capability further reduced the organization's ability to maximize the value of its operational information. Although management could determine the restaurant's total sales through manual reporting, obtaining more detailed information regarding best-selling menu items, customer purchasing trends, seasonal demand, peak operating hours, promotional effectiveness, and product performance required extensive manual computation and record verification. Consequently, important managerial decisions concerning menu planning, inventory replenishment, promotional activities, pricing strategies, and resource allocation were frequently based on personal experience and manual observation rather than comprehensive business analysis.

Evidence-based decision-making has become increasingly important for organizations operating in highly competitive industries. Restaurant managers require timely and reliable operational information to evaluate organizational performance, anticipate customer demand, allocate resources efficiently, and respond quickly to changing market conditions. Without an integrated analytical system capable of presenting operational data through meaningful reports and graphical summaries, management's ability to implement proactive business strategies becomes significantly limited. The researchers recognized that the absence of analytical capabilities represented one of the most significant operational limitations affecting the long-term competitiveness of Crates N' Plates.

Furthermore, manual operational procedures affected organizational productivity by increasing the administrative workload of restaurant personnel. Employees devoted considerable time to recording transactions, updating inventory logbooks, verifying reservation schedules, preparing reports, and correcting documentation errors. These routine administrative responsibilities reduced the amount of time available for customer service and other operational activities directly contributing to business performance. As customer demand continued to increase, maintaining these manual procedures became increasingly inefficient and unsustainable.

The researchers also observed that manual information management presented potential risks to data accuracy, record security, and information accessibility. Handwritten records were susceptible to physical damage, loss, duplication, unauthorized modification, and accidental misplacement. Retrieving historical business records likewise became increasingly difficult because employees were required to manually search through multiple files and logbooks. These limitations reduced the reliability of operational information while making long-term record management more complicated. As the business continued to expand, the need for a centralized and secure database became increasingly important to ensure the accuracy, consistency, confidentiality, and accessibility of organizational information.

Considering these operational concerns, the researchers recognized the necessity of developing an integrated information system capable of addressing the existing limitations of manual restaurant management. The proposed **Online Management System for Crates N' Plates with Sales Analytics and Reservation Scheduling** was therefore designed to automate the restaurant's major operational processes through a centralized web-based platform. The proposed system integrated sales transaction processing, reservation scheduling, inventory monitoring, sales analytics, report generation, and Role-Based Access Control (RBAC) into a single information system that supports efficient communication, accurate record management, and evidence-based managerial decision-making.

Through the implementation of the proposed system, sales transactions could be processed more efficiently using an integrated Point-of-Sale module capable of automatically recording completed transactions and generating electronic reports. Reservation scheduling could likewise be managed through a centralized database that minimized scheduling conflicts, prevented duplicate bookings, and improved coordination among restaurant personnel. Inventory records would be updated automatically as transactions occurred, enabling management to monitor stock movement accurately while reducing the possibility of inventory discrepancies and delayed replenishment.

The incorporation of a Sales Analytics Dashboard further distinguished the proposed system by transforming operational data into meaningful business information. Instead of manually preparing reports and performing repetitive computations, management would be provided with graphical summaries and analytical reports that present sales performance, customer purchasing patterns, inventory movement, revenue trends, and product demand in real time. These analytical capabilities were expected to strengthen managerial planning by enabling evidence-based decisions regarding marketing strategies, inventory management, pricing, menu planning, and business expansion.

Ultimately, the researchers sought to determine whether the development of an integrated Online Management System with Sales Analytics and Reservation Scheduling could effectively address the operational challenges encountered by Crates N' Plates. Specifically, the study aimed to evaluate whether the proposed system could improve operational efficiency, enhance data accuracy, strengthen organizational coordination, support managerial decision-making, improve customer service, and contribute to the restaurant's long-term operational sustainability.

To achieve these objectives, the study sought to answer the following research questions:

General Problem

How could the development and implementation of an Online Management System with Sales Analytics and Reservation Scheduling improve the operational efficiency and management processes of Crates N' Plates?

Specific Problems

Specifically, the study sought to answer the following questions:

1. What operational challenges were encountered by Crates N' Plates in terms of:

- **1.1** Sales transaction management;
- **1.2** Reservation scheduling; and
- **1.3** Inventory monitoring?

2. How was the Online Management System with Sales Analytics and Reservation Scheduling designed and developed to address the identified operational challenges of Crates N' Plates?

3. What features and functionalities were incorporated into the developed system in terms of:

3.1 Point-of-Sale (POS) Management;

3.2 Reservation Scheduling;

3.3 Inventory Monitoring;

3.4 Sales Analytics; and

3.5 Report Generation?

4. To what extent was the developed Online Management System evaluated by the respondents based on the **ISO/IEC 25010 Software Quality Model** in terms of:

4.1 Functional Suitability;

4.2 Performance Efficiency;

4.3 Usability;

4.4 Reliability;

4.5 Security;

4.6 Maintainability; and

4.7 Portability?

1.2.1 General Problem

Crates N' Plates experienced increasing operational challenges as the business expanded its products, services, and customer base while continuing to depend on traditional manual operational procedures. Although the existing paper-based system had initially supported the restaurant's daily activities during its early years of operation, it gradually became less effective in managing the increasing complexity of restaurant transactions, reservation requests, inventory records, and administrative responsibilities. As customer demand continued to grow, the limitations of manual record-keeping became more evident, affecting the efficiency, accuracy, reliability, and timeliness of the restaurant's operational processes.

The restaurant's day-to-day operations involved numerous interconnected activities that required continuous coordination among cashiers, kitchen personnel, managers, and other employees. These activities included processing customer orders, recording sales transactions, managing dine-in and catering reservations, monitoring inventory movement, preparing operational reports, and maintaining business records. Because these operational functions were performed manually and managed separately, employees were required to spend considerable time recording information, verifying documents, updating records, and coordinating activities across different departments. This manual workflow increased the administrative burden placed on employees while reducing the overall efficiency of restaurant operations.

Sales transaction processing represented one of the most significant operational concerns encountered by the restaurant. Customer orders and sales information were manually recorded and processed before being consolidated into daily, weekly, and monthly financial reports. This procedure increased the possibility of computational errors, duplicate entries, incomplete transaction records, misplaced sales documents, and delayed report preparation. During busy operating hours, employees were often required to process multiple customer transactions simultaneously, making it increasingly difficult to maintain complete accuracy while providing prompt customer service. Consequently, management experienced delays in obtaining reliable financial information needed to monitor daily sales performance, evaluate revenue, and support operational decision-making.

Reservation scheduling also relied heavily on manual coordination and handwritten documentation. Reservations for dine-in services, private function rooms, and catering events were recorded separately and verified manually before customer bookings could be confirmed. As reservation requests increased, coordinating multiple schedules became increasingly difficult because employees needed to verify booking availability while simultaneously performing other operational responsibilities. The absence of a centralized scheduling system increased the possibility of reservation conflicts, duplicate bookings, delayed confirmations, overlooked reservations, and communication errors among employees. These operational limitations negatively affected service efficiency and created unnecessary inconvenience for customers, potentially reducing customer satisfaction and confidence in the restaurant's reservation process.

Inventory monitoring likewise presented significant operational challenges because stock levels were maintained through handwritten inventory logbooks and periodic physical inventory checking. Separate records for kitchen ingredients and beverage supplies required employees to manually update inventory information after transactions were completed. This process consumed valuable employee time and increased the possibility of recording inaccuracies, delayed inventory updates, misplaced records, and inconsistencies between physical inventory and documented stock levels. Furthermore, management experienced difficulties in accurately monitoring inventory movement, identifying low-stock items, determining reorder requirements, and evaluating inventory consumption. These limitations increased the risk of stock shortages, overstocking, unnecessary purchasing expenses, inventory discrepancies, and food wastage, all of which negatively affected operational efficiency and profitability.

Another major concern identified by the researchers was the absence of a centralized computerized management system capable of integrating the restaurant's major operational activities. Sales records, reservation schedules, inventory information, and operational reports were maintained independently using separate documents and manual records. Because operational information was fragmented across different record-keeping systems, retrieving accurate and up-to-date business information required considerable time and effort. Managers often needed to review multiple receipts, reservation logs, inventory records, and handwritten reports before obtaining the information necessary for monitoring restaurant performance. This fragmented approach to information management reduced operational efficiency and limited management's ability to respond promptly to emerging business concerns.

The lack of an integrated information system also limited the restaurant's capability to generate real-time operational reports. Preparing daily, weekly, monthly, and annual reports required employees to manually consolidate operational records collected from various sources. This labor-intensive reporting process delayed the availability of important business information needed for monitoring sales performance, evaluating operational efficiency, assessing inventory status, and supporting managerial planning. Because reports were prepared manually, management frequently relied on delayed or incomplete information when making operational and strategic decisions, reducing the effectiveness of organizational planning and resource allocation.

Moreover, the restaurant lacked an integrated sales analytics capability that could transform operational data into meaningful business information. Although customer transactions generated valuable information regarding sales performance, customer purchasing behavior, inventory consumption, and product demand, these data remained largely underutilized because they were maintained in manual form. Without automated analytical tools, management found it difficult to identify best-selling products, monitor revenue trends, evaluate promotional activities, analyze customer preferences, recognize seasonal demand patterns, and forecast future inventory requirements. Consequently, many managerial decisions regarding pricing strategies, menu planning, promotional campaigns, and inventory replenishment relied primarily on manual observation and previous experience rather than objective, data-driven analysis.

Collectively, these operational limitations significantly affected the overall performance of Crates N' Plates. Manual operational procedures increased employee workload, delayed the availability of accurate business information, reduced operational efficiency, and limited management's ability to implement timely and evidence-based decisions. As the restaurant continued to expand its operations, these challenges became increasingly difficult to manage using traditional methods alone. The researchers recognized that the continued reliance on manual procedures could potentially hinder the restaurant's long-term growth, competitiveness, and ability to provide consistently high-quality customer service.

Recognizing these operational concerns, the researchers identified the need for a comprehensive and integrated information system capable of automating the restaurant's major business processes while improving the accuracy, accessibility, and security of operational information. Consequently, this study sought to develop an **Online Management System for Crates N' Plates with Sales Analytics and Reservation Scheduling** that integrated Point-of-Sale (POS) transaction processing, reservation scheduling, inventory monitoring, sales analytics, report generation, and Role-Based Access Control (RBAC) into a centralized web-based platform. Through the implementation of the proposed system, the researchers aimed to improve operational efficiency, reduce dependence on manual procedures, enhance customer service, strengthen organizational coordination, support evidence-based managerial decision-making, and contribute to the long-term growth, sustainability, and competitiveness of Crates N' Plates.

1.2.2 Specific Problems

The study sought to identify and address the specific operational problems encountered by Crates N' Plates that affected the efficiency, accuracy, and effectiveness of its daily business operations. These operational concerns were observed during the analysis of the restaurant's existing workflow and were identified as significant factors limiting organizational productivity, service quality, and managerial decision-making. Specifically, the study focused on the following operational problems:

Sales Transaction Management

Sales transaction processing was primarily performed through manual recording and documentation, requiring employees to prepare handwritten order slips, compute customer bills, and record completed transactions before consolidating them into sales reports. Although these procedures enabled the restaurant to process daily transactions, they required considerable employee effort and became increasingly inefficient as customer volume increased.

The reliance on manual transaction processing increased the possibility of computational errors, duplicate entries, incomplete records, misplaced sales documents, and delayed updating of transaction information. During peak business hours, employees were required to process numerous customer transactions simultaneously, increasing the likelihood of inaccuracies and reducing transaction efficiency. These operational limitations affected the reliability of financial records and delayed the preparation of sales summaries required by management for monitoring daily business performance.

Sales Analytics

The restaurant lacked an integrated sales analytics capability capable of transforming operational data into meaningful business information. Although daily transactions generated valuable information regarding customer purchases, product demand, and revenue generation, these data were maintained primarily for record-keeping purposes rather than analytical evaluation.

Without automated analytical tools, management experienced difficulties in identifying best-selling products, evaluating customer purchasing behavior, monitoring seasonal demand, assessing promotional effectiveness, forecasting future sales, and measuring overall business performance. Consequently, important managerial decisions regarding pricing strategies, promotional activities, menu development, and inventory planning relied heavily on manual observation and previous experience rather than objective data analysis.

Reservation Scheduling

Reservation scheduling for dine-in customers, private function rooms, and catering services was managed through handwritten records and manual scheduling procedures. Employees manually verified reservation availability before confirming bookings, requiring continuous coordination among restaurant personnel.

As reservation requests increased, maintaining reservation schedules manually became increasingly difficult. The absence of a centralized reservation system increased the possibility of scheduling conflicts, duplicate bookings, delayed confirmations, overlooked reservations, and communication errors. These operational issues reduced employee efficiency and created inconvenience for customers, potentially affecting customer satisfaction and the restaurant's professional reputation.

Inventory Monitoring

Inventory monitoring relied on handwritten inventory logbooks and periodic physical inventory checking. Separate records were maintained for kitchen ingredients and beverage supplies, requiring employees to manually update inventory information following each operational activity.

Because inventory records were maintained manually, management experienced difficulties in accurately monitoring stock movement, identifying low-stock items, determining reorder schedules, and maintaining consistent inventory records. Delayed inventory updates and recording inaccuracies occasionally resulted in inventory discrepancies, stock shortages, overstocking, unnecessary purchasing costs, and avoidable wastage of perishable supplies, thereby reducing operational efficiency and increasing business expenses.

Report Generation

Operational reports, including daily, weekly, monthly, and annual sales summaries, were prepared manually by consolidating receipts and handwritten transaction records. This reporting process required significant time and administrative effort before accurate financial information could be presented to management.

The absence of an automated reporting mechanism delayed the availability of essential business information required for operational monitoring, financial evaluation, inventory planning, and strategic decision-making. Because reports were generated manually, management frequently encountered delays in obtaining accurate operational information needed to evaluate restaurant performance promptly.

Operational Integration

The restaurant did not utilize a centralized information system capable of integrating its essential business functions into a single operational platform. Sales transactions, inventory monitoring, reservation scheduling, catering management, reporting, and customer information were maintained independently, requiring employees to perform repetitive encoding and manual verification across multiple records.

This fragmented operational structure resulted in duplicate work, delayed information sharing, inconsistent documentation, communication gaps among departments, and reduced organizational productivity. The absence of system integration limited management's ability to monitor restaurant operations comprehensively and coordinate business activities efficiently.

Data Management and Security

The existing manual information management process provided limited support for the secure storage, retrieval, updating, and preservation of business records. Operational information was maintained through paper-based documentation that remained vulnerable to physical damage, accidental loss, duplication, unauthorized modification, and difficulties in retrieving historical records.

The absence of role-based access control and centralized digital storage also limited data confidentiality and organizational accountability. As the volume of operational records continued to increase, maintaining accurate, secure, and accessible business information became increasingly difficult, highlighting the need for a computerized information management system capable of protecting organizational data while improving accessibility for authorized personnel.

1.3 Objective of the Study

1.3.1 General Objective

The primary objective of this study was to design, develop, implement, and evaluate an **Online Management System for Crates N' Plates with Sales Analytics and Reservation Scheduling** that would automate and integrate the restaurant's essential operational processes into a centralized web-based platform. The study sought to provide a comprehensive information system capable of improving the efficiency, accuracy, reliability, and security of the restaurant's daily operations while addressing the limitations associated with its existing manual management procedures.

Specifically, the proposed system was designed to automate the processing of customer transactions, order management, reservation scheduling, inventory monitoring, sales analytics, report generation, and user access management. Through the integration of these operational functions into a single information system, the researchers intended to eliminate redundant manual procedures, reduce administrative workload, minimize human errors, and improve coordination among restaurant personnel. The centralized nature of the system was also expected to provide authorized users with immediate access to accurate and up-to-date operational information, thereby improving organizational communication and facilitating more efficient business management.

Another major objective of the study was to improve the management of restaurant transactions through the implementation of an integrated **Point-of-Sale (POS)** module capable of accurately recording customer orders, processing payments, maintaining transaction histories, and automatically updating operational records. The automation of transaction processing was expected to reduce computational errors, prevent duplicate entries, improve billing accuracy, and shorten customer waiting time while maintaining organized electronic records of completed transactions.

The study also aimed to improve the restaurant's reservation management process by developing a centralized reservation scheduling module capable of managing dine-in reservations, private function room bookings, and catering schedules. The researchers intended for this module to simplify reservation processing, minimize scheduling conflicts, eliminate duplicate bookings, and improve communication among employees responsible for managing customer reservations. By maintaining organized reservation records within a centralized database, the proposed system was expected to enhance customer convenience while improving the efficiency of reservation management.

In addition, the study sought to strengthen inventory management by implementing an automated inventory monitoring module capable of recording inventory movement, monitoring stock availability, identifying low-stock items, and generating inventory reports. Through the automation of inventory-related processes, the researchers intended to improve inventory accuracy, reduce discrepancies between physical and recorded stock levels, prevent stock shortages and overstocking, and assist management in making timely purchasing and replenishment decisions. The inventory module was likewise expected to contribute to better resource utilization and more effective control over restaurant supplies.

A significant objective of the study was the incorporation of a **Sales Analytics Dashboard** capable of transforming operational data into meaningful business information. The researchers intended for the analytics module to provide graphical representations and comprehensive reports regarding sales performance, customer purchasing behavior, product demand, inventory movement, revenue trends, and best-selling menu items. Through these analytical capabilities, the proposed system was expected to support evidence-based managerial decision-making by providing management with reliable information necessary for evaluating business performance, developing marketing strategies, forecasting customer demand, planning inventory replenishment, and identifying opportunities for organizational improvement.

The study further aimed to automate the generation of operational reports by enabling the system to produce accurate and real-time reports related to sales transactions, reservations, inventory status, and overall restaurant performance. Unlike the existing manual reporting procedures that required considerable time and effort, the proposed system was expected to generate reports automatically using information stored within the centralized database. This functionality was intended to improve the timeliness, consistency, and reliability of business reports while reducing the administrative workload associated with manual report preparation.

Another important objective of the study was to strengthen data management and information security through the implementation of a centralized database and **Role-Based Access Control (RBAC)**. The researchers sought to ensure that business information would be securely stored, efficiently organized, and readily accessible only to authorized personnel according to their designated roles and responsibilities. By assigning different access privileges to administrators, managers, cashiers, kitchen staff, and other authorized users, the system was expected to protect confidential business information, improve accountability, and maintain the integrity and confidentiality of operational records.

Furthermore, the study intended to improve the overall quality of customer service by enabling faster transaction processing, more organized reservation scheduling, improved order coordination, and accurate inventory monitoring. By reducing delays associated with manual operational procedures, the proposed system was expected to shorten customer waiting time, improve service reliability, minimize operational errors, and create a more organized dining experience. These improvements were anticipated to contribute positively to customer satisfaction and strengthen the restaurant's reputation for delivering efficient and reliable service.

Beyond addressing immediate operational concerns, the researchers also intended for the developed system to support the long-term growth and sustainability of Crates N' Plates. The proposed information system was envisioned not only as an operational tool but also as a strategic resource capable of assisting management in evaluating organizational performance, identifying business opportunities, optimizing operational resources, and responding proactively to changing customer demands. Through the integration of business automation and analytical capabilities, the study aimed to demonstrate how information technology could enhance organizational efficiency while supporting informed managerial planning and continuous business improvement.

Finally, the study sought to evaluate the quality and acceptability of the developed Online Management System using the **ISO/IEC 25010 Software Quality Model**, specifically in terms of functional suitability, performance efficiency, usability, reliability, security, maintainability, and portability. The evaluation was intended to determine whether the developed system effectively satisfied the operational requirements of Crates N' Plates and whether it met recognized software quality standards. Ultimately, the successful implementation and evaluation of the proposed system were expected to contribute to improved operational efficiency, enhanced managerial decision-making, increased employee productivity, better customer service, and the long-term competitiveness and sustainability of Crates N' Plates.

1.3.2 Specific Objectives

To accomplish the general objective of the study, the researchers pursued the following specific objectives:

1. **To design, develop, and implement** an **Online Management System for Crates N' Plates with Sales Analytics and Reservation Scheduling** that integrated the restaurant's core operational processes, including Point-of-Sale (POS) transaction processing, order management, reservation scheduling, inventory monitoring, sales analytics, report generation, and user management into a centralized web-based platform that supports efficient business operations and information management.
2. **To automate the restaurant's sales transaction process** by implementing an integrated Point-of-Sale (POS) module capable of accurately recording customer orders, processing payments, generating electronic receipts, maintaining complete transaction histories, and automatically updating sales records in order to improve transaction accuracy, reduce manual errors, shorten transaction processing time, and enhance customer service.
3. **To develop and implement a centralized reservation scheduling module** that efficiently manages dine-in reservations, private function room bookings, and catering schedules by maintaining organized reservation records, preventing scheduling conflicts and duplicate bookings, facilitating timely reservation confirmation, and improving coordination among restaurant personnel responsible for reservation management.

4. **To develop an automated inventory monitoring module** capable of recording inventory movement, monitoring product availability, identifying low-stock items, determining reorder requirements, and generating inventory reports in order to improve inventory accuracy, reduce inventory discrepancies, prevent stock shortages and overstocking, and support efficient inventory replenishment and resource utilization.
5. **To incorporate a comprehensive Sales Analytics Dashboard** capable of collecting, processing, analyzing, and presenting operational data through graphical reports and statistical summaries that enable management to monitor sales performance, identify best-selling products, analyze customer purchasing behavior, evaluate promotional activities, recognize seasonal sales trends, forecast product demand, and support evidence-based managerial decision-making.
6. **To automate the generation of operational and managerial reports** by enabling the system to produce accurate, organized, and real-time reports related to sales transactions, reservation records, inventory status, catering schedules, product performance, and overall business operations, thereby reducing the time and administrative effort required for manual report preparation while improving the availability of information needed for planning and decision-making.
7. **To implement Role-Based Access Control (RBAC)** that regulates user accessibility by assigning appropriate permissions and system privileges to administrators, managers, cashiers, kitchen personnel, and other authorized

users, thereby ensuring that each user can only access functions and information relevant to their designated responsibilities while maintaining the confidentiality, integrity, availability, and security of organizational data.

8. **To evaluate the developed Online Management System** using the **ISO/IEC 25010 Software Quality Model** in terms of functional suitability, performance efficiency, usability, reliability, security, maintainability, and portability in order to determine the system's overall quality, effectiveness, and acceptability among the identified respondents.
9. **To determine whether the implementation of the developed system** contributed to improving the operational efficiency of Crates N' Plates by reducing dependence on manual procedures, improving data accuracy, strengthening communication among restaurant personnel, enhancing customer service, supporting evidence-based managerial decision-making, and promoting the long-term productivity, competitiveness, and sustainability of the organization.

1.4 Scope and Delimitation

This study focused on the design, development, implementation, and evaluation of an **Online Management System for Crates N' Plates with Sales Analytics and Reservation Scheduling**. The study aimed to address the operational challenges encountered by the restaurant by developing a centralized web-based information system capable of automating essential business processes, improving operational efficiency, and supporting evidence-based managerial decision-making. The developed system was specifically designed to replace the restaurant's existing manual operational procedures with an integrated digital platform that improved the accuracy, consistency, accessibility, and security of operational information.

The study concentrated on the operational requirements of **Crates N' Plates**, located at **108 Delgado Street, Barangay Mabulo-Delgado, Iloilo City**, which served as the beneficiary and implementation site of the proposed information system. The restaurant's existing workflow, operational procedures, employee responsibilities, and business processes served as the basis for the design and development of the proposed application. The researchers conducted interviews, observations, document reviews, and system analysis to identify operational requirements that guided the development of the system.

The developed application was implemented as a **web-based restaurant management system**, allowing authorized personnel to access system functionalities through desktop computers, laptops, tablets, and other internet-enabled devices using standard web browsers. The web-based architecture was selected to provide flexibility, centralized information management, and convenient system accessibility without requiring the installation of specialized software on individual client devices. Through centralized deployment, operational information remained synchronized and immediately accessible to authorized users regardless of their location within the restaurant premises.

The system primarily focused on integrating the restaurant's major operational activities into a unified information system. These operational components included customer order processing, Point-of-Sale (POS) transaction management, reservation scheduling, inventory monitoring, sales analytics, report generation, user management, and Role-Based Access Control (RBAC). Rather than operating as independent modules, these components were designed to communicate through a centralized database, allowing information generated from one operational process to be utilized automatically by other related modules. This integration minimized duplicate data entry, reduced inconsistencies among operational records, and improved organizational coordination.

One of the major components developed during the study was the **Point-of-Sale (POS) Management Module**. This module automated customer order processing, payment computation, receipt generation, transaction recording, and sales history maintenance. Customer orders entered by authorized cashiers were automatically recorded within the centralized database while simultaneously updating sales records and supporting report generation. The automation of transaction processing minimized manual computations, reduced transaction errors, shortened customer waiting time, and improved the overall efficiency of restaurant operations.

The developed system also incorporated an **Order Management Module** that improved communication between front-of-house personnel and kitchen staff. Once customer orders were confirmed through the Point-of-Sale system, food orders were electronically transmitted to the kitchen for preparation, eliminating the need for handwritten order slips and reducing communication delays. Kitchen personnel were provided with organized order information that enabled them to monitor food preparation efficiently while minimizing the possibility of misplaced orders and duplicated preparation. The module also enabled employees to monitor the preparation status of customer orders, thereby improving operational coordination and service delivery.

A comprehensive **Reservation Scheduling Module** was likewise developed to manage the restaurant's reservation services. The module accommodated dine-in reservations, private function room bookings, and catering reservations while maintaining complete customer reservation records within the centralized database. Authorized personnel could verify reservation availability, approve booking requests, update reservation information, modify schedules, or cancel reservations whenever necessary. Through automated reservation management, the system minimized scheduling conflicts, reduced duplicate bookings, improved reservation monitoring, and strengthened communication among restaurant personnel responsible for handling customer reservations.

The study further included the development of an **Inventory Monitoring Module** responsible for maintaining accurate inventory records for both kitchen ingredients and beverage supplies. The module recorded inventory movement, monitored stock availability, identified low-stock items, generated inventory reports, and assisted management in monitoring inventory consumption. Inventory information was updated continuously as sales transactions were completed, enabling management to maintain more accurate inventory records while reducing discrepancies commonly associated with manual inventory monitoring. This functionality supported more efficient purchasing decisions and reduced the possibility of stock shortages, overstocking, and unnecessary inventory wastage.

To support managerial planning and business evaluation, the researchers incorporated a **Sales Analytics Dashboard** capable of transforming operational data into meaningful business information. The analytics module presented graphical and statistical summaries of business performance, including daily, weekly, monthly, and yearly sales, best-selling products, revenue trends, inventory movement, customer purchasing behavior, and product demand. Through these visual reports, management was provided with comprehensive operational information that supported evidence-based planning, inventory replenishment, pricing decisions, promotional activities, and strategic business development.

Another important component included in the developed application was the **Automated Report Generation Module**. Unlike the restaurant's previous manual reporting process, the proposed system generated reports automatically using information stored within the centralized database. Authorized users could generate reports related to sales transactions, inventory status, reservation activities, catering schedules, employee activities, and overall restaurant performance whenever necessary. The automation of report generation reduced administrative workload while improving the timeliness, consistency, and reliability of business reports required for operational monitoring and managerial decision-making.

The researchers also implemented **Role-Based Access Control (RBAC)** to strengthen system security and protect confidential organizational information. Different user roles were established according to employee responsibilities, including administrators, managers, cashiers, kitchen personnel, and other authorized users. Each user was granted only the permissions necessary for performing assigned operational tasks, thereby preventing unauthorized access to confidential business information while improving accountability and maintaining data integrity.

The study further included the evaluation of the developed system using the **ISO/IEC 25010 Software Quality Model**. The evaluation focused on determining the quality and acceptability of the proposed application based on seven software quality characteristics, namely **functional suitability, performance efficiency, usability, reliability, security, maintainability, and portability**. Evaluation respondents consisted of individuals directly involved in the restaurant's daily operations and selected information technology professionals who assessed whether the developed application satisfied the operational requirements of Crates N' Plates while meeting recognized software quality standards.

Delimitation of the Study

Although the developed system integrated several major operational functions of the restaurant, the study was subject to specific limitations that defined its scope.

The developed application was designed **exclusively for Crates N' Plates** and reflected the restaurant's existing organizational structure, operational workflow, and business requirements. Consequently, the system was not intended for immediate implementation in multi-branch restaurants, franchise operations, hotel restaurant chains, or other large-scale food service organizations without further system modification and customization.

The study focused primarily on automating internal restaurant operations. Therefore, the developed system did **not** include integration with third-party food delivery services such as **GrabFood**, **foodpanda**, or similar online delivery platforms. Likewise, integration with external accounting software, supplier management systems, procurement systems, customer relationship management (CRM) platforms, payroll systems, and enterprise resource planning (ERP) applications was beyond the scope of the study.

The proposed application also excluded online payment gateways, digital wallets, banking integration, cryptocurrency payment options, and automated payment verification services. Customer payments were processed according to the operational procedures currently practiced by Crates N' Plates during the conduct of the study.

Similarly, the study did not include Internet of Things (IoT)-enabled inventory monitoring devices, automated kitchen equipment, smart sensors, barcode scanners, RFID inventory tracking, artificial intelligence-based sales forecasting, machine learning recommendation systems, predictive analytics, facial recognition technologies, voice-assisted ordering, chatbot integration, or mobile application development. Although these technologies could further improve restaurant operations, their implementation was beyond the objectives and available resources of the present study.

Furthermore, the researchers limited the evaluation to the development and implementation period of the study. Long-term organizational performance, return on investment (ROI), post-deployment customer satisfaction, employee productivity after extended system usage, and long-term financial impact were not included in the evaluation. The assessment focused solely on determining the software quality and operational acceptability of the developed system based on the **ISO/IEC 25010 Software Quality Model**.

Finally, the study concentrated on the design, development, implementation, and evaluation of the proposed Online Management System and did not attempt to redesign the restaurant's organizational policies, staffing structure, management practices, pricing strategies, menu composition, or overall business model. Any future enhancements beyond the functionalities developed during the study were considered recommendations for subsequent research and system development initiatives.

1.5 Significance of the Study

The development of the **Online Management System for Crates N' Plates with Sales Analytics and Reservation Scheduling** was expected to provide significant contributions to the restaurant and its various stakeholders by improving operational efficiency, strengthening information management, enhancing service quality, and supporting evidence-based managerial decision-making. Through the integration of sales transaction processing, reservation scheduling, inventory monitoring, sales analytics, automated reporting, and secure user management into a centralized web-based platform, the study sought to address the operational limitations associated with manual restaurant management. Furthermore, the implementation of the proposed system was expected to contribute not only to the operational improvement of Crates N' Plates but also to the broader application of information technology within the food service industry.

Specifically, the findings and implementation of this study were expected to benefit the following:

Crates N' Plates

Crates N' Plates was the primary beneficiary of this study because the proposed information system was specifically designed according to its operational requirements and business processes. The implementation of the developed system was expected to improve the overall efficiency of restaurant operations by replacing manual procedures with an integrated digital management platform capable of automating sales transaction processing, reservation scheduling, inventory monitoring, report generation, and user management.

The centralized database developed for the restaurant was expected to improve the accuracy, consistency, accessibility, and security of operational information while reducing the occurrence of duplicate records, misplaced documents, and manual recording errors. Through the integration of Sales Analytics, management was also expected to gain meaningful insights into customer purchasing behavior, product demand, inventory movement, and overall business performance. These analytical capabilities would support evidence-based managerial planning, improve resource utilization, strengthen operational control, and contribute to the restaurant's long-term growth, competitiveness, and sustainability.

Administrators and Managers

The developed system was expected to provide administrators and managers with a comprehensive information management tool capable of supporting daily operational supervision and long-term strategic planning. Through the integration of Point-of-Sale management, reservation scheduling, inventory monitoring, automated reporting, and sales analytics, management would be provided with timely and reliable operational information necessary for monitoring restaurant performance and making informed business decisions.

The Sales Analytics Dashboard was expected to assist administrators in evaluating revenue trends, identifying best-selling products, monitoring customer purchasing patterns, assessing promotional effectiveness, and analyzing business performance through graphical reports and statistical summaries. These analytical capabilities would allow management to develop more effective operational strategies, optimize inventory planning, improve financial management, and allocate organizational resources more efficiently. Furthermore, the availability of real-time reports would reduce delays associated with manual report preparation while strengthening management's ability to respond promptly to operational concerns.

Employees, Cashiers, and Kitchen Personnel

The proposed system was expected to improve employee productivity by reducing the administrative workload associated with manual operational procedures. Cashiers would benefit from the implementation of the integrated Point-of-Sale (POS) module, which automated customer order processing, payment computation, receipt generation, and sales recording. Through automation, transaction processing would become faster, more accurate, and less susceptible to computational errors.

Kitchen personnel would likewise benefit from the Order Management Module, which electronically transmitted customer orders from the cashier to the kitchen, thereby reducing communication delays and minimizing the possibility of misplaced or duplicated order slips. Reservation scheduling and inventory monitoring modules would further simplify employee responsibilities by eliminating the need for repetitive manual record-keeping. Consequently, employees would be able to devote more time to customer service while improving coordination, productivity, and operational efficiency during both regular and peak business hours.

Customers

Customers were expected to benefit from improved service quality resulting from the automation of restaurant operations. The implementation of the Point-of-Sale system was expected to reduce customer waiting time by enabling faster transaction processing, accurate billing, and more efficient order handling. Likewise, the Reservation Scheduling Module would provide a more organized and reliable reservation process by minimizing scheduling conflicts, delayed confirmations, and duplicate bookings.

The improved coordination between cashiers and kitchen personnel would also contribute to more efficient food preparation and order fulfillment, resulting in shorter waiting times and more accurate service delivery. Overall, the implementation of the proposed system was expected to create a more organized, responsive, and customer-oriented dining experience that enhanced customer satisfaction while strengthening customer confidence in the restaurant's services.

The Restaurant Industry

Although the developed system was specifically designed for Crates N' Plates, the findings of this study may also provide valuable insights for other small and medium-sized restaurants that continue to experience operational challenges associated with manual management practices. The study demonstrates how integrated information systems can improve transaction processing, reservation management, inventory monitoring, reporting, and business analytics without requiring highly complex enterprise-level solutions.

Restaurant owners and managers may utilize the concepts and system architecture presented in this study as references when planning similar digital transformation initiatives for their own establishments. Consequently, the study contributes to the continuing modernization of restaurant operations through the effective utilization of information technology.

The Academic Community

This study contributes to the growing body of knowledge in the fields of Information Technology, Information Systems, Business Process Automation, and Restaurant Management Systems. It demonstrates the practical application of software engineering principles, systems analysis and design, database management, web development, and software quality evaluation within a real-world business environment.

The documentation of the research process, system architecture, development methodology, implementation procedures, and software evaluation provides educators, students, and academic institutions with additional reference material that may support future instruction and research involving information systems development.

Future Researchers

The findings of this study may serve as a useful reference for future researchers interested in conducting similar investigations involving restaurant management systems, inventory management, sales analytics, reservation scheduling, business intelligence, and digital transformation within small and medium-sized enterprises.

Future researchers may build upon the methodologies, conceptual framework, system architecture, research instruments, and evaluation procedures presented in this study. Likewise, the limitations identified during system development may provide opportunities for conducting more advanced investigations involving emerging technologies applicable to restaurant information systems.

Future Software Developers

The developed system may serve as a foundation for future software developers interested in improving restaurant management applications. The modular architecture adopted during system development allows future enhancements without requiring complete system redevelopment.

Future developers may integrate additional technologies and functionalities, including mobile application support, cloud deployment, online payment gateways, customer loyalty and rewards systems, supplier management, procurement automation, barcode and QR-code integration, RFID-based inventory tracking, artificial intelligence-assisted sales forecasting, predictive analytics, Internet of Things (IoT)-enabled inventory monitoring, and integration with third-party food delivery platforms. These enhancements may further improve the operational capabilities of restaurant management systems while addressing the evolving technological requirements of the food service industry.

Information Technology Students

The study may also benefit students pursuing degrees in Information Technology, Computer Science, and related disciplines by providing a practical example of how theoretical concepts may be applied to solve real-world organizational problems. The system development process demonstrates the application of software engineering methodologies, systems analysis and design techniques, database management, web development, software testing, and information security within an actual business environment.

Students may utilize this study as a reference when conducting similar capstone projects, software development research, or feasibility studies involving business automation and information systems implementation.

In summary, this study emphasized the significant role of information technology in improving organizational performance through the automation and integration of essential restaurant operations. The development and implementation of the **Online Management System for Crates N' Plates with Sales Analytics and Reservation**

Scheduling was expected to improve operational efficiency, strengthen information management, enhance customer service, support evidence-based managerial decision-making, and promote sustainable business growth. Beyond its immediate benefits to Crates N' Plates, the study also contributes to the broader advancement of digital transformation initiatives within the food service industry and provides valuable references for educators, researchers, software developers, students, and organizations pursuing similar technological innovations.

1.6 Definition of Terms

To provide a clearer understanding of the concepts used throughout the study, the following terms are defined conceptually and operationally as applied in this research.

Analytics Dashboard

Conceptual Definition. An analytics dashboard is a visual information interface that consolidates and presents organizational data through charts, graphs, tables, and other graphical representations to facilitate performance monitoring, trend analysis, and informed decision-making *(Author, Year)*.

Operational Definition. In this study, an analytics dashboard refers to the graphical interface integrated into the developed Online Management System that presents sales trends, revenue summaries, best-selling products, customer purchasing patterns, inventory movement, and other business performance indicators for Crates N' Plates.

Catering Service

Conceptual Definition. Catering service refers to the preparation, delivery, and provision of food and beverages for meetings, celebrations, social gatherings, corporate functions, and other special events conducted either within or outside a business establishment *(Author, Year)*.

Operational Definition. In this study, catering service refers to one of the primary business operations of Crates N' Plates in which customer bookings, schedules, event details, and reservation records are managed through the developed Online Management System.

Inventory Monitoring

Conceptual Definition. Inventory monitoring is the continuous process of recording, tracking, controlling, and evaluating inventory movement to ensure that adequate stock levels are maintained while preventing shortages, overstocking, and inventory losses *(Author, Year)*.

Operational Definition. In this study, inventory monitoring refers to the system module responsible for recording stock movement, monitoring kitchen ingredients and beverage supplies, identifying low-stock items, maintaining inventory records, and generating inventory reports that support inventory management and replenishment planning.

Online Management System

Conceptual Definition. An online management system is an integrated information system that utilizes internet-based technologies to automate organizational processes, manage business information, facilitate communication, and support operational and managerial activities through centralized data processing *(Author, Year)*.

Operational Definition. In this study, the Online Management System refers to the web-based application developed specifically for Crates N' Plates that integrates Point-of-Sale management, reservation scheduling, inventory monitoring, sales analytics, report generation, and user management into a centralized platform.

Online Reservation System

Conceptual Definition. An online reservation system is a computerized scheduling platform that enables users to create, manage, modify, monitor, and organize reservations electronically while improving scheduling accuracy and resource allocation *(Author, Year)*.

Operational Definition. In this study, the online reservation system refers to the reservation scheduling module responsible for managing dine-in reservations, private function room bookings, and catering reservations while reducing scheduling conflicts, duplicate bookings, and delayed confirmations.

Point-of-Sale (POS) System

Conceptual Definition. A Point-of-Sale (POS) system is a computerized transaction processing system used to record customer purchases, process payments, generate receipts, maintain sales records, and support business reporting *(Author, Year)*.

Operational Definition. In this study, the Point-of-Sale (POS) system refers to the integrated transaction processing module that records customer orders, computes payments, generates electronic receipts, updates sales records, and maintains transaction histories within the developed Online Management System.

Report Generation

Conceptual Definition. Report generation refers to the systematic process of collecting, organizing, processing, and presenting business information in the form of reports that support operational monitoring and managerial decision-making *(Author, Year)*.

Operational Definition. In this study, report generation refers to the automated process performed by the developed system in producing sales reports, reservation reports, inventory reports, catering reports, and other operational summaries in real time.

Restaurant Management System

Conceptual Definition. A Restaurant Management System is an integrated software application designed to automate and coordinate restaurant operations, including customer order processing, sales transactions, reservations, inventory management, employee management, reporting, and customer information management *(Author, Year)*.

Operational Definition. In this study, the Restaurant Management System refers to the web-based information system developed for Crates N' Plates to automate and integrate its daily operational activities through a centralized digital platform.

Role-Based Access Control (RBAC)

Conceptual Definition. Role-Based Access Control (RBAC) is an information security model that restricts system access by assigning permissions according to predefined organizational roles and responsibilities rather than individual users *(Author, Year)*.

Operational Definition. In this study, Role-Based Access Control refers to the security mechanism implemented within the developed system that assigns specific access privileges to administrators, managers, cashiers, kitchen personnel, and other authorized users according to their designated responsibilities.

Sales Analytics

Conceptual Definition. Sales analytics is the process of collecting, organizing, analyzing, interpreting, and presenting sales-related information to evaluate business performance, identify trends, forecast demand, and support managerial decision-making *(Author, Year)*.

Operational Definition. In this study, sales analytics refers to the integrated dashboard that analyzes restaurant sales data, identifies best-selling products, monitors customer purchasing behavior, evaluates revenue performance, and generates graphical reports for managerial use.

Scheduling System

Conceptual Definition. A scheduling system is an information system designed to organize, coordinate, and manage appointments, reservations, bookings, and events according to predetermined dates, times, and available resources *(Author, Year)*.

Operational Definition. In this study, the scheduling system refers to the reservation management module responsible for organizing dine-in reservations, catering schedules, and private function room bookings while preventing scheduling conflicts and improving reservation coordination.

Web-Based Application

Conceptual Definition. A web-based application is a software application that operates through a web browser using internet or network technologies, allowing users to access system functionalities without installing software on individual client devices *(Author, Year)*.

Operational Definition. In this study, the web-based application refers to the Online Management System developed for Crates N' Plates that enables authorized users to manage restaurant operations, sales transactions, reservations, inventory, reports, and sales analytics through internet-connected devices using a standard web browser.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Study 1. Restaurant Revenue Management: A Systematic Literature Review and Future Challenges

Bujalance-López et al. (2025) conducted a systematic literature review entitled **"Restaurant Revenue Management: A Systematic Literature Review and Future Challenges."** The primary objective of the study was to examine the evolution of restaurant revenue management (RRM) by synthesizing previous scholarly publications related to reservation systems, customer behavior, revenue optimization, and technological innovations within restaurant operations. The researchers also sought to identify emerging trends and future research opportunities that could further improve restaurant management practices.

The researchers employed the **Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)** methodology in reviewing published literature. Using multiple academic databases, the study analyzed peer-reviewed articles focusing on revenue optimization, reservation management, digital technologies, customer behavior, and business analytics within restaurant environments. The collected literature was evaluated through qualitative synthesis to identify recurring themes, research gaps, and technological developments affecting restaurant operations.

The findings revealed that modern restaurant businesses increasingly depend on integrated information systems capable of combining reservation management, customer relationship management, sales analytics, and business intelligence to improve organizational performance. Digital transformation has enabled restaurants to optimize seating utilization, analyze customer purchasing behavior, forecast demand, and improve decision-making through data-driven management. The researchers emphasized that reservation systems, analytical dashboards, and integrated operational platforms have become essential technologies for maintaining competitiveness within the hospitality industry. They further recommended that future restaurant management systems incorporate advanced analytics and intelligent decision-support capabilities to improve operational efficiency and long-term sustainability.

This study is highly relevant to the present research because it highlights the growing importance of integrating reservation scheduling and sales analytics into restaurant management systems. Similar to the present study, it recognizes that modern restaurant operations require centralized information systems capable of supporting operational management and strategic decision-making. However, unlike the systematic review, the present study focuses on the actual design, development, and evaluation of an Online Management System specifically tailored to the operational requirements of Crates N' Plates.

Study 2. Innovating Restaurant Inventory Management Application Using the Extreme Programming Method

Sugianto and colleagues (2023) conducted a study entitled **"Innovating Restaurant Inventory Management Application Using the Extreme Programming Method."** The study aimed to develop a computerized inventory management application capable of addressing common inventory problems encountered by restaurants, including inaccurate stock recording, inventory shortages, delayed replenishment, and material wastage. The researchers recognized that manual inventory management frequently resulted in operational inefficiencies that affected restaurant productivity and profitability.

The researchers adopted the **Extreme Programming (XP)** software development methodology, an Agile software development approach emphasizing iterative development, continuous testing, and active user participation throughout system development. The application was designed to automate inventory recording, monitor stock movement, generate inventory reports, and assist restaurant personnel in maintaining accurate inventory information. The completed system was evaluated based on its functionality, usability, and operational effectiveness within restaurant environments.

The study demonstrated that the computerized inventory application significantly improved inventory accuracy while reducing manual record-keeping and administrative workload. Automated inventory monitoring enabled restaurant personnel to identify low-stock items promptly, monitor inventory movement more effectively, and minimize inventory discrepancies caused by manual recording errors. The researchers concluded that integrating inventory management into computerized information systems improves organizational efficiency while reducing inventory losses and unnecessary purchasing costs.

The findings are closely related to the present study because the proposed Online Management System likewise incorporates an automated inventory monitoring module designed to maintain accurate stock records, monitor inventory movement, identify low-stock products, and generate inventory reports. However, the present study extends beyond inventory management by integrating Point-of-Sale operations, reservation scheduling, sales analytics, and report generation into a centralized web-based platform.

Study 3. Integrated Restaurant Management System (IRMS)

Pasalkar et al. (2023) conducted a study entitled **"Restaurant Management System."** The primary objective of the research was to develop an Integrated Restaurant Management System (IRMS) capable of modernizing restaurant operations through the integration of customer ordering, inventory management, staff scheduling, customer engagement, and operational reporting within a centralized information system. The researchers observed that traditional restaurant management practices remained dependent on fragmented systems and manual procedures that limited operational efficiency and managerial control.

The researchers designed the proposed Integrated Restaurant Management System using a centralized software architecture that connected various operational modules within a single application. The developed system integrated customer order processing, inventory management, employee scheduling, customer engagement, and reporting functionalities. By centralizing operational information, the system enabled restaurant managers to monitor restaurant activities more efficiently while reducing repetitive administrative tasks.

The findings indicated that integrating multiple restaurant operations into a unified information system significantly improved communication among employees, strengthened inventory control, enhanced customer service, and provided managers with real-time operational information necessary for effective decision-making. The researchers emphasized that centralized restaurant management systems eliminate fragmented workflows and improve organizational productivity by allowing operational information to flow automatically among different business functions.

The present study is closely related because it also develops an integrated restaurant management system that centralizes sales transactions, reservation scheduling, inventory monitoring, reporting, and user management. However, the proposed system for Crates N' Plates further distinguishes itself by incorporating a **Sales Analytics Dashboard** and **Role-Based Access Control (RBAC)**, features that provide enhanced managerial support and information security beyond the capabilities discussed in the Integrated Restaurant Management System.

2.2 Local Studies

Study 1. Analysis of Inventory Management Systems of Selected Small-Sized Restaurants in Quezon Province

Baylen (2020) conducted a study entitled **"Analysis of Inventory Management Systems of Selected Small-Sized Restaurants in Quezon Province: Basis for an Inventory System Manual."** The primary objective of the study was to examine the inventory management practices of selected small-sized restaurants in Quezon Province and determine how existing inventory procedures affected business operations. Specifically, the study evaluated inventory recording, stock monitoring, purchasing practices, and inventory control measures used by restaurant owners. The researcher also sought to develop an inventory system manual that could improve inventory management among small restaurant businesses.

The researcher employed a **descriptive research design** using survey questionnaires and interviews with restaurant owners and managers. Data collected from participating restaurants were analyzed to determine the effectiveness of their inventory management practices and identify common operational problems associated with manual inventory recording. The study focused on inventory accuracy, inventory control procedures, stock replenishment, and inventory documentation.

The findings revealed that many small restaurants continued to depend on manual inventory management practices, resulting in inventory discrepancies, delayed replenishment of supplies, inaccurate inventory records, and operational inefficiencies. The researcher concluded that computerized inventory systems could significantly improve inventory monitoring, reduce recording errors, strengthen inventory control, and support better purchasing decisions. Furthermore, the study emphasized that systematic inventory management contributes directly to improved operational efficiency and organizational productivity.

The present study is closely related because it likewise recognizes inventory management as one of the critical operational functions requiring automation. Similar to Baylen's (2020) study, the proposed Online Management System incorporates an inventory monitoring module designed to record inventory movement, identify low-stock products, generate inventory reports, and improve inventory accuracy. However, unlike Baylen's research, which primarily focused on inventory management, the present study integrates inventory monitoring with Point-of-Sale transactions, reservation scheduling, sales analytics, and report generation within a centralized web-based information system.

Study 2. Steffi's Online Reservation Management System (SORMS)

Villanueva (2025) developed **Steffi's Online Reservation Management System (SORMS)** to improve reservation processing through a web-based platform. The study aimed to replace traditional reservation methods with an automated system capable of managing customer bookings efficiently while improving communication between customers and business personnel. The proposed system focused on reservation scheduling, booking confirmation, reservation modification, and record management within a centralized information system.

The researcher utilized a **developmental research methodology**, incorporating system analysis, software development, implementation, and evaluation. The application was designed to automate reservation processing while maintaining customer information and reservation schedules within a centralized database. System evaluation focused on functionality, usability, efficiency, and user satisfaction among participating respondents.

The study demonstrated that implementing an online reservation management system significantly improved reservation accuracy, minimized scheduling conflicts, reduced duplicate bookings, and accelerated reservation confirmation. Respondents also reported improved customer satisfaction because reservation information became easier to access, update, and monitor. The centralized reservation database improved communication among personnel responsible for reservation management while reducing administrative workload.

The findings strongly support the present study because the proposed Online Management System likewise incorporates a reservation scheduling module that manages dine-in reservations, private function room bookings, and catering reservations. However, unlike the SORMS application, the proposed system further integrates Point-of-Sale management, inventory monitoring, sales analytics, and automated reporting into a single restaurant management platform.

Study 3. Web-Based Case Management Information System

Grepon (2021) developed a **Web-Based Case Management Information System** for trial courts in the Philippines. Although the application was implemented within the judicial sector, the study examined the effectiveness of centralized web-based information systems in improving organizational efficiency, information management, and operational coordination. The objective of the research was to replace manual record management with a centralized digital platform capable of improving information accessibility and reducing administrative workload.

The researcher adopted the **Software Development Life Cycle (SDLC)** in designing and implementing the proposed information system. Functional testing and user evaluation were conducted to determine the effectiveness of the application in improving organizational productivity, information retrieval, and operational coordination. The study emphasized the importance of centralized databases, secure information management, and web-based accessibility.

The findings indicated that the implementation of the web-based information system significantly reduced document retrieval time, improved operational efficiency, strengthened information security, and enhanced organizational productivity. The researcher concluded that centralized information systems contribute substantially to organizational effectiveness by improving information accessibility and reducing manual administrative procedures.

Although the application was developed for judicial case management, the concepts presented are highly relevant to the present study because both systems utilize centralized web-based information management to improve operational efficiency. Similar to Grepon's study, the proposed Online Management System provides centralized access to operational information while improving coordination among authorized users. However, the present study specifically addresses restaurant management through the integration of sales transactions, reservation scheduling, inventory monitoring, analytics, and reporting.

2.3 Related Literature

Information Systems

Information systems have become one of the fundamental technological resources utilized by organizations to improve operational efficiency, facilitate communication, support managerial decision-making, and achieve organizational objectives. As organizations continue to operate in increasingly dynamic and competitive environments, the demand for timely, accurate, and relevant information has significantly increased. Information systems provide the technological infrastructure that enables organizations to collect, process, store, retrieve, and distribute information necessary for both operational and strategic activities. Rather than functioning solely as data repositories, modern information systems integrate people, business processes, software applications, hardware resources, and communication networks into a unified environment that supports organizational performance and continuous improvement (Laudon & Laudon, 2022).

The evolution of information systems has transformed the manner in which organizations perform their daily operations. Traditional business processes that relied heavily on paper-based documentation and manual procedures have gradually been replaced by computerized systems capable of processing large volumes of information with greater speed, consistency, and accuracy. This transformation has enabled organizations to reduce operational costs, minimize human error, improve customer service, and strengthen managerial control through real-time access to organizational information. Consequently, information systems have become strategic organizational assets that support innovation, operational excellence, and sustainable competitive advantage (Laudon & Laudon, 2022).

Modern organizations increasingly depend on integrated information systems because business operations are no longer isolated activities. Sales transactions, inventory management, customer services, financial reporting, procurement, and employee management are interconnected functions that require continuous information exchange. Integrated information systems eliminate fragmented workflows by enabling operational data generated from one business process to become immediately available to other related functions. This integration improves organizational coordination while reducing duplicate work, inconsistencies in business records, and delays in decision-making. According to Laudon and Laudon (2022), enterprise-wide information systems improve organizational performance by facilitating seamless communication among departments while providing management with comprehensive information necessary for planning, controlling, and evaluating business operations.

The significance of information systems extends beyond operational efficiency because they also support organizational decision-making at different managerial levels. Managers utilize information systems to monitor organizational performance, evaluate operational activities, identify emerging trends, forecast future demands, and formulate strategic decisions based on reliable information rather than intuition alone. Decision-support capabilities incorporated into contemporary information systems enable organizations to transform operational data into meaningful information through analytical reports, graphical dashboards, and performance indicators. These capabilities improve organizational responsiveness while allowing management to identify operational problems before they significantly affect business performance (Laudon & Laudon, 2022).

Recent systematic literature reviews further emphasized the growing importance of information systems in organizational transformation. Aziz (2024) explained that the quality and effectiveness of information systems depend not only on technological infrastructure but also on information quality, system quality, service quality, organizational processes, and user acceptance. The review synthesized numerous empirical studies and concluded that successful information systems improve organizational productivity by providing reliable, accessible, and relevant information that supports operational efficiency and managerial decision-making. The study further highlighted that organizations increasingly recognize information quality as a critical factor influencing system effectiveness and user satisfaction.

Similarly, a systematic literature review conducted on information systems implementation reported that organizational performance is significantly influenced by the successful integration of information systems into business processes. The review emphasized that organizations implementing well-designed information systems experienced improvements in communication, coordination, information accessibility, operational performance, and service delivery. However, the researchers also noted that the effectiveness of information systems depends on organizational readiness, user participation, management support, and continuous system improvement. These findings demonstrate that information systems should be viewed not merely as technological tools but as organizational resources that facilitate business transformation and continuous process improvement.

Information systems have likewise become essential within service-oriented industries such as hospitality and food service. Restaurants generate substantial amounts of operational information daily, including customer orders, inventory transactions, reservation schedules, sales records, employee activities, and financial reports. Managing these information resources manually often results in delayed processing, inaccurate records, communication gaps, and reduced operational efficiency. Consequently, restaurant businesses increasingly adopt integrated information systems capable of automating operational activities while maintaining centralized databases that support real-time information sharing among authorized personnel.

Within restaurant operations, information systems integrate several functional components including Point-of-Sale (POS) systems, inventory management, reservation scheduling, customer information management, sales reporting, and business analytics. These integrated applications improve operational coordination by ensuring that information entered into one functional module is automatically reflected in other related modules. For example, completed customer transactions automatically update inventory records, generate financial reports, and contribute to sales analytics without requiring repetitive manual data entry. Such integration minimizes operational inconsistencies while improving information accuracy and managerial decision-making.

Another significant contribution of information systems is their ability to support organizational adaptability in rapidly changing business environments. Digital transformation has increased customer expectations regarding service speed, transaction accuracy, and personalized experiences. Organizations capable of utilizing integrated information systems are generally more responsive to market changes because they possess immediate access to operational information necessary for timely decision-making. Information systems therefore contribute not only to operational efficiency but also to organizational resilience, innovation, and long-term competitiveness.

The relevance of information systems to the present study is highly significant because the proposed **Online Restaurant Management System with Sales Analytics and Reservation Scheduling** primarily functions as an integrated information system specifically developed for Crates N' Plates. Similar to the concepts presented in the reviewed literature, the proposed system integrates sales transaction processing, reservation scheduling, inventory monitoring, report generation, and sales analytics into a centralized web-based platform. Through this integration, the system seeks to improve operational efficiency, strengthen information management, minimize manual errors, and support evidence-based managerial decision-making. The reviewed literature therefore provides a strong theoretical foundation supporting the development of an integrated information system capable of addressing the operational challenges experienced by Crates N' Plates.

Restaurant Management Systems

Restaurant Management Systems (RMS) have become indispensable technological solutions within the food service industry because they automate and integrate various operational processes into a centralized information system. As restaurants experience increasing customer demands, diverse service offerings, and complex operational activities, the implementation of computerized management systems has become essential in maintaining efficiency, accuracy, and service quality. Unlike traditional manual procedures that rely heavily on paper-based records, Restaurant Management Systems provide an integrated platform that manages customer orders, Point-of-Sale (POS) transactions, reservation scheduling, inventory monitoring, employee management, kitchen operations, financial reporting, and business analytics within a single application (Laudon & Laudon, 2022).

The evolution of Restaurant Management Systems reflects the broader digital transformation occurring within the hospitality industry. Initially, restaurant computerization focused primarily on electronic billing and Point-of-Sale functions. However, advances in information technology have expanded the capabilities of RMS by integrating customer relationship management, inventory control, reservation management, online ordering, business intelligence, cloud computing, and mobile accessibility. These technological developments have enabled restaurants to streamline daily operations, reduce administrative workload, improve communication among departments, and provide customers with faster and more reliable services (Buhalis & Leung, 2018).

One of the primary advantages of Restaurant Management Systems is the integration of operational activities into a centralized database. Rather than maintaining separate records for customer orders, inventory, reservations, and financial transactions, RMS allows all operational information to be stored, processed, and updated automatically. This integration reduces duplicate data entry, minimizes human errors, and ensures that information remains consistent throughout the organization. For example, when a customer order is processed through the Point-of-Sale system, inventory records are automatically updated, sales reports are generated, and management dashboards reflect the transaction in real time. Such automation significantly improves organizational coordination while allowing managers to monitor restaurant performance more efficiently (Prakash et al., 2025).

Recent studies have demonstrated that integrated Restaurant Management Systems significantly improve operational efficiency, customer satisfaction, and managerial decision-making. Prakash et al. (2025) developed a cloud-integrated Restaurant Point-of-Sale System that combined billing, order processing, inventory management, and reporting into a centralized platform. The researchers found that integrating these operational functions reduced transaction errors, accelerated customer service, improved inventory accuracy, and provided restaurant managers with real-time access to operational information. Furthermore, the study emphasized that cloud-enabled restaurant management systems improve organizational flexibility because managers can remotely monitor business performance while maintaining secure and centralized operational records. These findings indicate that modern Restaurant Management Systems contribute not only to operational automation but also to improved organizational responsiveness and strategic management.

Similarly, Pasalkar et al. (2023) proposed an Integrated Restaurant Management System that connected customer ordering, employee scheduling, inventory management, customer engagement, and reporting into a unified software platform. The researchers observed that many restaurants continued to experience operational inefficiencies due to fragmented workflows and independent management of operational activities. Their findings demonstrated that integrating restaurant operations into a centralized information system improved communication among personnel, reduced administrative workload, enhanced inventory control, and enabled managers to monitor restaurant performance more effectively. The study further emphasized that centralized restaurant management systems eliminate unnecessary duplication of work while improving organizational productivity through efficient information sharing.

Another important characteristic of Restaurant Management Systems is their ability to support data-driven managerial decision-making. Modern RMS applications continuously collect operational data generated from customer transactions, reservations, inventory movement, and financial activities. These data are transformed into meaningful business information through analytical dashboards and automated reports, allowing restaurant managers to evaluate sales performance, identify best-selling menu items, monitor customer preferences, forecast inventory requirements, and assess employee productivity. Consequently, restaurant owners no longer rely solely on intuition or manual observations but instead make strategic decisions based on comprehensive operational information (Buhalis & Leung, 2018).

Restaurant Management Systems also contribute significantly to improving customer experience. Through automated reservation scheduling, digital order processing, and accurate billing, customers receive faster, more reliable, and more convenient services. Integrated customer databases likewise enable restaurants to maintain customer profiles, reservation histories, and purchasing records that support personalized service and stronger customer relationships. As customer expectations continue to evolve, restaurants increasingly recognize that information technology plays a vital role in maintaining competitiveness and delivering high-quality customer experiences.

Furthermore, Restaurant Management Systems strengthen internal organizational control by implementing security mechanisms that regulate access to confidential business information. Modern systems commonly incorporate authentication procedures, user account management, audit logs, and Role-Based Access Control (RBAC) to ensure that employees access only those system functions relevant to their assigned responsibilities. Such security measures improve data integrity, accountability, and organizational governance while protecting sensitive business information from unauthorized access or accidental modification.

The adoption of Restaurant Management Systems has become particularly important for small and medium-sized restaurants seeking to improve operational performance while managing limited financial and human resources. Integrated management systems reduce manual administrative work, minimize operational errors, improve information accuracy, and optimize resource utilization without requiring substantial increases in staffing. As a result, restaurant owners are better positioned to respond to changing customer demands while sustaining business growth and long-term competitiveness.

The concepts presented in the reviewed literature are directly relevant to the present study because the proposed **Online Restaurant Management System with Sales Analytics and Reservation Scheduling** was designed using the same principles of integration, automation, and centralized information management. Similar to the Restaurant Management Systems discussed by previous authors, the proposed system combines Point-of-Sale transactions, reservation scheduling, inventory monitoring, sales analytics, report generation, and user management within a unified web-based platform. However, unlike many existing restaurant systems that emphasize transaction processing alone, the proposed system incorporates a comprehensive Sales Analytics Dashboard that enables management to monitor business performance, identify sales trends, evaluate customer purchasing behavior, and support evidence-based managerial decision-making. Therefore, the reviewed literature provides a strong theoretical foundation supporting the development of an integrated Restaurant Management System capable of addressing the operational challenges encountered by Crates N' Plates.

Digital Transformation

Digital transformation has become one of the most significant developments influencing organizations across various industries, including the hospitality and food service sectors. It refers to the integration of digital technologies into business processes, organizational strategies, and customer interactions to improve operational efficiency, service quality, and organizational competitiveness. Unlike simple computerization, digital transformation involves redesigning business operations by utilizing emerging technologies such as cloud computing, artificial intelligence, mobile applications, data analytics, and integrated information systems. Through digital transformation, organizations are able to automate routine tasks, improve collaboration, optimize resource utilization, and respond more effectively to changing market conditions (Vial, 2019).

The increasing adoption of digital technologies has fundamentally changed the way organizations conduct business. Traditional business operations that relied heavily on manual documentation and face-to-face transactions have gradually shifted toward digital platforms capable of providing real-time information, automated workflows, and data-driven decision-making. According to Vial (2019), digital transformation enables organizations to create value by improving business processes, enhancing customer experiences, and developing new organizational capabilities that support innovation and long-term sustainability. Rather than viewing technology merely as an operational tool, organizations increasingly recognize digital transformation as a strategic initiative that contributes to organizational resilience and competitive advantage.

Within the hospitality industry, digital transformation has significantly influenced restaurant operations by introducing technologies that improve both customer service and internal management. Restaurants have adopted digital ordering systems, electronic payment platforms, online reservation systems, inventory management software, cloud-based Point-of-Sale systems, and sales analytics dashboards to improve operational efficiency while responding to customers' increasing expectations for speed, convenience, and service quality. These technologies allow restaurant managers to coordinate daily operations more effectively while reducing reliance on manual procedures (Buhalis & Leung, 2018).

The COVID-19 pandemic further accelerated digital transformation within the food service industry. As restaurants adapted to changing consumer behavior, businesses increasingly relied on online ordering platforms, contactless payment systems, digital reservations, and cloud-based management applications to maintain operations despite mobility restrictions. According to Verhoef et al. (2021), digital transformation became essential for organizational survival because digital technologies enabled businesses to maintain customer engagement, improve operational flexibility, and respond rapidly to environmental uncertainties. Organizations that successfully adopted digital technologies demonstrated greater resilience and adaptability compared with businesses that continued to depend on traditional operational practices.

Another important aspect of digital transformation is the generation and utilization of organizational data. Modern organizations continuously collect operational information from customer transactions, inventory activities, reservation records, employee performance, and financial reports. Through digital transformation, these operational data are converted into valuable business intelligence that supports evidence-based decision-making. Managers utilize analytical tools to identify customer purchasing behavior, evaluate sales performance, forecast demand, monitor inventory consumption, and assess operational efficiency. Consequently, digital transformation improves not only business automation but also organizational learning and strategic planning (Verhoef et al., 2021).

Digital transformation also encourages greater integration among organizational functions. Instead of maintaining independent systems for sales, inventory, reservations, accounting, and customer management, integrated digital platforms allow these operational activities to function as interconnected components within a centralized information system. Such integration improves communication among departments, reduces duplicate data entry, minimizes human error, and provides managers with a comprehensive view of organizational performance. This integrated approach has become increasingly important for service-oriented organizations where operational efficiency directly influences customer satisfaction and business profitability.

Small and medium-sized enterprises (SMEs), including independently owned restaurants, have likewise benefited from digital transformation despite resource limitations. Cloud-based applications, web-based information systems, and affordable business management software have made digital technologies more accessible to smaller organizations. These technologies enable SMEs to automate business operations without requiring expensive enterprise-level software or extensive information technology infrastructure. As a result, digital transformation has become an attainable strategy for improving productivity, operational efficiency, and business competitiveness among local restaurant establishments.

Despite its numerous benefits, successful digital transformation requires organizational readiness, management support, employee acceptance, and continuous technological improvement. Organizations that merely implement digital technologies without redesigning business processes often fail to realize the full benefits of digital transformation. Vial (2019) emphasized that effective digital transformation involves aligning technology with organizational objectives, business strategies, and operational requirements. Therefore, successful implementation depends not only on technological resources but also on organizational commitment, employee competence, and continuous process improvement.

The concepts discussed in this literature are highly relevant to the present study because the proposed **Online Restaurant Management System with Sales Analytics and Reservation Scheduling** represents a digital transformation initiative for Crates N' Plates. The developed system replaces manual operational procedures with an integrated web-based platform that automates sales transactions, reservation scheduling, inventory monitoring, report generation, and sales analytics. By centralizing operational information and providing management with real-time access to business data, the proposed system supports the restaurant's digital transformation while improving operational efficiency, customer service, and managerial decision-making. Consequently, the reviewed literature provides a strong theoretical foundation for understanding the role of digital transformation in modern restaurant management and reinforces the need for implementing integrated information systems within the food service industry.

Business Process Automation (BPA)

Business Process Automation (BPA) has become one of the most significant applications of information technology in modern organizations because it enables businesses to automate repetitive, time-consuming, and manual operational activities through integrated information systems. Rather than relying on paper-based documentation and human intervention, BPA utilizes software applications, databases, and digital technologies to execute organizational workflows efficiently and consistently. The implementation of business process automation minimizes human error, reduces operational costs, improves productivity, accelerates information processing, and enhances service quality. As organizations continue to embrace digital transformation, BPA has become a strategic initiative that supports organizational competitiveness, operational excellence, and long-term sustainability (Dumas et al., 2018).

Business Process Automation extends beyond the simple computerization of individual tasks. Instead, it focuses on redesigning organizational workflows by integrating multiple business functions into a unified process capable of operating with minimal manual intervention. According to Dumas et al. (2018), effective business process automation requires organizations to analyze existing workflows, identify inefficiencies, redesign operational procedures, and implement information technologies that optimize business performance. Consequently, automation not only improves operational efficiency but also enables organizations to standardize business procedures, strengthen quality management, and improve organizational governance.

Recent systematic literature reviews have further emphasized the growing importance of BPA across different industries. Fernández-Ropero et al. (2024) conducted a comprehensive systematic literature review on Business Process Automation in Small and Medium-sized Enterprises (SMEs). The review analyzed more than 300 scholarly publications and concluded that BPA significantly improves organizational productivity, operational flexibility, service quality, and resource utilization. The researchers observed that automation enables SMEs to reduce repetitive administrative work, improve information accuracy, strengthen organizational coordination, and respond more efficiently to changing business environments. Furthermore, the review highlighted that business process automation has become increasingly accessible to SMEs because of advances in cloud computing, web-based technologies, and affordable enterprise applications.

Another significant finding reported by Fernández-Ropero et al. (2024) is that organizations implementing BPA experience improvements in decision-making because automated systems continuously generate reliable operational information. Rather than depending on manually prepared reports, managers are provided with real-time business information that supports planning, monitoring, and performance evaluation. The researchers emphasized that automation contributes not only to operational efficiency but also to organizational intelligence by transforming business data into meaningful managerial information that supports evidence-based decision-making.

Business Process Automation has become particularly valuable within service-oriented industries such as hospitality and restaurant management. Restaurant operations involve numerous repetitive activities including order processing, billing, reservation scheduling, inventory recording, report generation, customer information management, and employee coordination. When these activities are performed manually, organizations frequently encounter delays, duplicated work, communication problems, and recording errors that reduce operational efficiency and customer satisfaction. BPA addresses these challenges by integrating operational functions into centralized information systems capable of automatically processing transactions and updating organizational records in real time.

Automation likewise contributes significantly to improving customer service. By reducing the amount of manual intervention required during customer transactions, organizations can provide faster service while maintaining greater consistency and accuracy. Automated reservation systems reduce scheduling conflicts, computerized Point-of-Sale systems improve billing accuracy, and automated inventory monitoring enables businesses to maintain product availability more effectively. Consequently, customers experience shorter waiting times, fewer transaction errors, and improved overall service quality. These improvements strengthen customer satisfaction and contribute to the long-term competitiveness of service-oriented organizations.

Modern Business Process Automation also supports organizational transparency and accountability. Automated systems maintain comprehensive transaction histories, user activity logs, and operational records that allow management to monitor employee performance and organizational activities more effectively. Audit trails generated by automated systems facilitate organizational control by documenting every transaction performed within the system. These capabilities improve data integrity, strengthen internal control mechanisms, and support organizational compliance with established business procedures (Dumas et al., 2018).

The implementation of BPA further enables organizations to optimize the utilization of available resources. Automated workflows reduce unnecessary paperwork, eliminate duplicate data entry, minimize administrative workload, and improve communication among organizational units. Employees are consequently able to devote more time to customer-oriented activities rather than repetitive clerical work. This improved allocation of human resources contributes directly to increased productivity while reducing operational expenses associated with inefficient business processes.

Another important characteristic of Business Process Automation is its integration with Business Process Management (BPM). While BPM focuses on analyzing, modeling, improving, and monitoring organizational workflows, BPA provides the technological mechanisms that automate these redesigned processes. Together, BPM and BPA enable organizations to achieve continuous process improvement by combining organizational analysis with digital technologies. According to Dumas et al. (2018), organizations that successfully integrate BPM and BPA are better positioned to improve operational efficiency while adapting to evolving customer requirements and technological innovations.

The relevance of Business Process Automation to the present study is highly evident because the proposed **Online Restaurant Management System with Sales Analytics**

and Reservation Scheduling was specifically developed to automate the manual operational procedures of Crates N' Plates. The system automates sales transaction processing, reservation scheduling, inventory monitoring, report generation, and sales analytics through a centralized web-based platform. Similar to the concepts discussed in the reviewed literature, the proposed system seeks to minimize manual intervention, reduce operational errors, improve organizational productivity, and provide management with timely business information that supports effective decision-making. Therefore, Business Process Automation serves as one of the primary theoretical foundations supporting the design and implementation of the proposed Online Restaurant Management System.

Point-of-Sale (POS) Systems

Point-of-Sale (POS) systems have become an integral component of modern restaurant operations because they automate transaction processing, improve operational efficiency, and facilitate the management of sales information. Traditionally, restaurants relied on manual cash registers and handwritten receipts to record customer purchases. While these methods were adequate for small-scale operations, they often resulted in computational errors, duplicate entries, misplaced receipts, delayed reporting, and inefficient transaction processing. As the food service industry became increasingly competitive, restaurants adopted computerized Point-of-Sale systems to improve the accuracy, speed, and reliability of sales transactions while enhancing overall customer service (Laudon & Laudon, 2022).

A Point-of-Sale system is more than a computerized cash register. Modern POS applications integrate various operational functions such as order management, payment processing, receipt generation, inventory monitoring, customer information management, employee monitoring, and business reporting into a centralized platform. Every customer transaction processed through the POS system automatically updates the sales database, inventory records, and financial reports, eliminating the need for repetitive manual data entry. This integration improves operational coordination while reducing the likelihood of human errors and inconsistencies among business records (Dumas et al., 2018).

The implementation of POS systems has significantly transformed restaurant management by providing managers with real-time operational information. Instead of waiting until the end of the business day to manually summarize transactions, managers can immediately monitor sales performance, transaction volume, employee productivity, and product demand through computerized dashboards. According to Laudon and Laudon (2022), integrated POS systems provide organizations with immediate access to operational information that supports timely managerial decision-making, inventory planning, and financial monitoring.

Recent studies have likewise demonstrated that Point-of-Sale technologies contribute significantly to improving restaurant productivity and customer satisfaction. Prakash et al. (2025) developed a cloud-integrated Restaurant Point-of-Sale System that combined transaction processing, billing, inventory management, and reporting into a centralized platform. The researchers found that the implementation of the computerized POS system significantly reduced transaction errors, accelerated customer service, improved inventory accuracy, and enabled restaurant managers to monitor business operations remotely through cloud technology. The study concluded that integrating POS systems with inventory management and reporting modules enhances operational efficiency while improving organizational decision-making.

Similarly, Pasalkar et al. (2023) emphasized that Point-of-Sale systems function as the operational backbone of integrated restaurant management systems because they connect customer ordering, inventory management, employee activities, and reporting within a unified information system. The researchers observed that automated POS systems reduce administrative workload, eliminate duplicate transaction recording, and strengthen communication among restaurant personnel. Furthermore, centralized transaction records generated by the POS system provide managers with reliable business information that supports planning, operational monitoring, and organizational control.

One of the most important advantages of modern POS systems is their integration with inventory management. Every completed sales transaction automatically updates inventory records by deducting the corresponding quantities of ingredients or products sold. This real-time synchronization eliminates the need for separate inventory recording and enables restaurant managers to monitor stock movement continuously. Consequently, businesses can identify low-stock items promptly, prevent inventory shortages, and improve purchasing decisions based on accurate inventory information. This capability contributes significantly to reducing food waste, avoiding overstocking, and maintaining operational continuity.

POS systems also improve customer service by shortening transaction time and reducing waiting periods. Through automated billing, electronic receipt generation, and digital payment processing, restaurants are able to provide customers with faster and more accurate services. In addition, integrated POS applications reduce billing discrepancies and computational mistakes that commonly occur during manual transaction processing. Improved transaction accuracy enhances customer trust and satisfaction while strengthening the professional image of the restaurant.

Another important feature of contemporary POS systems is the generation of comprehensive business reports. Information collected from daily customer transactions is automatically organized into sales summaries, financial reports, inventory movement reports, employee performance records, and analytical dashboards. These reports enable restaurant managers to identify best-selling menu items, evaluate sales performance, monitor peak operating hours, assess employee productivity, and analyze customer purchasing behavior. Such information supports evidence-based managerial decision-making while allowing organizations to formulate more effective operational and marketing strategies.

Security has also become a critical consideration in Point-of-Sale systems. Because POS applications process financial transactions and confidential customer information, modern systems implement authentication mechanisms, user account management, encrypted data transmission, and Role-Based Access Control (RBAC) to protect sensitive organizational data. These security features minimize unauthorized access while ensuring accountability for every transaction performed within the system.

Cloud computing has further expanded the capabilities of Point-of-Sale systems by enabling remote accessibility and centralized information management. Cloud-based POS applications allow restaurant owners and managers to monitor sales performance, inventory status, and business reports using internet-connected devices regardless of location. This capability improves managerial responsiveness while simplifying software maintenance, database backup, and system scalability. As cloud technologies continue to evolve, cloud-enabled POS systems are becoming increasingly popular among small and medium-sized restaurants because of their affordability and operational flexibility (Prakash et al., 2025).

The reviewed literature strongly supports the present study because the proposed **Online Restaurant Management System with Sales Analytics and Reservation Scheduling** incorporates an integrated Point-of-Sale module responsible for processing customer orders, recording transactions, generating receipts, updating inventory records, and maintaining sales histories. Similar to the POS systems discussed in previous studies, the proposed system aims to improve transaction accuracy, accelerate customer service, reduce manual workload, and provide restaurant management with real-time operational information. However, the proposed system extends beyond traditional POS functionalities by integrating reservation scheduling, inventory monitoring, sales analytics, and automated reporting into a centralized web-based platform specifically designed for the operational requirements of Crates N' Plates.

Inventory Management

Inventory management is one of the most critical operational functions within the restaurant industry because it ensures the continuous availability of ingredients, beverages, and other supplies necessary for daily business operations. Restaurants operate in a highly dynamic environment where inventory is consumed rapidly, and many products are perishable. Consequently, effective inventory management is essential to maintain food quality, minimize waste, control operational costs, and ensure uninterrupted service delivery. Without an organized inventory management system, restaurants become susceptible to stock shortages, overstocking, spoilage, inaccurate inventory records, and increased operating expenses (Heizer et al., 2020).

Traditionally, many restaurants managed inventory through handwritten logbooks, manual stock counting, and paper-based inventory reports. Although these methods allowed businesses to monitor available supplies, they were often labor-intensive, time-consuming, and prone to human error. Manual inventory systems frequently resulted in inaccurate stock records, delayed inventory updates, misplaced documents, and inconsistencies between recorded and actual stock levels. These challenges became more pronounced as restaurant operations expanded and customer demand increased, making manual inventory management increasingly inefficient (Laudon & Laudon, 2022).

The adoption of computerized inventory management systems has significantly improved inventory control by automating stock recording, inventory tracking, and replenishment planning. Modern inventory systems continuously monitor stock movement by recording inventory inflows and outflows while automatically updating inventory records whenever transactions occur. This automation provides managers with real-time information regarding product availability, inventory consumption, reorder levels, and inventory valuation. Consequently, restaurant managers are able to make more informed purchasing decisions while reducing inventory losses associated with spoilage, theft, and inaccurate record-keeping (Heizer et al., 2020).

One of the most significant advantages of computerized inventory management is its integration with Point-of-Sale systems. Every completed customer transaction automatically deducts the corresponding inventory items used in preparing the ordered products. This integration eliminates duplicate inventory recording while ensuring that inventory information remains accurate and current. Through real-time synchronization, restaurant managers can immediately identify low-stock ingredients, monitor fast-moving and slow-moving products, and determine appropriate reorder quantities based on actual inventory consumption. Such capabilities improve inventory accuracy while reducing unnecessary purchasing costs and operational delays (Dumas et al., 2018).

Inventory management also contributes directly to financial performance because inventory represents one of the largest operational investments within restaurant businesses. Poor inventory management often results in excessive purchasing, expired products, inventory shortages, and increased food waste, all of which reduce organizational profitability. Effective inventory control enables organizations to optimize inventory levels while maintaining sufficient stock to meet customer demand. According to Heizer et al. (2020), inventory management seeks to achieve a balance between inventory availability and inventory cost by ensuring that organizations maintain the appropriate quantity of inventory at the appropriate time.

Recent research has further emphasized the importance of inventory management in improving restaurant operations. Baylen (2020) examined the inventory management practices of selected small-sized restaurants in Quezon Province and found that many establishments continued to rely on manual inventory procedures. The study revealed that manual inventory recording frequently resulted in inventory discrepancies, delayed replenishment, inaccurate stock monitoring, and operational inefficiencies. The researcher recommended the implementation of computerized inventory systems capable of improving inventory accuracy, strengthening inventory control, and supporting more effective purchasing decisions. These findings demonstrate that inventory automation significantly contributes to organizational productivity and operational efficiency.

Similarly, Sugianto et al. (2023) developed a computerized inventory management application for restaurant operations using the Extreme Programming software development methodology. The study aimed to address common inventory-related problems such as inaccurate stock recording, delayed inventory updates, and inefficient inventory monitoring. The developed application automated inventory recording, monitored stock movement in real time, generated inventory reports, and provided inventory notifications for low-stock items. The researchers concluded that the implementation of computerized inventory management significantly improved inventory accuracy, reduced administrative workload, and strengthened organizational efficiency.

The study also highlighted that integrating inventory management with other restaurant operational modules further enhanced overall business performance.

Another important function of inventory management is supporting managerial decision-making. Through automated inventory reports and analytical summaries, restaurant managers are able to monitor inventory turnover, identify frequently consumed ingredients, evaluate supplier performance, forecast purchasing requirements, and analyze inventory-related costs. These analytical capabilities enable organizations to optimize inventory investment while ensuring that operational requirements are continuously satisfied. Furthermore, historical inventory data provide valuable information for identifying seasonal demand patterns and planning future inventory procurement strategies.

Inventory management systems likewise contribute to organizational accountability and internal control. Automated inventory systems maintain complete records of inventory transactions, including receiving, issuing, transferring, and adjusting stock quantities. These transaction histories improve transparency by allowing managers to monitor inventory movement while identifying discrepancies resulting from theft, wastage, or recording errors. User authentication and audit logs further strengthen inventory security by recording the identity of employees responsible for each inventory transaction.

The integration of inventory management with sales analytics has become increasingly important in modern restaurant operations. Sales information generated through Point-of-Sale systems provides valuable insights regarding product demand and inventory consumption. By analyzing sales trends together with inventory movement, managers can accurately forecast inventory requirements, minimize food waste, and improve menu planning. Such integration enables restaurants to respond proactively to changing customer preferences while maintaining adequate inventory levels.

The concepts discussed in the reviewed literature are highly relevant to the present study because the proposed **Online Restaurant Management System with Sales Analytics and Reservation Scheduling** incorporates an integrated inventory monitoring module designed to automate stock recording, monitor inventory movement, identify low-stock products, and generate inventory reports. Similar to the inventory management systems described in previous studies, the proposed system seeks to eliminate manual inventory procedures while improving inventory accuracy, operational efficiency, and managerial decision-making. Unlike traditional inventory systems, however, the proposed application integrates inventory management with Point-of-Sale transactions, reservation scheduling, and sales analytics, thereby providing Crates N' Plates with a centralized platform capable of supporting efficient restaurant operations and long-term business sustainability.

Reservation Scheduling Systems

Reservation Scheduling Systems have become an essential component of modern restaurant operations because they enable businesses to organize customer bookings efficiently while improving service quality and resource utilization. As restaurants experience increasing customer demand, managing reservations manually becomes more challenging due to the growing volume of bookings, frequent schedule modifications, and the need to coordinate available seating or event spaces. Traditional reservation methods, such as handwritten reservation books, telephone logs, and paper-based schedules, are often prone to scheduling conflicts, duplicate bookings, delayed confirmations, and communication errors. Consequently, many restaurants have adopted computerized reservation systems to improve scheduling accuracy and enhance customer satisfaction (Bujalance-López et al., 2025).

Reservation scheduling systems are designed to automate the entire reservation process by allowing businesses to record, monitor, modify, and manage reservations through a centralized database. These systems provide real-time information regarding reservation availability, customer details, booking schedules, and seating capacity, enabling restaurant personnel to respond quickly to customer inquiries and reservation requests. According to Laudon and Laudon (2022), integrating reservation management into an organization's information system improves operational coordination because reservation information becomes immediately accessible to authorized personnel, reducing communication delays and administrative workload.

The development of reservation scheduling technologies has also contributed significantly to customer convenience. Modern reservation systems allow customers to reserve tables or event spaces through online platforms without the need to visit the restaurant personally or communicate through lengthy telephone conversations. Automated confirmation messages, reservation reminders, and booking notifications further improve customer satisfaction by providing immediate feedback regarding reservation status. Such capabilities have become increasingly important as customer expectations continue to shift toward digital service platforms and contactless transactions.

Another significant advantage of computerized reservation systems is the prevention of scheduling conflicts and double bookings. Through automated schedule validation, reservation systems verify table availability before confirming reservations, ensuring that multiple customers cannot reserve the same resource during identical time periods. This functionality significantly improves operational efficiency while protecting the restaurant from service disruptions resulting from inaccurate reservation records. Bujalance-López et al. (2025) emphasized that reservation management technologies have become fundamental components of restaurant revenue management because they optimize seating utilization, improve demand forecasting, and contribute to organizational profitability by maximizing available dining capacity. These findings demonstrate that reservation systems contribute not only to operational management but also to strategic business planning.

Reservation scheduling systems likewise improve organizational communication by allowing different departments to access identical reservation information through a centralized database. Front-desk personnel, cashiers, managers, and kitchen staff can coordinate customer reservations more effectively because reservation updates become immediately visible throughout the system. This eliminates the need for repeated manual verification while reducing communication errors among restaurant personnel. Such coordination becomes particularly important for restaurants offering multiple services such as dine-in reservations, catering events, private function rooms, and banquet facilities.

The integration of reservation systems with other operational modules further enhances organizational efficiency. Modern restaurant management systems commonly connect reservation scheduling with Point-of-Sale systems, customer databases, sales analytics, and inventory management. This integration enables restaurants to monitor reservation trends, evaluate customer preferences, forecast demand during peak periods, and prepare sufficient inventory and staffing resources based on scheduled reservations. Consequently, reservation information becomes valuable operational data supporting managerial decision-making and long-term planning.

Reservation scheduling systems also contribute significantly to customer relationship management. By maintaining customer reservation histories, contact information, special requests, and previous dining preferences, restaurants are able to provide personalized customer service while strengthening long-term customer relationships. Historical reservation records enable management to recognize loyal customers, evaluate reservation trends, and develop targeted promotional activities that improve customer retention. These capabilities demonstrate that reservation systems serve not only as scheduling tools but also as valuable customer information management platforms.

Recent Philippine research further supports the effectiveness of computerized reservation management systems. Villanueva (2025) developed **Steffi's Online Reservation Management System (SORMS)** to replace manual reservation procedures through a web-based platform. Using a developmental research methodology, the study designed and evaluated a computerized reservation system capable of managing customer bookings, reservation schedules, and booking confirmations through a centralized database. The results demonstrated that the developed system significantly improved reservation accuracy, minimized scheduling conflicts, reduced duplicate bookings, and accelerated reservation processing. Respondents also reported improved operational efficiency because reservation information became more organized and readily accessible. The researcher concluded that computerized reservation systems improve both customer satisfaction and administrative efficiency by replacing traditional paper-based reservation procedures with automated scheduling technologies.

Another important contribution of reservation scheduling systems is their ability to support managerial decision-making through analytical reporting. Reservation records provide managers with valuable information regarding customer demand, peak operating periods, reservation cancellations, booking frequency, and seasonal trends. By analyzing these data, restaurant managers can improve workforce scheduling, optimize seating arrangements, allocate resources more efficiently, and implement marketing strategies targeting periods of low customer demand. These analytical capabilities strengthen organizational planning while contributing to improved operational performance and revenue generation.

Reservation scheduling systems likewise improve organizational security by protecting customer reservation information through authentication mechanisms and controlled user access. Modern reservation applications commonly implement user authentication, encrypted databases, and Role-Based Access Control (RBAC) to ensure that only authorized personnel can access, modify, or approve reservation records. Such security measures strengthen data confidentiality while maintaining the integrity of customer information stored within the reservation database.

The reviewed literature strongly supports the present study because the proposed **Online Restaurant Management System with Sales Analytics and Reservation Scheduling** incorporates a reservation scheduling module specifically designed to manage dine-in reservations, catering bookings, and private function room reservations for Crates N' Plates. Similar to the systems discussed in previous literature, the proposed application automates reservation processing, maintains centralized reservation records, minimizes scheduling conflicts, and improves operational coordination among authorized personnel. However, unlike stand-alone reservation applications, the proposed system integrates reservation scheduling with Point-of-Sale transactions, inventory monitoring, sales analytics, report generation, and user management within a single web-based platform. Such integration enables Crates N' Plates to manage its expanding operations more efficiently while improving customer service, organizational productivity, and managerial decision-making.

Sales Analytics

Sales analytics has become an indispensable component of modern business information systems because it enables organizations to transform raw transactional data into meaningful insights that support operational and strategic decision-making. As businesses generate increasing volumes of sales data through Point-of-Sale systems, online transactions, and customer interactions, organizations have recognized the importance of analyzing these data to understand customer behavior, monitor business performance, forecast demand, and identify opportunities for organizational improvement. Rather than relying solely on intuition or manual observations, sales analytics provides managers with objective, data-driven information that supports evidence-based decision-making (Sharda et al., 2023).

Sales analytics refers to the systematic process of collecting, organizing, analyzing, and interpreting sales-related data to evaluate business performance and support organizational planning. Through statistical analysis, visualization techniques, and reporting tools, businesses are able to identify sales trends, measure revenue growth, monitor product performance, evaluate promotional activities, and analyze customer purchasing behavior. According to Sharda et al. (2023), sales analytics enables organizations to convert operational data into actionable knowledge that improves managerial decision-making and organizational competitiveness. By utilizing analytical dashboards and business reports, managers can respond more effectively to changing market conditions and customer preferences.

The growing adoption of Business Intelligence (BI) technologies has significantly enhanced the capabilities of sales analytics. Business Intelligence integrates data collection, reporting, visualization, and analytical tools that allow organizations to examine historical and current business performance from multiple perspectives. Through interactive dashboards, graphical reports, and performance indicators, managers can monitor sales activities in real time and identify emerging trends that may influence future business decisions. Wixom and Watson (2010) explained that Business Intelligence systems improve organizational performance by providing timely, accurate, and relevant information that supports planning, forecasting, and strategic management. Although BI has evolved considerably over the past decade, its fundamental purpose remains the transformation of organizational data into meaningful business intelligence.

One of the primary advantages of sales analytics is its ability to identify best-selling and slow-moving products. By continuously analyzing transaction histories, organizations can determine which products generate the highest revenue and which products experience declining customer demand. This information enables managers to improve menu planning, optimize inventory purchasing, revise pricing strategies, and eliminate products that contribute minimally to organizational profitability. Sales analytics therefore supports more efficient resource allocation while reducing unnecessary operational expenses associated with low-performing products.

Sales analytics also contributes significantly to customer behavior analysis. Every customer transaction provides valuable information regarding purchasing patterns, product preferences, transaction frequency, seasonal demand, and spending behavior. Through analytical processing, organizations are able to segment customers according to purchasing habits and identify factors influencing customer demand. Such information enables businesses to develop targeted promotional campaigns, improve customer engagement, and strengthen long-term customer relationships. Consequently, sales analytics supports not only operational management but also customer relationship management and marketing effectiveness.

Forecasting is another important application of sales analytics. Historical sales information allows organizations to predict future product demand, estimate inventory requirements, and prepare operational resources during periods of increased customer activity. According to Heizer et al. (2020), forecasting supports effective inventory management by enabling organizations to anticipate future demand while minimizing inventory shortages and overstocking. Within restaurant operations, accurate forecasting contributes to better purchasing decisions, improved workforce scheduling, reduced food waste, and enhanced service quality.

Recent studies have further demonstrated the importance of sales analytics within restaurant management. Posch et al. (2020) developed a Bayesian forecasting model using historical Point-of-Sale transaction data collected from restaurant operations. The researchers analyzed food and beverage sales to determine whether historical transaction data could accurately predict future customer demand. Their findings demonstrated that sales analytics significantly improved forecasting accuracy, enabling restaurant managers to optimize inventory purchasing, reduce food waste, and improve operational profitability. The study concluded that transaction data generated through Point-of-Sale systems represent valuable organizational assets capable of supporting managerial planning and evidence-based decision-making. These findings highlight the growing role of analytical technologies within modern restaurant management.

Similarly, Bujalance-López et al. (2025) emphasized that sales analytics has become an essential component of restaurant revenue management because it provides managers with detailed information regarding customer demand, reservation patterns, sales performance, and operational efficiency. Their systematic literature review revealed that restaurants implementing analytical tools achieved better resource utilization, improved demand forecasting, enhanced customer satisfaction, and stronger financial performance compared with organizations relying solely on manual managerial observations. The researchers further emphasized that integrating sales analytics with reservation systems and customer relationship management improves organizational competitiveness within the hospitality industry.

Sales analytics likewise improves organizational transparency by generating comprehensive business reports that summarize daily, weekly, monthly, and annual sales performance. These reports allow managers to monitor revenue growth, evaluate employee productivity, assess marketing effectiveness, and compare organizational performance across different operational periods. Analytical dashboards present these information through graphical visualizations such as bar charts, line graphs, pie charts, and trend analyses, making complex business data easier to interpret and utilize during managerial planning. The availability of automated reports significantly reduces the time required to prepare financial summaries while improving the accuracy and consistency of organizational reporting.

Another important contribution of sales analytics is its integration with other business functions. Modern restaurant management systems connect sales analytics with Point-of-Sale operations, inventory management, reservation scheduling, and customer databases. This integration enables managers to analyze the relationship between sales performance, inventory consumption, reservation demand, and customer purchasing behavior within a single information system. Such interconnected analysis provides a more comprehensive understanding of business operations and supports strategic decision-making across multiple organizational functions.

The reviewed literature strongly supports the present study because the proposed **Online Restaurant Management System with Sales Analytics and Reservation Scheduling** incorporates a dedicated Sales Analytics Dashboard that enables Crates N' Plates to monitor daily, weekly, monthly, and annual sales performance through interactive graphical reports. Similar to the concepts discussed in previous literature, the proposed system transforms operational transaction records into meaningful business information that supports managerial planning and decision-making. Through the integration of Point-of-Sale transactions, reservation scheduling, inventory monitoring, and analytical reporting, the proposed system provides restaurant management with timely and reliable information necessary for evaluating business performance, identifying sales trends, monitoring customer purchasing behavior, and supporting long-term organizational growth. Therefore, sales analytics represents one of the most significant technological features incorporated into the proposed system and serves as a major contribution of the present study to restaurant management practices.

Business Intelligence (BI)

Business Intelligence (BI) has become an essential component of modern information systems because it enables organizations to transform large volumes of operational data into meaningful information that supports strategic and operational decision-making. As organizations generate increasing amounts of data through daily transactions, customer interactions, inventory movements, and financial activities, the ability to organize, analyze, and interpret these data has become a critical organizational capability. Business Intelligence integrates technologies, applications, databases, and analytical tools that allow decision-makers to access accurate, timely, and relevant information necessary for improving organizational performance and competitiveness (Sharda et al., 2023).

The concept of Business Intelligence extends beyond the collection of business data. It encompasses the entire process of extracting, organizing, analyzing, and presenting information through reports, dashboards, and visualization tools that facilitate informed decision-making. According to Sharda et al. (2023), Business Intelligence enables organizations to convert raw operational data into actionable knowledge by applying analytical techniques, performance indicators, and visualization technologies. This transformation allows managers to understand business performance comprehensively while identifying patterns, opportunities, and potential operational issues that require immediate attention.

Modern organizations increasingly rely on Business Intelligence because business environments have become more dynamic and competitive. Managers are expected to make timely decisions based on reliable information rather than assumptions or intuition. Business Intelligence systems provide decision-makers with real-time access to organizational data through interactive dashboards, statistical summaries, graphical reports, and performance indicators. These tools enable management to monitor key performance metrics, evaluate operational efficiency, identify emerging trends, and measure organizational progress toward established objectives (Laudon & Laudon, 2022).

Within the restaurant industry, Business Intelligence has significantly improved the ability of managers to monitor and evaluate business operations. Restaurant businesses generate extensive operational data from Point-of-Sale transactions, reservation records, inventory movement, employee activities, and customer purchasing behavior. Business Intelligence systems consolidate these information sources into centralized dashboards that provide comprehensive insights regarding restaurant performance. Through visual reports and analytical summaries, managers can monitor daily sales, identify best-selling menu items, evaluate customer preferences, assess inventory consumption, and determine periods of peak business activity. Such information supports evidence-based managerial planning while improving operational efficiency.

One of the primary benefits of Business Intelligence is its capability to integrate data from multiple operational modules. Instead of analyzing sales, inventory, reservations, and customer information independently, BI systems combine these data sources into a unified analytical environment. This integration enables managers to identify relationships among various business functions. For example, managers can determine how reservation volume influences inventory consumption, evaluate how promotional campaigns affect customer purchasing behavior, or examine the relationship between seasonal demand and sales performance. Such integrated analysis provides a more comprehensive understanding of organizational performance than isolated operational reports.

Business Intelligence also enhances organizational responsiveness by providing real-time information. Traditional reporting methods often require manual consolidation of records before management can evaluate organizational performance. In contrast, BI systems automatically collect operational data and present updated information through dashboards and visual reports. This capability allows organizations to identify operational problems immediately, respond promptly to changing customer demands, and implement corrective actions before problems significantly affect business performance. Real-time reporting likewise improves communication among managers by ensuring that organizational decisions are based on consistent and current information.

Data visualization represents another significant feature of Business Intelligence. Modern BI applications utilize charts, graphs, heat maps, trend analyses, scorecards, and performance dashboards to present complex business information in formats that are easier to interpret. Visualization enables managers to identify sales trends, seasonal fluctuations, customer purchasing patterns, and inventory movement more efficiently than traditional tabular reports. According to Few (2013), effective data visualization improves decision-making because graphical representations allow users to recognize relationships and patterns that may not be immediately apparent through numerical data alone.

Recent literature emphasizes that Business Intelligence contributes significantly to organizational competitiveness by improving evidence-based decision-making. Wixom and Watson (2010) explained that organizations implementing Business Intelligence systems experience improvements in strategic planning, operational monitoring, and organizational performance because decision-makers are provided with reliable information generated from integrated organizational databases. The researchers further emphasized that Business Intelligence should be viewed as an organizational capability rather than merely a technological application, since its effectiveness depends upon both technological infrastructure and managerial utilization of information.

Similarly, Sharda et al. (2023) emphasized that Business Intelligence has evolved beyond traditional reporting by incorporating advanced analytics, predictive modeling, artificial intelligence, and machine learning techniques. Although many organizations initially adopted BI for reporting purposes, contemporary Business Intelligence platforms now support forecasting, customer segmentation, performance measurement, and predictive decision-making. These developments have expanded the role of Business Intelligence from descriptive reporting toward proactive organizational planning and continuous performance improvement.

Another important contribution of Business Intelligence is its support for organizational transparency and accountability. Because BI systems obtain information directly from centralized operational databases, managers are able to monitor business performance objectively through standardized performance indicators. This reduces dependence on manually prepared reports while minimizing opportunities for data manipulation or reporting inconsistencies. Furthermore, centralized reporting improves organizational governance by ensuring that managers utilize consistent information when evaluating business operations and organizational performance.

The concepts discussed in the reviewed literature are highly relevant to the present study because the proposed **Online Restaurant Management System with Sales Analytics and Reservation Scheduling** incorporates a Business Intelligence-oriented dashboard that consolidates operational information generated from Point-of-Sale transactions, inventory monitoring, reservation scheduling, and sales activities. Similar to the Business Intelligence systems discussed by previous authors, the proposed application transforms operational data into graphical reports and analytical summaries that support managerial planning and evidence-based decision-making. Through the integration of multiple operational modules, the system enables Crates N' Plates to monitor organizational performance more effectively while improving operational efficiency, customer service, and long-term business sustainability. Consequently, Business Intelligence serves as one of the major theoretical foundations supporting the implementation of the proposed Online Restaurant Management System.

Database Management Systems (DBMS)

A Database Management System (DBMS) is a software application that enables organizations to create, store, organize, retrieve, update, and manage large volumes of data in a structured and efficient manner. In modern organizations, databases serve as the foundation of information systems because they provide centralized repositories that allow multiple users and applications to access consistent and reliable information simultaneously. Unlike traditional file-based systems that store information separately in different files, DBMS integrates organizational data into a unified environment, reducing redundancy, improving consistency, and supporting efficient information management (Coronel & Morris, 2023).

The increasing dependence on digital technologies has made database management systems indispensable across various industries, including healthcare, education, finance, retail, manufacturing, and hospitality. Organizations generate substantial amounts of operational data daily, including customer information, transaction records, inventory details, employee records, and financial reports. Managing these data manually or through isolated files often leads to duplication, inconsistency, data loss, and inefficient information retrieval. According to Elmasri and Navathe (2017), a DBMS addresses these challenges by providing mechanisms for centralized storage, data integrity, security, concurrency control, and efficient retrieval of information.

One of the most significant advantages of a DBMS is the reduction of data redundancy. In traditional manual or file-processing systems, identical information is often stored repeatedly in different records, increasing storage requirements and creating inconsistencies whenever updates are made. Database systems eliminate unnecessary duplication by maintaining a single, centralized version of organizational data that can be accessed by multiple system modules simultaneously. This centralized approach improves data accuracy while ensuring that all users access consistent information regardless of the operational function they perform (Coronel & Morris, 2023).

Data integrity is another essential characteristic of Database Management Systems. Data integrity refers to the accuracy, consistency, and reliability of information stored within the database throughout its entire life cycle. Modern DBMS applications implement validation rules, constraints, and transaction management mechanisms that prevent invalid or inconsistent information from being stored. These mechanisms ensure that organizational records remain accurate even when multiple users simultaneously perform data entry, updates, or retrieval. Maintaining data integrity is particularly important in business environments where inaccurate information may result in financial losses, operational errors, or poor managerial decisions (Elmasri & Navathe, 2017).

Database Management Systems also provide efficient data retrieval capabilities. Through structured query languages and indexing mechanisms, users can quickly retrieve specific information without manually searching through large volumes of records. This capability significantly improves organizational productivity by reducing the time required to locate customer records, inventory information, reservation schedules, transaction histories, and business reports. Managers are therefore able to access relevant information immediately whenever operational or strategic decisions must be made.

Security represents another critical function of a DBMS. Organizational databases often contain confidential information including customer profiles, financial records, employee information, and business transactions. Consequently, database systems implement authentication mechanisms, user authorization, encryption, backup procedures, and access control policies to protect sensitive information from unauthorized access, modification, or loss. According to Connolly and Begg (2015), database security is essential for maintaining confidentiality, integrity, and availability of organizational information, thereby ensuring that business operations remain secure and reliable.

Another important feature of modern Database Management Systems is concurrency control. In multi-user environments, numerous employees may access the database simultaneously while performing different organizational tasks. Without proper concurrency management, simultaneous updates may create inconsistencies or corrupt organizational data. DBMS implements transaction management and concurrency control mechanisms that coordinate simultaneous operations while preserving data consistency and preventing conflicts among users. This capability is particularly important for restaurants where cashiers, managers, kitchen personnel, and administrators may simultaneously access operational information during peak business hours.

Cloud database technologies have further transformed database management by providing scalable, accessible, and highly available data storage solutions. Unlike traditional on-premises databases, cloud-based databases enable organizations to access operational information from any location through internet-connected devices. Cloud databases also simplify maintenance, automatic backup, disaster recovery, and scalability, making them particularly suitable for small and medium-sized enterprises. Google Firebase, for example, provides a cloud-hosted NoSQL database that supports real-time synchronization, allowing multiple users to access updated information simultaneously. This capability has made cloud databases increasingly popular among web-based information systems because they improve system availability while reducing infrastructure costs (Google, 2024).

Within restaurant operations, Database Management Systems play a crucial role in integrating operational information generated from various functional modules. Customer transactions, reservation records, inventory movements, employee information, catering schedules, and analytical reports are stored within a centralized database that supports efficient information sharing among authorized personnel. Because all operational modules utilize the same database, information entered into one module becomes immediately available throughout the entire system. For example, completed sales transactions automatically update inventory records, while reservation approvals become immediately visible to authorized users responsible for scheduling and customer service. Such integration significantly improves operational coordination while minimizing duplicate data entry and communication delays.

Database Management Systems likewise support organizational reporting and business analytics. Historical transaction records stored within the database provide valuable information for generating sales reports, monitoring inventory consumption, evaluating customer purchasing behavior, and identifying business trends. Analytical modules retrieve these historical data to produce graphical dashboards and performance summaries that support managerial planning and evidence-based decision-making. Consequently, the database functions not merely as a storage repository but also as the primary source of organizational knowledge that supports operational improvement and strategic planning.

The reviewed literature strongly supports the present study because the proposed **Online Restaurant Management System with Sales Analytics and Reservation Scheduling** utilizes a centralized cloud-based database to manage customer information, Point-of-Sale transactions, reservation schedules, inventory records, sales reports, and user accounts. Similar to the concepts discussed in previous literature, the proposed system aims to improve data accuracy, eliminate redundant information, strengthen security, and facilitate real-time information retrieval through a centralized database environment. Furthermore, the implementation of a cloud-based database enables authorized users to access updated operational information efficiently while supporting data consistency across all modules of the system. Therefore, the principles of Database Management Systems provide a strong theoretical foundation for the development and implementation of the proposed Online Restaurant Management System for Crates N' Plates.

Web-Based Information Systems

Web-Based Information Systems (WBIS) have become one of the most widely adopted technologies in modern organizations because they enable users to access organizational information through web browsers using internet or local network connections. Unlike traditional desktop applications that require installation on individual computers, web-based systems operate on centralized servers, allowing authorized users to access information from various devices, including desktop computers, laptops, tablets, and smartphones. This flexibility has significantly transformed organizational operations by improving accessibility, collaboration, and real-time information sharing across different departments and locations (Laudon & Laudon, 2022).

The rapid advancement of internet technologies has accelerated the development and adoption of web-based information systems across numerous industries, including education, healthcare, banking, retail, manufacturing, and hospitality. Organizations increasingly prefer web-based applications because they provide centralized information management, simplified software maintenance, lower deployment costs, and improved scalability compared with traditional standalone applications. Since all users interact with a single centralized system, software updates, bug fixes, and new functionalities can be implemented without requiring installation on individual client devices. Consequently, web-based systems reduce maintenance costs while ensuring that all users operate using the latest version of the application (Connolly & Begg, 2015).

One of the primary advantages of Web-Based Information Systems is centralized information management. Rather than storing organizational data separately on multiple computers, web-based applications maintain a centralized database that enables authorized users to retrieve and update information in real time. This centralized architecture minimizes data duplication, reduces inconsistencies, and ensures that organizational records remain synchronized across different operational modules. According to Pressman and Maxim (2020), centralized web applications improve organizational efficiency because they facilitate seamless communication among users while maintaining consistent access to organizational information regardless of location.

Accessibility represents another significant benefit of web-based systems. Employees, managers, and administrators can securely access organizational information from any internet-connected device provided they possess appropriate authorization credentials. This capability has become increasingly valuable in organizations requiring continuous monitoring of business operations beyond traditional office environments. Restaurant owners, for example, can review sales reports, inventory levels, reservation schedules, and business performance remotely without being physically present within the establishment. Such flexibility supports faster managerial responses while improving organizational productivity and operational control.

Web-Based Information Systems also improve organizational collaboration by enabling multiple users to access identical information simultaneously. Employees from different departments can perform their respective responsibilities while sharing the same centralized database. For example, cashiers may process customer transactions while inventory personnel monitor stock levels, kitchen staff receive customer orders, and managers review sales reports—all within the same integrated information system. Because information is updated in real time, operational coordination becomes more efficient while minimizing communication delays and duplicate data entry (Laudon & Laudon, 2022).

Another important characteristic of web-based systems is scalability. Organizations continuously evolve as customer demand, operational complexity, and information requirements increase. Web-based architectures allow businesses to expand system functionalities, accommodate additional users, and integrate new operational modules without completely redesigning the existing system. This scalability makes web-based applications particularly suitable for small and medium-sized enterprises that anticipate future organizational growth. As restaurants expand their services by introducing online reservations, delivery services, customer loyalty programs, or additional business branches, web-based systems provide a flexible technological foundation capable of supporting these organizational developments.

Security has likewise become a major consideration in web-based information systems because organizational data are transmitted across communication networks. Modern web applications implement multiple security mechanisms including user authentication, encrypted communication protocols, secure database connections, session management, and Role-Based Access Control (RBAC) to protect sensitive organizational information. These security measures ensure that only authorized users can access confidential business records while minimizing the risks associated with unauthorized access, cyber threats, and data breaches. According to Stallings (2021), implementing comprehensive security mechanisms is essential for maintaining confidentiality, integrity, and availability within web-based information systems.

Cloud computing has further enhanced the capabilities of Web-Based Information Systems by enabling organizations to deploy applications without maintaining expensive local server infrastructures. Cloud-hosted web applications provide automatic backup, high system availability, remote accessibility, and simplified maintenance while reducing hardware costs. Organizations utilizing cloud-based technologies benefit from scalable computing resources that adjust according to operational requirements. Google Firebase, Amazon Web Services (AWS), and Microsoft Azure are examples of cloud platforms commonly utilized in modern web-based application development because they provide reliable database management, authentication services, and cloud storage facilities suitable for organizational information systems (Google, 2024).

Within the hospitality industry, web-based information systems have become increasingly important because restaurant operations require continuous coordination among various functional areas, including sales transaction processing, reservation management, inventory monitoring, customer service, financial reporting, and managerial supervision. Through web-based applications, restaurant personnel can efficiently manage customer reservations, process transactions, monitor inventory levels, generate analytical reports, and coordinate daily operations regardless of their physical location. Customers likewise benefit from web-based technologies through online reservation platforms that provide greater convenience, accessibility, and faster communication with restaurant personnel.

Recent studies have demonstrated that web-based information systems significantly improve organizational performance by enhancing operational efficiency, reducing administrative workload, and strengthening information accessibility. Grepon (2021) developed a web-based case management information system that centralized organizational records and automated administrative processes. Although implemented within the Philippine judicial system, the study concluded that centralized web-based applications substantially improved information retrieval, operational coordination, record management, and organizational productivity. The findings further demonstrated that web-based technologies provide organizations with more efficient mechanisms for managing large volumes of information while reducing reliance on manual documentation. These principles are equally applicable to restaurant management because both environments require timely access to accurate organizational information to support operational activities and managerial decision-making.

Web-Based Information Systems also support organizational sustainability by reducing paper consumption and minimizing dependence on manual documentation. Electronic forms, digital reports, online transactions, and automated record management significantly reduce the need for printed documents while improving information accuracy and environmental sustainability. Furthermore, automated data backup mechanisms improve business continuity by protecting organizational information from accidental loss resulting from hardware failures, natural disasters, or human error.

The concepts discussed in the reviewed literature strongly support the present study because the proposed **Online Restaurant Management System with Sales Analytics and Reservation Scheduling** was developed as a web-based application accessible through standard web browsers. Similar to the characteristics described in previous literature, the proposed system utilizes centralized data management, real-time information processing, cloud-based technologies, and secure user authentication to improve restaurant operations. Authorized users—including administrators, managers, cashiers, and kitchen personnel—can efficiently access system functionalities according to their assigned responsibilities while maintaining accurate and synchronized business information. Through the implementation of a web-based architecture, the proposed system enhances operational efficiency, facilitates organizational collaboration, supports evidence-based decision-making, and provides Crates N' Plates with a scalable technological platform capable of supporting future organizational growth.

Role-Based Access Control (RBAC)

Role-Based Access Control (RBAC) is one of the most widely adopted access control models in modern information systems because it regulates user access to system resources according to predefined organizational roles and responsibilities. Rather than assigning permissions individually to every user, RBAC associates permissions with specific roles, and users are subsequently assigned to these roles based on their job functions. This approach simplifies access management while ensuring that employees can only access the information and perform the operations necessary to fulfill their assigned responsibilities. As organizations continue to implement integrated information systems that process sensitive operational and financial information, RBAC has become an essential security mechanism for protecting organizational data and maintaining information integrity (Sandhu et al., 1996).

Access control has become increasingly important as organizations transition from manual record-keeping to computerized information systems. Traditional paper-based records provided limited accessibility because documents were physically stored and could only be accessed by authorized personnel. However, digital information systems allow multiple users to access organizational data simultaneously through computer networks. Without appropriate access control mechanisms, confidential information becomes vulnerable to unauthorized viewing, modification, deletion, or misuse. Consequently, organizations implement RBAC to ensure that users only have access to system resources relevant to their designated responsibilities (Stallings, 2021).

The fundamental principle of RBAC is the **principle of least privilege**, which states that users should only receive the minimum permissions necessary to perform their assigned tasks. This principle minimizes security risks by preventing employees from accessing confidential information unrelated to their responsibilities. According to Sandhu et al. (1996), limiting user privileges reduces the likelihood of unauthorized data modification, accidental deletion of information, and internal security breaches while improving organizational accountability. The implementation of least privilege has become a standard practice in modern information security because it balances operational efficiency with data protection.

One of the primary advantages of RBAC is simplified user administration. In organizations with numerous employees, manually assigning permissions to individual users becomes time-consuming and difficult to manage. RBAC addresses this challenge by assigning permissions to organizational roles rather than individual accounts. When a new employee joins the organization, the administrator simply assigns the appropriate role, automatically granting all permissions associated with that role. Similarly, changes in employee responsibilities require only role modification instead of manually updating numerous individual permissions. This significantly reduces administrative complexity while maintaining consistent security policies throughout the organization (Ferraiolo et al., 2001).

RBAC also improves organizational accountability through user authentication and activity monitoring. Every authorized user accesses the system using individual credentials linked to specific organizational roles. Consequently, every transaction performed within the system can be associated with the responsible user, enabling administrators to maintain audit trails and monitor system activities effectively. Audit logs generated by RBAC systems support organizational transparency by recording login attempts, data modifications, transaction processing, and other system activities. These records assist management in identifying operational issues, investigating security incidents, and maintaining compliance with organizational policies (NIST, 2020).

Another important characteristic of RBAC is its ability to strengthen data confidentiality. Organizations frequently maintain sensitive information including customer records, employee information, financial transactions, inventory reports, and managerial documents. Through RBAC, confidential information remains accessible only to authorized personnel whose responsibilities require such access. Employees performing routine operational tasks cannot access strategic reports or administrative functions beyond their assigned roles. This segregation of responsibilities significantly reduces opportunities for unauthorized information disclosure while strengthening organizational governance.

Modern web-based information systems commonly integrate RBAC with authentication mechanisms such as usernames, passwords, encrypted sessions, and multi-factor authentication. These security measures ensure that only authenticated users gain access to organizational resources while RBAC determines the specific functionalities available after successful authentication. According to Stallings (2021), combining authentication and authorization mechanisms creates multiple layers of security that improve the overall protection of organizational information systems.

Within restaurant management systems, RBAC plays an essential role because different employees perform distinct operational responsibilities requiring different levels of system access. Restaurant administrators typically manage employee accounts, system configuration, and overall business reports. Managers monitor sales performance, inventory status, reservation schedules, and operational activities. Cashiers process customer transactions and generate receipts, while kitchen personnel receive customer orders and update food preparation status. Customers or online users may only access reservation services and personal booking information. Implementing RBAC ensures that each category of user accesses only the system functionalities necessary for performing their assigned responsibilities while preventing unauthorized access to confidential business information.

Recent developments in cloud computing and web-based applications have further emphasized the importance of Role-Based Access Control. Cloud-hosted systems allow users to access organizational information remotely through internet-connected devices. While this accessibility improves organizational flexibility, it also increases potential security risks associated with unauthorized access and cyber threats. Consequently, RBAC has become a fundamental security mechanism integrated into modern cloud applications to protect organizational data while maintaining controlled access across distributed computing environments.

Another significant benefit of RBAC is its contribution to organizational efficiency. Because users interact only with system modules relevant to their assigned responsibilities, the user interface becomes less complex and easier to navigate. Employees can perform operational tasks more efficiently without unnecessary distractions created by unrelated system functions. Simplified interfaces also reduce training requirements while minimizing operational errors resulting from unauthorized or inappropriate system usage.

The concepts discussed in the reviewed literature are highly relevant to the present study because the proposed **Online Restaurant Management System with Sales Analytics and Reservation Scheduling** implements **Role-Based Access Control** to regulate access among administrators, managers, cashiers, kitchen personnel, and Google users. Similar to the RBAC model discussed by previous authors, the proposed system assigns system permissions according to organizational responsibilities, ensuring that each user can access only the functionalities necessary for performing assigned operational tasks. The implementation of RBAC strengthens data confidentiality, improves system security, supports organizational accountability through user activity monitoring, and protects sensitive business information from unauthorized access. Consequently, Role-Based Access Control serves as one of the primary security mechanisms incorporated into the proposed system and provides a strong theoretical foundation for secure restaurant information management.

ISO/IEC 25010 Software Quality Model

Software quality has become one of the most important considerations in software engineering because it determines the extent to which a software product satisfies user requirements, performs its intended functions, and maintains reliable performance throughout its operational life cycle. As software systems continue to support critical business operations across various industries, evaluating software quality has become essential to ensure that developed applications meet acceptable standards of functionality, efficiency, security, reliability, maintainability, and user satisfaction. Software quality evaluation enables developers and organizations to identify strengths, recognize areas requiring improvement, and verify that a system effectively addresses the operational requirements for which it was developed (Pressman & Maxim, 2020).

To establish internationally recognized criteria for software quality evaluation, the **International Organization for Standardization (ISO)** and the **International Electrotechnical Commission (IEC)** developed the **ISO/IEC 25010 Systems and Software Quality Requirements and Evaluation (SQuaRE) Model**. ISO/IEC 25010 provides a comprehensive framework for evaluating software products based on defined quality characteristics that measure both the internal and external quality of software systems. The standard serves as one of the most widely accepted software quality models in software engineering because it provides systematic guidelines for assessing software performance, usability, security, maintainability, and overall quality (ISO/IEC, 2023).

According to the ISO/IEC 25010 standard, software quality refers to the degree to which a software product satisfies stated and implied user needs under specified conditions. Rather than focusing solely on technical correctness, the model recognizes that software quality encompasses multiple characteristics that collectively determine the effectiveness of a software application in supporting user activities and organizational objectives. These quality characteristics enable software developers, researchers, and evaluators to assess software performance comprehensively while ensuring consistency during software evaluation and acceptance testing (ISO/IEC, 2023).

One of the primary quality characteristics identified in ISO/IEC 25010 is **Functional Suitability**. Functional suitability measures the degree to which software provides functions that satisfy stated and implied user requirements. It evaluates whether the implemented functionalities are complete, accurate, and appropriate for accomplishing specified tasks. A software application demonstrates high functional suitability when it performs intended operations correctly while addressing user requirements without unnecessary complexity. Within restaurant management systems, functional suitability determines whether modules such as Point-of-Sale, reservation scheduling, inventory monitoring, sales analytics, and report generation operate according to business requirements.

Another important quality characteristic is **Performance Efficiency**, which evaluates how efficiently software utilizes system resources while providing acceptable response times under specified operating conditions. Performance efficiency includes considerations related to processing speed, resource utilization, system responsiveness, and operational capacity. Efficient software minimizes waiting time during transaction processing while maintaining stable performance even when multiple users simultaneously access the system. For web-based restaurant management systems, performance efficiency directly influences customer service quality because slow transaction processing may increase customer waiting time and reduce organizational productivity (ISO/IEC, 2023).

Usability refers to the extent to which software can be understood, learned, operated, and found attractive by intended users under specified conditions. Software with high usability enables users to perform operational tasks efficiently without requiring extensive technical knowledge or prolonged training. According to Pressman and Maxim (2020), user-centered software design significantly improves user satisfaction because intuitive interfaces reduce operational errors while increasing productivity. Within restaurant operations, usability is particularly important because employees with varying levels of technological experience must efficiently utilize the system during busy operating hours.

Another essential software quality characteristic is **Reliability**, which measures the ability of software to perform specified functions consistently under stated conditions for a specified period. Reliable software maintains stable operation despite continuous use while minimizing system failures, data loss, or unexpected interruptions. Reliability is particularly important in business environments where system failures may disrupt daily operations, affect customer service, and result in financial losses. Restaurant management systems require high reliability because interruptions during transaction processing, reservation management, or inventory monitoring may significantly affect operational efficiency and customer satisfaction.

Security has likewise become one of the most significant quality characteristics of modern software systems because organizations increasingly manage confidential information through digital platforms. Security evaluates the software's ability to protect information and data from unauthorized access while ensuring confidentiality, integrity, authenticity, accountability, and availability. ISO/IEC 25010 emphasizes that software should implement appropriate security mechanisms capable of preventing unauthorized access, protecting sensitive organizational information, and maintaining user accountability. Within restaurant management systems, security mechanisms such as user authentication, encrypted communication, audit logs, and Role-Based Access Control (RBAC) protect customer information, financial records, inventory data, and administrative functions from unauthorized use (ISO/IEC, 2023).

The quality model also identifies **Maintainability** as an important characteristic that measures the ease with which software can be modified to correct faults, improve performance, or adapt to changing operational requirements. Maintainable software possesses well-organized program structures, modular components, and comprehensive documentation that facilitate future system enhancement and maintenance activities. According to Sommerville (2021), maintainability significantly influences the long-term sustainability of software because organizational requirements continuously evolve, requiring software applications to accommodate new functionalities and technological advancements.

Another quality characteristic included in ISO/IEC 25010 is **Portability**, which refers to the ability of software to be transferred from one environment to another with minimal modification. Portable software can operate effectively across different hardware configurations, operating systems, web browsers, and computing environments. The growing adoption of web-based applications has increased the importance of portability because users frequently access software using various devices, including desktop computers, laptops, tablets, and smartphones. High software portability enables organizations to maximize system accessibility while reducing compatibility-related operational problems.

The ISO/IEC 25010 quality model has been widely adopted within software engineering research because it provides standardized evaluation criteria applicable to diverse software applications. Numerous information system development studies utilize ISO/IEC 25010 to evaluate web-based applications, mobile applications, information management systems, healthcare systems, educational systems, and business management software. The standardized quality characteristics facilitate objective software evaluation while enabling researchers to compare software quality across different systems using internationally recognized criteria.

Within the context of restaurant management systems, ISO/IEC 25010 provides an appropriate framework for evaluating software applications because restaurant operations require systems that are functionally complete, operationally efficient, user-friendly, secure, reliable, maintainable, and accessible across different computing environments. These quality characteristics collectively determine whether a restaurant management system effectively supports daily business operations while satisfying user expectations and organizational requirements.

The reviewed literature strongly supports the present study because the developed **Online Restaurant Management System with Sales Analytics and Reservation Scheduling** was evaluated using the **ISO/IEC 25010 Software Quality Model**. Specifically, the developed system was assessed in terms of **Functional Suitability, Performance Efficiency, Usability, Reliability, Security, Maintainability, and Portability**. These quality characteristics provided standardized criteria for determining the overall quality and acceptability of the developed application as evaluated by administrators, managers, cashiers, kitchen personnel, and other authorized users. By adopting the ISO/IEC 25010 model, the present study utilized an internationally recognized framework that strengthened the credibility, objectivity, and reliability of the software evaluation process.

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CHAPTER III

METHODOLOGY

3.1 Research Design

This study employed a **developmental research design** to design, develop, implement, and evaluate an **Online Restaurant Management System with Sales Analytics and Reservation Scheduling** for Crates N' Plates. Developmental research is a systematic research approach that focuses on the creation, improvement, and evaluation of products, processes, or systems intended to address identified problems or operational needs. Unlike descriptive or experimental research, developmental research emphasizes the design and construction of practical solutions that are validated through testing and evaluation before implementation (Richey & Klein, 2014).

The developmental research design was considered appropriate because the primary objective of the study was not merely to observe or describe existing restaurant operations but to develop a computerized system capable of improving the efficiency and effectiveness of the establishment's daily business processes. The researchers identified several operational challenges encountered by Crates N' Plates, including manual sales transaction recording, reservation scheduling, inventory monitoring, report generation, and sales analysis. These operational limitations served as the basis for developing an integrated web-based restaurant management system that automated and centralized essential business functions.

The study followed a systematic software development process using the **Agile Software Development Methodology**. Agile emphasizes iterative and incremental development, continuous collaboration with stakeholders, and the ability to adapt to changing business requirements throughout the software development life cycle (Pressman & Maxim, 2020). The methodology allowed the researchers to gather requirements from the client, design appropriate system features, develop functional modules incrementally, conduct continuous testing, and incorporate client feedback throughout the development process. This iterative approach ensured that the developed system closely aligned with the operational requirements of Crates N' Plates while allowing improvements to be implemented whenever necessary during development.

During the planning phase, the researchers conducted interviews and direct observations with the owner of Crates N' Plates to determine the restaurant's existing operational procedures, identify current challenges, and gather the functional and non-functional requirements of the proposed system. Information obtained during this phase became the foundation for the system's architecture, database design, user interface, and module development.

Following the planning and requirements gathering activities, the researchers designed the overall system architecture, database structure, user interface layouts, and functional modules. The development process focused on integrating major operational components, including the Point-of-Sale (POS) module, reservation scheduling, inventory monitoring, kitchen order management, sales analytics dashboard, report generation, and Role-Based Access Control (RBAC). Each module was developed and tested incrementally to ensure that individual functionalities operated correctly before integration into the complete system.

Throughout the development process, continuous communication with the client was maintained to validate whether the implemented functionalities satisfied the operational requirements of the restaurant. Feedback gathered during each development iteration enabled the researchers to improve system usability, modify specific functionalities, and resolve identified issues before proceeding to subsequent development phases. This collaborative process reflected one of the fundamental principles of Agile development, which emphasizes customer involvement and continuous improvement.

Upon completion of the system, comprehensive testing procedures were conducted to verify that all modules functioned according to their intended specifications. Functional testing ensured that each feature operated correctly, while integration testing verified that different modules communicated effectively with one another. User Acceptance Testing (UAT) was also performed with the client to determine whether the completed system successfully addressed the operational needs identified during the initial requirements gathering phase. The client evaluated the usability, functionality, and overall performance of the developed system based on actual business processes and confirmed that the implemented features aligned with the operational requirements of Crates N' Plates.

The developed system was subsequently evaluated using the **ISO/IEC 25010 Software Quality Model**, which served as the framework for assessing the overall quality and acceptability of the application. The evaluation focused on the software quality characteristics of **Functional Suitability, Performance Efficiency, Usability, Reliability, Security, Maintainability, and Portability**. These internationally recognized quality characteristics provided standardized criteria for determining whether the developed system met acceptable software quality standards and effectively addressed the operational challenges encountered by Crates N' Plates.

Overall, the developmental research design provided an appropriate framework for achieving the objectives of the study. It enabled the researchers to systematically identify operational problems, develop an integrated web-based solution, validate the developed application through iterative development and testing, and evaluate its quality using recognized software engineering standards. Through this research design, the study successfully demonstrated how information technology could be utilized to improve restaurant operations, support data-driven decision-making, and enhance service delivery through the implementation of an Online Restaurant Management System with Sales Analytics and Reservation Scheduling.

3.2 System Development Methodology

This study adopted the **Agile Software Development Methodology** in designing and developing the **Online Restaurant Management System with Sales Analytics and Reservation Scheduling** for Crates N' Plates. Agile is an iterative and incremental software development methodology that emphasizes continuous collaboration with stakeholders, rapid delivery of functional software, adaptability to changing requirements, and continuous improvement throughout the development process. Unlike traditional software development methodologies that complete each phase sequentially, Agile encourages repeated development cycles known as iterations or sprints, allowing developers to continuously evaluate, improve, and refine the system based on stakeholder feedback (Pressman & Maxim, 2020).

The Agile methodology was selected because it provided flexibility throughout the software development process. Since Crates N' Plates continuously expanded its operations and operational requirements evolved during system development, the researchers required a methodology capable of accommodating changes without disrupting the overall development process. Through Agile, system features were developed incrementally, allowing the researchers to consult the client regularly, validate completed modules, and implement revisions whenever necessary. This iterative approach ensured that the developed system remained aligned with the operational needs of the restaurant while minimizing development risks and improving software quality.

The Agile Software Development Methodology followed in this study consisted of six major phases: **Planning, Requirements Analysis, System Design, Development, Testing, and Deployment & Maintenance**. Each phase played a significant role in ensuring the successful completion of the proposed Online Restaurant Management System.

3.2.1 Planning

The planning phase served as the foundation of the entire software development process. During this phase, the researchers identified the overall objectives of the project, determined the scope of the proposed system, established the project timeline, and identified the necessary hardware, software, and human resources required for development.

The researchers initially coordinated with the owner of Crates N' Plates to discuss the existing operational procedures and determine the feasibility of developing a computerized restaurant management system. Preliminary meetings were conducted to identify the restaurant's daily workflow, including customer ordering, reservation handling, inventory monitoring, report generation, and sales management. The researchers also discussed the expected functionalities, limitations, and priorities of the proposed system.

Based on these discussions, the researchers prepared the project development plan, identified the major system modules, and established development milestones. The planning phase also included identifying potential risks that could affect project implementation, such as changing client requirements, data migration concerns, schedule adjustments, and technical constraints. Appropriate mitigation strategies were established to minimize the impact of these risks throughout the development process.

3.2.2 Requirements Analysis

The requirements analysis phase involved gathering detailed information regarding the operational needs of Crates N' Plates. The primary objective of this phase was to identify the problems encountered in the existing manual system and determine the functional and non-functional requirements that the proposed system should satisfy.

The researchers utilized interviews, direct observations, and document analysis during this phase. Interviews were conducted with the restaurant owner to obtain comprehensive information regarding existing business processes, operational challenges, and expected system functionalities. Direct observations enabled the researchers to understand the actual workflow involved in order processing, reservation scheduling, inventory monitoring, and report preparation. Existing operational documents, including reservation forms, inventory records, sales receipts, and manually prepared reports, were also reviewed to understand the restaurant's documentation procedures.

Information gathered during this phase was analyzed and organized into system requirements. Functional requirements included Point-of-Sale transaction processing, reservation scheduling, inventory monitoring, kitchen order management, report generation, user management, and sales analytics. Non-functional requirements included usability, security, reliability, maintainability, performance efficiency, and accessibility.

The requirements gathered during this phase served as the primary reference for designing the system architecture and database structure.

3.2.3 System Design

Following the completion of the requirements analysis, the researchers proceeded to the system design phase. This phase focused on translating the identified requirements into a detailed system blueprint that guided the development of the application.

The researchers designed the overall system architecture, database structure, user interface layouts, navigation flow, and module interactions. Various software engineering tools such as Use Case Diagrams, Data Flow Diagrams (DFD), Entity Relationship Diagrams (ERD), and system flowcharts were prepared to visualize the interaction between users, system processes, and stored data.

The user interface was designed with simplicity and usability as primary considerations. Navigation menus, dashboards, transaction forms, inventory screens, reservation pages, and sales reports were organized to allow users to complete tasks efficiently with minimal training. Color consistency, readability, and accessibility were also considered during interface development.

Database design focused on maintaining data integrity and minimizing redundancy through the use of Firebase Realtime Database. Separate collections were designed for users, products, reservations, inventory records, sales transactions, reports, and kitchen orders while maintaining relationships among these data through unique identifiers.

3.2.4 System Development

During the development phase, the researchers translated the completed system design into a functional web-based application. Development was performed incrementally following Agile principles, wherein individual modules were completed and evaluated before proceeding to subsequent components.

The system was developed using **PHP**, **HTML5**, **CSS3**, and **JavaScript** as the primary programming languages. Visual Studio Code served as the integrated development environment, while XAMPP was utilized as the local development server during testing. Firebase Realtime Database functioned as the centralized cloud database for storing operational data, while Firebase Authentication managed user authentication and secure login sessions.

The researchers developed the system module by module. The Point-of-Sale module was developed to process customer transactions efficiently while automatically recording sales information. The reservation module managed dine-in reservations, catering bookings, and private function room schedules. Inventory management monitored stock movement and product availability, while the kitchen module displayed incoming customer orders for food preparation. The sales analytics dashboard generated graphical reports that assisted management in evaluating sales performance and customer purchasing trends.

Throughout development, completed modules were demonstrated to the client for validation. Client feedback was immediately incorporated into succeeding iterations, allowing continuous improvement of the system throughout the development cycle.

3.2.5 Testing

System testing was conducted continuously throughout the Agile development process. Instead of postponing testing until system completion, individual modules were tested immediately after development to identify errors early and minimize software defects.

Functional testing verified whether each module operated according to the specified requirements. The researchers tested customer transactions, reservation scheduling, inventory monitoring, kitchen order processing, user authentication, report generation, and sales analytics individually before integrating them into the complete system.

Integration testing was subsequently performed to ensure proper communication among different modules. Particular attention was given to verifying that completed sales transactions automatically updated inventory records, approved reservations synchronized with scheduling records, and analytical dashboards correctly reflected operational data stored within the database.

User Acceptance Testing (UAT) was conducted with the owner of Crates N' Plates after system completion. The client evaluated the system by performing actual operational tasks and verified whether the developed application satisfied the restaurant's business requirements. Issues identified during testing were corrected before final deployment.

3.2.6 Deployment and Maintenance

Following successful testing and client approval, the completed Online Restaurant Management System was deployed for operational use at Crates N' Plates. The deployment phase involved configuring the application, preparing the Firebase database, creating user accounts, and verifying system accessibility.

The researchers also provided orientation regarding the operation of major system modules, including transaction processing, reservation management, inventory monitoring, report generation, and sales analytics.

Maintenance activities continued after deployment to ensure stable system performance. Minor software issues identified during actual usage were corrected through system updates. As new operational requirements emerged, the Agile methodology allowed additional improvements and feature enhancements to be incorporated without significantly disrupting existing functionalities.

The maintenance phase ensured that the developed system remained reliable, secure, and responsive to the evolving operational requirements of Crates N' Plates.

Figure 3.1 Agile Software Development Methodology

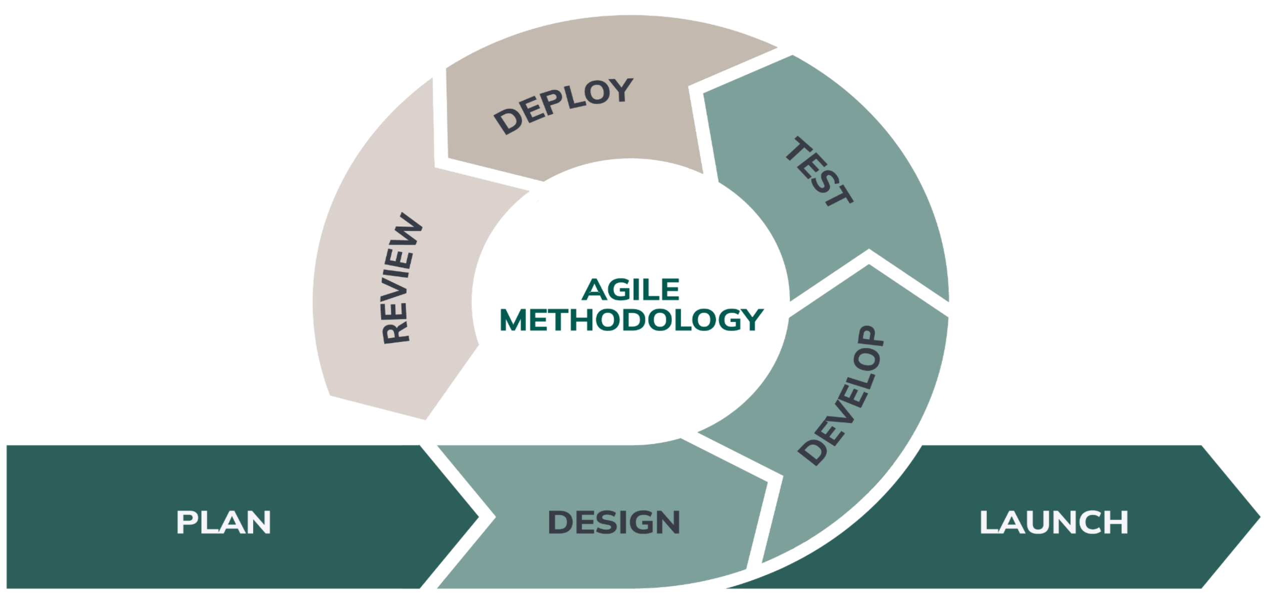

Figure 3.1 illustrates the Agile Software Development Methodology adopted in the study. Unlike traditional linear development models, Agile promotes continuous feedback and iterative improvement. Each completed module was evaluated by the client, allowing the researchers to implement revisions and enhancements throughout the software development life cycle.

3.3 System Overview

The **Online Restaurant Management System with Sales Analytics and Reservation Scheduling** was developed to improve the operational efficiency of Crates N' Plates by automating and integrating the restaurant's core business processes into a centralized web-based platform. Prior to the development of the system, the establishment relied heavily on manual procedures in processing customer transactions, recording reservations, monitoring inventory, and preparing business reports. Although these manual methods had been sufficient during the restaurant's early years of operation, they became increasingly inefficient as customer demand, business transactions, and operational activities expanded. Consequently, the researchers designed and developed a computerized restaurant management system capable of addressing these operational challenges through automation, centralized data management, and real-time information processing.

The developed system served as an integrated platform that connected the restaurant's major operational functions, including Point-of-Sale (POS) transaction processing, reservation scheduling, inventory monitoring, kitchen order management, sales analytics, report generation, and user management. Rather than maintaining separate paper-based records for each business function, all operational information was stored within a centralized Firebase Realtime Database, allowing authorized users to access accurate and up-to-date information whenever necessary. This centralized architecture minimized duplicate data entry, reduced human errors, and improved coordination among restaurant personnel.

The system was implemented as a web-based application, allowing authorized users to access its functionalities through desktop computers, laptops, tablets, or mobile devices connected to the internet or local network. Because the application was browser-based, users were not required to install specialized software on individual devices, making system deployment and maintenance more efficient. The centralized nature of the application also ensured that all operational data remained synchronized across different system modules, thereby improving consistency and reliability throughout restaurant operations.

The developed system was designed to accommodate multiple categories of users, each assigned specific responsibilities according to their operational roles within the restaurant. To maintain information security and operational integrity, the application implemented **Role-Based Access Control (RBAC)**, ensuring that users could only access functionalities relevant to their assigned responsibilities. Administrators were responsible for managing user accounts, maintaining system settings, monitoring reports, and overseeing the overall operation of the application. Managers were provided access to sales analytics, inventory reports, reservation schedules, and other managerial information necessary for decision-making. Cashiers primarily handled customer transactions through the Point-of-Sale module, processed payments, managed reservations, and generated transaction receipts. Kitchen personnel received customer orders electronically through the kitchen management interface, allowing food preparation to begin immediately after customer transactions were confirmed. Customers, through their Google accounts, could submit reservation requests and monitor the status of their bookings.

One of the primary modules incorporated into the system was the **Point-of-Sale (POS) Module**, which automated customer transaction processing. This module enabled cashiers to record customer orders, calculate transaction totals, process payments, generate electronic receipts, and automatically store sales information within the centralized database. Completed transactions simultaneously updated inventory records and contributed to the sales analytics dashboard, thereby eliminating duplicate encoding and reducing computational errors commonly encountered in manual transaction processing.

Another major component of the system was the **Reservation Scheduling Module**, which managed dine-in reservations, private function room bookings, and catering reservations. Authorized personnel could verify reservation requests, approve or reject bookings, update reservation schedules, and monitor booking availability through a centralized scheduling interface. By maintaining reservation information within a single database, the module minimized scheduling conflicts, prevented duplicate bookings, and improved communication among restaurant personnel responsible for customer reservations.

The **Inventory Monitoring Module** automated the recording and monitoring of inventory movement for both kitchen ingredients and beverage supplies. Product quantities were automatically updated whenever customer transactions were completed, allowing management to monitor current stock levels accurately. The module also assisted authorized users in identifying low-stock items, monitoring inventory consumption, and generating inventory reports that supported purchasing decisions and inventory replenishment planning. Through automated inventory tracking, the restaurant reduced the occurrence of inventory discrepancies, stock shortages, and unnecessary overstocking.

To improve communication between the front-of-house and kitchen personnel, the system incorporated a **Kitchen Order Management Module**. Customer food orders entered by the cashier were transmitted electronically to the kitchen interface immediately after transaction confirmation. Kitchen staff could monitor incoming orders, update preparation status, and coordinate food preparation more efficiently without relying on handwritten order slips or verbal communication. This electronic workflow reduced communication delays, minimized order errors, and improved overall service efficiency during peak operating hours.

A distinguishing feature of the developed application was its **Sales Analytics Dashboard**, which transformed operational transaction data into meaningful business information. Through graphical reports, charts, and statistical summaries, management could monitor daily, weekly, monthly, and annual sales performance, identify best-selling menu items, analyze customer purchasing behavior, evaluate promotional activities, and monitor revenue trends. The analytical reports generated by the dashboard supported evidence-based managerial decision-making, allowing the restaurant to optimize inventory planning, improve marketing strategies, and enhance overall business performance.

The system also incorporated an **Automated Report Generation Module**, which generated real-time reports related to sales transactions, reservations, inventory status, customer bookings, and overall restaurant performance. Unlike manual reporting procedures that required considerable time and effort, the automated reporting module enabled authorized personnel to generate comprehensive reports instantly whenever required. These reports assisted management in monitoring daily operations, evaluating employee performance, and preparing business summaries necessary for strategic planning.

To ensure the confidentiality, integrity, and security of operational information, the developed system implemented multiple security mechanisms. User authentication was managed through secure login procedures integrated with Firebase Authentication, while Role-Based Access Control regulated user permissions according to assigned organizational roles. Sensitive operational data stored within the Firebase Realtime Database were accessible only to authorized users, thereby protecting business information from unauthorized access or accidental modification. These security measures strengthened organizational data protection while maintaining accountability through individual user accounts.

Overall, the Online Restaurant Management System with Sales Analytics and Reservation Scheduling functioned as a comprehensive business management platform capable of integrating multiple operational processes into a single web-based application. By automating sales transaction processing, reservation scheduling, inventory monitoring, kitchen order management, business reporting, and sales analytics, the system significantly reduced reliance on manual procedures while improving operational efficiency, data accuracy, customer service, and managerial decision-making. The developed application therefore provided Crates N' Plates with a centralized, secure, and scalable information system capable of supporting its present operational requirements and future business growth.

Figure 3.2 Overall System Overview

(Insert ang diagram di erd or data flow.)

Figure 3.2 illustrates the overall operational flow of the proposed Online Restaurant Management System. Customer transactions, reservations, inventory monitoring, kitchen order processing, sales analytics, and report generation are integrated through a centralized Firebase Realtime Database. This architecture enables authorized users to access real-time operational information while improving coordination, reducing manual processes, and supporting efficient restaurant management.

3.4 System Architecture

The **Online Restaurant Management System with Sales Analytics and Reservation Scheduling** was designed using a centralized web-based architecture that enables multiple users to access and manage restaurant operations through a secure and integrated platform. The architecture consists of four major user groups: **Customers**

(Users), Cashiers, Kitchen Staff, and Administrators. Each user interacts with the system through a web browser, while all operational data are stored and synchronized using the **Firebase Realtime Database**. This architecture ensures that information is updated in real time, allowing authorized personnel to access accurate and current operational data whenever necessary.

The developed system follows a client-server architecture in which users access the application through desktop computers, laptops, tablets, or mobile devices connected to the internet or a local network. The client-side interface is responsible for displaying information, accepting user input, and transmitting requests to the application. The server-side components process these requests, execute the necessary business logic, communicate with the Firebase Realtime Database, and return the requested information to the user interface. This architecture enables efficient communication between different system modules while maintaining centralized control over business data.

The **Customer Module** serves as the primary interface for restaurant customers. Through this module, customers may register for an account, log in securely, browse available food, beverages, and catering services, create reservations, complete online payments, and receive electronic receipts through their registered email addresses. Customers may also monitor the status of their reservations after submission. Reservation requests remain pending until they are reviewed by the cashier, ensuring that only verified bookings proceed to the preparation stage.

The **Cashier Module** is responsible for managing customer transactions and reservation processing. Cashiers review incoming online reservations and determine whether they should be approved or rejected based on restaurant availability and operational requirements. The module also functions as the restaurant's Point-of-Sale (POS) system, allowing cashiers to process walk-in transactions, generate receipts, verify customer payments, and maintain accurate sales records. Once a reservation or customer order has been approved, the cashier forwards the order electronically to the kitchen module for preparation.

The **Kitchen Module** receives all approved customer orders transmitted by the cashier. Kitchen personnel use this module to monitor incoming food orders and update the status of each order throughout the preparation process. Order status progresses through several stages, including **Preparing**, **Cooking**, and **Done**. Status updates are reflected immediately within the system, enabling the cashier to monitor order progress and provide customers with accurate information regarding their orders.

The **Administrator Module** provides complete control over the system. Administrators manage employee accounts, assign user roles, maintain menu information, monitor reservations, supervise inventory, review sales transactions, generate business reports, and analyze organizational performance using graphical sales analytics. This module also enables administrators to create accounts for restaurant personnel, ensuring that only authorized users have access to specific system functionalities through Role-Based Access Control (RBAC).

The system utilizes **Firebase Realtime Database** as its centralized cloud database. Firebase stores user accounts, reservation records, customer transactions, inventory information, catering bookings, kitchen orders, reports, and sales analytics. One of the primary advantages of Firebase Realtime Database is its ability to synchronize information instantly across all connected users. Whenever a cashier approves a reservation, the kitchen immediately receives the corresponding order. Similarly, when the kitchen updates an order's preparation status, the cashier can instantly monitor the progress without requiring manual communication. This real-time synchronization significantly improves coordination among restaurant personnel while minimizing communication delays.

Security was incorporated into the architecture through user authentication and Role-Based Access Control (RBAC). Each user must log into the system using authorized credentials before accessing the application. System permissions are assigned according to organizational roles, ensuring that customers, cashiers, kitchen staff, and administrators only access the modules and information necessary to perform their respective responsibilities. This approach strengthens data confidentiality, minimizes unauthorized access, and protects sensitive business information.

The architecture also supports the integration of multiple operational modules within a single platform. Customer reservations, Point-of-Sale transactions, inventory monitoring, kitchen order management, report generation, and sales analytics are interconnected through the centralized database. Consequently, completed transactions automatically update inventory records, contribute to sales reports, and generate analytical information used by management for business planning and decision-making. This integration eliminates duplicate data entry, improves operational efficiency, and enhances the reliability of restaurant records.

Overall, the developed system architecture provides a scalable, secure, and centralized framework capable of supporting the day-to-day operations of Crates N' Plates. Through the integration of real-time data synchronization, role-based user management, centralized information storage, and automated business processes, the architecture enables the restaurant to improve operational efficiency, strengthen communication among personnel, and provide better customer service while supporting future system expansion.

Figure 3.3 System Architecture

(Insert ang architecture diagram diri (example lng ni))

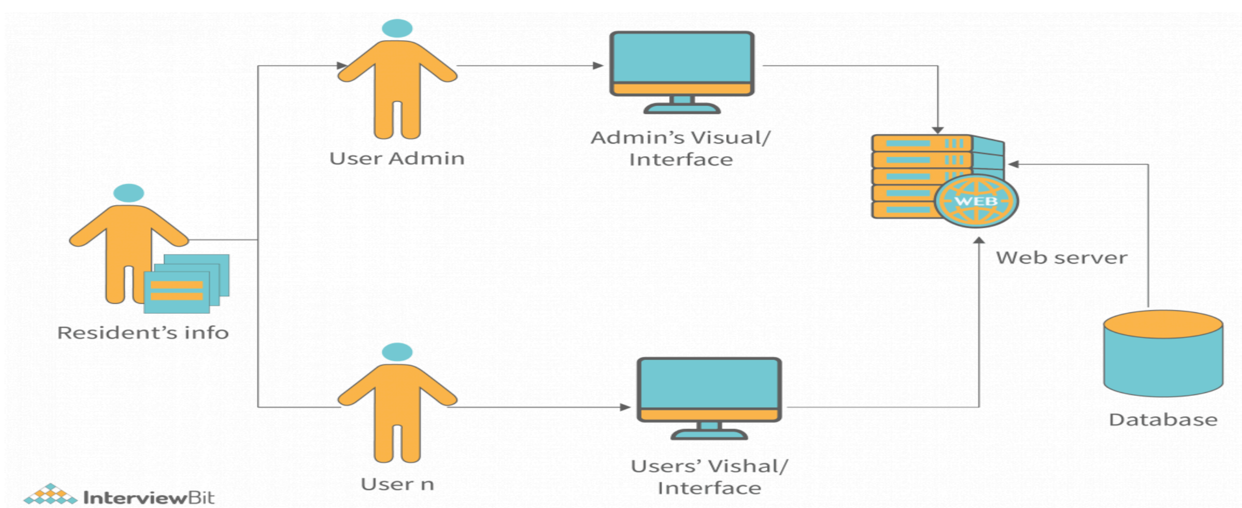

Figure 3.3 illustrates the overall architecture of the Online Restaurant Management System. Customers access the system through a web browser to submit reservations and payments. Reservation requests are processed by the cashier, approved orders are

forwarded to the kitchen for preparation, and administrators oversee the entire operation through the management dashboard. All system modules communicate with the Firebase Realtime Database, ensuring real-time synchronization of operational data throughout the restaurant.

3.5 System Components and Modules

The **Online Restaurant Management System with Sales Analytics and Reservation Scheduling** was developed as an integrated web-based application composed of several interconnected modules. Each module was designed to perform specific operational functions while communicating with other components through the centralized Firebase Realtime Database. The modular design improves system organization, simplifies maintenance, and allows each operational process to function independently while remaining integrated with the overall restaurant management system.

The major system components include the **User Management Module**, **Reservation Management Module**, **Point-of-Sale (POS) Module**, **Kitchen Management Module**, **Inventory Monitoring Module**, **Sales Analytics Module**, **Report Generation Module**, and **Administration Module**. Together, these modules automate the daily business operations of Crates N' Plates and provide management with accurate, real-time operational information.

3.5.1 User Management Module

The User Management Module is responsible for managing customer and employee accounts within the system. It provides secure authentication and ensures that only authorized individuals can access the application's functionalities. Customers create their own accounts by registering through the online registration page, after which they may log in using their registered credentials to access reservation services. Employees, including cashiers, kitchen personnel, and administrators, are assigned accounts by the system administrator. This approach allows management to regulate employee access while preventing unauthorized individuals from accessing confidential business information.

The module implements **Role-Based Access Control (RBAC)**, which assigns specific permissions according to each user's organizational role. Customers can only access reservation-related functions, while cashiers, kitchen staff, and administrators are provided access only to the modules necessary for their assigned responsibilities. By limiting user privileges, the system enhances security, protects sensitive operational data, and maintains accountability for all transactions performed within the application.

3.5.2 Reservation Management Module

The Reservation Management Module enables customers to reserve tables, food packages, beverages, and catering services before their scheduled visit to the restaurant. Customers complete the reservation form by selecting their preferred date, time, number of guests, and required services. Catering reservations also allow customers to reserve additional items such as tables, chairs, and skirting cloths according to their event requirements.

After completing the reservation details, customers proceed with the required online payment. Upon successful payment, the system automatically generates an electronic receipt and sends it to the customer's registered email address. Reservation requests are initially marked as **Pending**, awaiting verification by the cashier. This verification process allows restaurant personnel to confirm reservation availability before final approval.

Cashiers review all incoming reservation requests through the reservation management dashboard. Reservations may either be **approved** or **declined** depending on restaurant availability and operational considerations. Approved reservations proceed to the kitchen for order preparation when necessary, while declined reservations notify customers of the reservation status. This process minimizes scheduling conflicts, prevents duplicate bookings, and improves reservation organization.

3.5.3 Point-of-Sale (POS) Module

The Point-of-Sale (POS) Module automates the processing of customer transactions within the restaurant. Cashiers utilize this module to process walk-in customer orders, calculate transaction totals, record payments, and generate official transaction receipts. Unlike manual transaction recording, the POS module automatically stores transaction details within the Firebase Realtime Database, thereby reducing computational errors and improving record accuracy.

The module also manages online reservation payments that have been approved by the cashier. Completed transactions immediately update the restaurant's sales records and contribute to the Sales Analytics Module for reporting purposes. The automation of sales recording significantly reduces administrative workload while improving financial reporting accuracy.

3.5.4 Kitchen Management Module

The Kitchen Management Module facilitates communication between the cashier and kitchen personnel. Once the cashier approves a reservation or confirms a customer order, the corresponding order information is transmitted electronically to the kitchen interface. This eliminates the need for handwritten order slips and verbal communication, reducing the likelihood of order misinterpretation.

Kitchen personnel monitor all incoming orders through the module and update each order according to its current preparation stage. The available status indicators include **Preparing**, **Cooking**, and **Done**. These status updates are reflected immediately within the system, allowing the cashier to monitor order progress and provide customers with accurate information regarding food preparation.

The Kitchen Management Module improves coordination among restaurant personnel while reducing communication delays during busy operating hours.

3.5.5 Inventory Monitoring Module

The Inventory Monitoring Module manages the restaurant's available stocks for kitchen ingredients, beverages, and catering equipment. The module automatically updates inventory quantities whenever customer transactions are completed or catering reservations are confirmed. By integrating inventory management with the Point-of-Sale module, the system minimizes manual stock recording and improves inventory accuracy.

Authorized personnel can monitor product availability, identify low-stock items, review inventory movement, and generate inventory reports through this module. Separate inventory categories are maintained for kitchen ingredients, beverages, and rentable catering equipment to reflect the actual operational practices of Crates N' Plates. Automated inventory monitoring assists management in preventing stock shortages, reducing overstocking, and improving purchasing decisions.

3.5.6 Sales Analytics Module

The Sales Analytics Module transforms operational transaction records into meaningful business information through graphical reports and statistical summaries. Every completed transaction processed through the Point-of-Sale module automatically contributes to the analytical database, allowing management to monitor restaurant performance in real time.

The module generates reports illustrating daily, weekly, monthly, and yearly sales performance. It also identifies best-selling products, monitors customer purchasing trends, analyzes revenue generation, and evaluates overall business performance. These analytical capabilities support evidence-based managerial decision-making by enabling administrators to recognize operational patterns and develop strategies for improving profitability.

Charts and graphical dashboards generated by this module provide management with a clear visualization of business performance without requiring manual computation.

3.5.7 Report Generation Module

The Report Generation Module automatically produces comprehensive business reports based on information stored within the Firebase Realtime Database. Authorized users may generate reports related to sales transactions, reservation activities, inventory records, customer bookings, and overall operational performance.

Unlike manual reporting procedures that require considerable time and effort, the automated reporting module allows reports to be generated instantly with minimal user intervention. Generated reports support operational monitoring, financial analysis, inventory management, and strategic planning while improving the accuracy and consistency of business documentation.

3.5.8 Administration Module

The Administration Module serves as the central management component of the entire application. Administrators are granted complete system privileges that allow them to manage employee accounts, monitor reservations, supervise inventory, maintain menu information, generate business reports, and review sales analytics.

Through this module, administrators create accounts for cashiers and kitchen personnel while assigning appropriate user roles according to organizational responsibilities. The module also provides access to business performance dashboards that display profit generation, sales summaries, reservation statistics, and inventory reports.

Because administrators oversee all system activities, the module incorporates Role-Based Access Control and authentication mechanisms that protect confidential organizational information while ensuring that all user activities remain properly documented.

Module Integration

Although each module performs specific operational functions, all modules are interconnected through the Firebase Realtime Database. Customer reservations automatically become available to cashiers for verification. Approved reservations generate kitchen orders, completed transactions update inventory quantities, and sales records immediately contribute to the Sales Analytics and Report Generation modules. This integrated architecture eliminates duplicate data entry, improves operational coordination, and ensures that all authorized users access accurate and consistent information throughout restaurant operations.

The modular architecture adopted by the researchers enhances system maintainability by allowing individual modules to be modified or expanded without significantly affecting the remaining components. Future enhancements such as online delivery integration, customer loyalty programs, supplier management, or artificial intelligence-based sales forecasting may therefore be incorporated with minimal modification to the existing system architecture.

3.6 User Interface Design

The user interface (UI) of the **Online Restaurant Management System with Sales Analytics and Reservation Scheduling** was designed to provide users with an intuitive, responsive, and user-friendly environment that supports efficient interaction with the system. The interface was developed following fundamental usability principles, emphasizing simplicity, consistency, readability, accessibility, and ease of navigation. Since the system is utilized by customers, cashiers, kitchen personnel, and administrators, each interface was specifically designed according to the responsibilities and operational requirements of its intended users.

The graphical user interface was implemented as a web-based application, allowing authorized users to access the system using desktop computers, laptops, tablets, or smartphones through a web browser. Common interface elements such as navigation menus, buttons, icons, forms, tables, and dashboards were consistently arranged throughout the application to minimize user confusion and reduce the learning curve for first-time users.

The user interface also incorporated responsive web design principles, enabling the application to automatically adjust its layout according to different screen sizes and resolutions. This flexibility improved accessibility while ensuring that users could perform their tasks efficiently regardless of the device being used.

3.6.1 Login Interface

The Login Interface serves as the primary gateway to the Online Restaurant Management System. It requires users to authenticate themselves before accessing any system functionality. Customers may log in using their registered account, while employees—including cashiers, kitchen staff, and administrators—log in using credentials assigned by the administrator.

The login interface validates user credentials and determines the user's assigned role before granting access to the appropriate dashboard. Invalid login attempts display appropriate error messages to guide users without exposing sensitive system information. By requiring authentication before access, the system protects confidential business data and prevents unauthorized use of administrative functions.

(Insert Figure 3.4 ss sng Login Interface.)

3.6.2 Customer Dashboard

The Customer Dashboard provides customers with access to the primary services offered by Crates N' Plates. After successful authentication, customers may browse available menu items, reserve food and beverage packages, book catering services, monitor reservation status, and review previous transactions.

The dashboard presents available services through organized navigation menus and clearly labeled options that allow customers to complete reservations with minimal effort. The interface prioritizes usability by minimizing unnecessary navigation steps and presenting information in an organized manner.

(Insert Figure 3.5 ss sng User/Customer Dashboard.)

3.6.3 Reservation Interface

The Reservation Interface allows customers to submit advance reservations for food, beverages, and catering services. Customers specify reservation details such as date, time, number of guests, selected menu items, catering requirements, and additional rental equipment including tables, chairs, and skirting cloths.

The interface validates required information before allowing customers to proceed with payment. Upon successful submission, reservation information is stored within the centralized database and forwarded to the cashier for verification.

(Insert Figure 3.6 ss sng Reservation/Order Interface.)

3.6.4 Payment Interface

The Payment Interface enables customers to complete online payments for confirmed reservations. The interface securely displays payment information, calculates reservation totals, and confirms successful payment transactions.

Following payment confirmation, the system automatically generates an electronic receipt and sends it to the customer's registered email address. The interface also notifies customers that their reservation remains subject to cashier approval before final confirmation.

(Insert Figure 3.7 ss sng Payment Interface)

3.6.5 Cashier Dashboard

The Cashier Dashboard serves as the operational workspace for restaurant cashiers. Through this interface, cashiers process Point-of-Sale transactions, review reservation requests, approve or decline customer bookings, process customer payments, and generate transaction receipts.

Reservation requests awaiting verification are clearly displayed together with customer information and reservation details. Approved reservations are automatically forwarded to the kitchen module, while declined reservations generate notifications informing customers of the reservation outcome.

The dashboard also provides quick access to current sales transactions and reservation records, enabling cashiers to perform daily operations efficiently.

(Insert Figure 3.8 ss sng Cashier Dashboard)

3.6.6 Kitchen Dashboard

The Kitchen Dashboard displays all customer orders approved by the cashier. Kitchen personnel utilize this interface to monitor incoming orders and update preparation progress.

Order status may be updated according to predefined preparation stages including:

- *Preparing*
- *Cooking*
- *Done*

Status updates are immediately reflected within the system, allowing cashiers to monitor preparation progress without requiring verbal communication with kitchen personnel.

The interface displays orders chronologically to improve organization during peak operating hours.

(Insert Figure 3.9 ss sng Kitchen Dashboard)

3.6.7 Inventory Management Interface

The Inventory Management Interface enables authorized users to monitor inventory quantities, update stock records, review inventory movement, and identify low-stock products.

Separate inventory categories are maintained for kitchen ingredients, beverages, and rentable catering equipment. Inventory quantities are automatically updated whenever customer transactions are completed, reducing manual inventory recording while improving stock accuracy.

The interface also allows administrators to add new products, update inventory information, and review historical stock movement records.

(Insert Figure 3.10 Inventory Management Interface here.)

3.6.8 Sales Analytics Dashboard

The Sales Analytics Dashboard provides administrators with graphical representations of restaurant performance. Interactive charts display daily, weekly, monthly, and yearly sales summaries together with information regarding best-selling products and customer purchasing trends.

Through this interface, management may quickly evaluate restaurant performance without manually preparing reports. Graphical visualization supports faster interpretation of business information while facilitating evidence-based decision-making.

(Insert Figure 3.11 ss sng Sales Analytics Dashboard)

3.6.9 Report Generation Interface

The Report Generation Interface enables authorized users to generate comprehensive reports related to sales transactions, reservation records, inventory status, and operational performance.

Users may select report categories, specify reporting periods, and generate printable reports instantly. The interface minimizes the time required to prepare managerial reports while improving reporting accuracy through automated data retrieval from the centralized database.

(Insert Figure 3.12 ss sng Report Generation Interface)

3.6.10 Administration Dashboard

The Administration Dashboard functions as the central control panel of the entire system. It provides administrators with complete access to system management functionalities including employee account creation, user role assignment, product management, reservation monitoring, inventory supervision, report generation, and sales analytics.

The dashboard also allows administrators to manage restaurant personnel by assigning appropriate user roles through the Role-Based Access Control (RBAC) mechanism. Through centralized administration, management maintains complete oversight of restaurant operations while ensuring that each employee only accesses functions appropriate to their assigned responsibilities.

(Insert Figure 3.13 ss sng Administration Dashboard)

Summary

The user interface of the Online Restaurant Management System was developed to provide a consistent, responsive, and user-friendly experience for all categories of users. Each interface was carefully designed according to its operational purpose while maintaining visual consistency throughout the application. By simplifying navigation, reducing unnecessary user actions, and providing organized access to system functions, the interface contributes significantly to the overall usability and efficiency of the developed system.

3.7 Tools and Technologies Used

The development of the ***Online Restaurant Management System with Sales Analytics and Reservation Scheduling*** *required the integration of various programming languages, software development tools, database technologies, and supporting software applications. Each technology was selected based on its compatibility with the project requirements, ease of implementation, reliability, scalability, and ability to support the operational needs of Crates N' Plates. The combination of these technologies enabled the researchers to develop a secure, efficient, and responsive web-based application capable of automating the restaurant's daily business processes.*

3.7.1 Programming Languages

The researchers utilized multiple programming languages to develop both the front-end and back-end components of the system. Each language performed a specific role in the overall implementation of the application.

PHP (Hypertext Preprocessor)

PHP served as the primary server-side scripting language of the system. It was responsible for processing user requests, implementing the system's business logic, validating user inputs, communicating with the Firebase Realtime Database through the Firebase API, managing user authentication, processing reservations, handling Point-of-Sale transactions, generating reports, and controlling system operations. PHP was selected because of its reliability, extensive documentation, ease of integration with web technologies, and widespread use in developing dynamic web applications.

HTML5 (HyperText Markup Language)

HTML5 was used to create the structural components of the web application. It defined the layout and organization of web pages, including forms, tables, navigation menus, buttons, images, and other interface elements. HTML5 provided the framework upon which the user interface was built and ensured compatibility across modern web browsers.

CSS3 (Cascading Style Sheets)

CSS3 was utilized to enhance the visual appearance of the application by controlling page layout, typography, colors, spacing, responsiveness, and overall interface design. The use of CSS3 allowed the researchers to create a consistent and user-friendly interface while ensuring that the application adapted properly to different screen sizes and devices.

JavaScript

JavaScript was employed to improve the interactivity of the web application. It was used to validate user input, update page contents dynamically, display notifications, manage asynchronous requests, and improve the overall responsiveness of the user interface. JavaScript enabled users to interact with the application without requiring frequent page reloads, thereby enhancing system usability and user experience.

3.7.2 Database

Firebase Realtime Database

Firebase Realtime Database served as the centralized cloud database of the Online Restaurant Management System. It stored customer information, employee accounts, reservation records, sales transactions, inventory information, kitchen orders, reports, and sales analytics. One of Firebase's major advantages is its ability to synchronize data instantly across all connected users, allowing updates made by one user to become immediately available to other authorized users.

The use of Firebase Realtime Database eliminated the need for manual database synchronization while improving data consistency and reducing the possibility of conflicting records. Since the database is cloud-based, authorized users could access updated information in real time regardless of their physical location, provided internet connectivity was available.

3.7.3 Development Tools

Visual Studio Code

Visual Studio Code (VS Code) served as the primary Integrated Development Environment (IDE) throughout the development of the application. It provided source code editing, syntax highlighting, debugging support, extension integration, and project management tools that improved development efficiency. The lightweight architecture and extensive extension library of VS Code enabled the researchers to manage project files effectively while supporting PHP, HTML, CSS, JavaScript, and Firebase integration.

XAMPP

XAMPP functioned as the local web server during the development and testing phases of the project. It provided the Apache web server and PHP runtime environment necessary for executing the web application before deployment. XAMPP allowed the researchers to test application functionalities in a local environment without requiring immediate online deployment.

Firebase Console

The Firebase Console was used to configure and manage the Firebase Realtime Database and Firebase Authentication services. Through the console, the researchers monitored database contents, configured authentication settings, maintained database security rules, and observed application activity during development and testing.

3.7.4 Other Technologies

Firebase Authentication

Firebase Authentication was implemented to provide secure user authentication within the application. It managed user login sessions, verified user identities, and controlled access to protected system resources. Authentication ensured that only authorized users could access system functionalities according to their assigned roles.

Bootstrap

Bootstrap was utilized as the primary front-end framework for developing responsive web pages. It provided reusable interface components, responsive grid layouts, navigation menus, buttons, forms, tables, and other visual elements that improved both development speed and interface consistency.

Chart.js

Chart.js was integrated into the Sales Analytics Module to generate graphical reports and visual representations of restaurant performance. The library was used to display *sales trends, revenue summaries, best-selling products, and other business performance indicators through interactive charts. These visualizations assisted management in interpreting operational data more effectively than traditional numerical reports.*

JSON (JavaScript Object Notation)

JSON served as the primary data interchange format between the PHP application and Firebase Realtime Database. It enabled efficient transmission and storage of structured data while simplifying communication between different system components.

Google Mail Service

The system utilized Google Mail services to automatically send electronic receipts and reservation confirmations to customers after successful online payments. This feature enhanced customer convenience by providing immediate transaction confirmation while maintaining electronic records of reservations.

Summary

The combination of PHP, HTML5, CSS3, JavaScript, Firebase Realtime Database, Firebase Authentication, Visual Studio Code, XAMPP, Bootstrap, Chart.js, and JSON provided a stable and efficient technological foundation for the development of the Online Restaurant Management System. Each technology contributed specific functionalities that collectively enabled the system to automate restaurant operations, improve data management, enhance user experience, and support real-time business *monitoring. The integration of these technologies ensured that the developed application remained reliable, secure, scalable, and capable of supporting the operational requirements of Crates N' Plates.*

3.8 Data Gathering

The development of the **Online Restaurant Management System with Sales Analytics and Reservation Scheduling** required a thorough understanding of the existing operational procedures of Crates N' Plates. To ensure that the proposed system accurately addressed the restaurant's operational needs, the researchers employed several data gathering techniques. These techniques enabled the researchers to identify the limitations of the existing manual processes, determine the functional and non-functional requirements of the proposed system, and establish the basis for the design and development of the application.

The researchers utilized **interviews**, **direct observations**, and **document analysis** to obtain comprehensive information regarding the restaurant's day-to-day operations. The combination of these methods provided sufficient information to understand the restaurant's workflow, identify operational challenges, and formulate appropriate system solutions.

3.8.1 Interview

The interview served as the primary method of gathering information for the study. The researchers conducted a face-to-face interview with the owner of Crates N' Plates, who acted as the primary client and source of information regarding the restaurant's existing business operations. The interview aimed to identify the operational challenges encountered by the establishment and determine the specific functionalities required for the proposed Online Restaurant Management System.

During the interview, discussions focused on the restaurant's existing workflow for sales transactions, reservation scheduling, catering services, inventory monitoring, kitchen operations, report preparation, and overall business management. The owner explained how customer reservations were processed, how sales transactions were recorded, how inventory was monitored, and how reports were manually prepared. Information regarding existing operational problems, including manual record keeping, reservation conflicts, delayed reporting, inventory discrepancies, and communication issues between restaurant personnel, was also obtained.

The interview further enabled the researchers to identify additional system requirements that would improve restaurant operations. These included online reservation scheduling, Point-of-Sale (POS) integration, electronic receipt generation, sales analytics, inventory monitoring, kitchen order management, and Role-Based Access Control (RBAC). The information gathered during the interview became the primary basis for defining the system requirements and determining the major components of the developed application.

3.8.2 Observation

Direct observation was conducted to obtain a better understanding of the restaurant's actual operational procedures. The researchers observed the daily workflow of Crates N' Plates, including customer order processing, reservation handling, inventory monitoring, kitchen coordination, and cashier operations. This allowed the researchers to compare the actual business processes with the information obtained during the interview and identify additional operational challenges that were not immediately evident.

During the observation period, the researchers noted that customer orders and reservations were processed manually, requiring employees to record information on paper before forwarding it to the appropriate personnel. Communication between the cashier and kitchen staff relied primarily on manual order slips, while inventory records were maintained separately using handwritten logbooks. The preparation of business reports also required manual consolidation of transaction records, resulting in additional administrative workload.

Through direct observation, the researchers gained a clearer understanding of the restaurant's operational environment, employee responsibilities, and workflow. These observations helped ensure that the developed system accurately reflected the actual business processes of Crates N' Plates and addressed the operational challenges encountered during daily operations.

3.8.3 Document Analysis

The researchers also conducted document analysis to examine the business records currently used by Crates N' Plates. Existing documents such as sales receipts, reservation forms, inventory records, catering reservation documents, and manually prepared reports were reviewed to understand how business information was recorded, organized, and maintained.

The analysis of these documents enabled the researchers to identify the types of information required by the proposed system, determine the structure of the database, and design electronic forms that closely resembled the restaurant's existing documentation process. Reviewing these records also allowed the researchers to identify redundant procedures, duplicated information, and manual tasks that could be automated through the proposed system.

The information obtained from document analysis was integrated with the results of the interviews and observations to produce a comprehensive set of system requirements. These requirements guided the design of the database structure, user interfaces, system modules, and reporting functionalities incorporated into the Online Restaurant Management System.

Summary

The combination of interviews, direct observations, and document analysis provided the researchers with sufficient information to understand the operational requirements of Crates N' Plates. These data gathering techniques enabled the researchers to identify the restaurant's existing operational challenges, determine the functional requirements of the proposed system, and ensure that the developed application addressed the actual needs of the establishment. The information gathered through these methods served as the foundation for the design, development, implementation, and testing of the Online Restaurant Management System with Sales Analytics and Reservation Scheduling.

3.9 Testing Procedures

Testing is an essential phase of software development that ensures the developed system performs according to its intended requirements while minimizing software defects prior to deployment. Throughout the development of the **Online Restaurant Management System with Sales Analytics and Reservation Scheduling**, the researchers conducted several testing activities to verify the correctness, reliability, functionality, and usability of the application. These testing procedures enabled the researchers to identify software defects, validate system functionalities, and ensure that all modules operated properly before implementation at Crates N' Plates.

Consistent with the Agile Software Development Methodology, testing was performed continuously during each development iteration rather than being conducted only after the completion of the entire system. Every completed module was tested individually before being integrated with other system components. This iterative testing approach minimized software errors and allowed improvements to be incorporated immediately whenever problems were identified.

The researchers employed **Unit Testing, Integration Testing, System Testing, Black-Box Testing, and User Acceptance Testing (UAT)** to verify the quality and reliability of the developed application.

3.9.1 Unit Testing

Unit Testing was performed to verify the functionality of individual modules independently before they were integrated into the complete application. Each module was tested separately to determine whether it performed its intended functions according to the specified system requirements.

The researchers conducted unit testing on the major components of the system, including the User Management Module, Reservation Management Module, Point-of-Sale (POS) Module, Kitchen Management Module, Inventory Monitoring Module, Sales Analytics Module, Report Generation Module, and Administration Module. Each module was tested by providing valid and invalid input values to determine whether the application correctly processed transactions, validated user inputs, stored information accurately, and displayed appropriate responses.

Errors identified during unit testing were corrected immediately before proceeding to the next stage of software development.

3.9.2 Integration Testing

Following successful unit testing, Integration Testing was conducted to verify the communication and interaction among different system modules. Since the developed application consisted of multiple interconnected components, it was necessary to ensure that information flowed correctly throughout the entire system.

The researchers verified that customer reservations submitted through the User Module became available within the Cashier Module for approval. Approved reservations were subsequently transmitted to the Kitchen Module, where kitchen personnel updated order preparation status. Likewise, completed Point-of-Sale transactions automatically updated inventory records, sales reports, and analytical dashboards stored within the Firebase Realtime Database.

Integration testing confirmed that all modules exchanged information correctly and maintained consistent operational data throughout the application.

3.9.3 System Testing

System Testing was performed after the successful integration of all software modules. The objective of this testing procedure was to determine whether the completed application functioned correctly as a complete and integrated system.

The researchers evaluated all major system functionalities, including customer registration, user authentication, reservation scheduling, online payment processing, Point-of-Sale transactions, inventory monitoring, kitchen order management, report generation, sales analytics, and administrative functions. Each operational process was executed under normal working conditions to verify that the system performed according to its intended design.

System testing also ensured that the application maintained consistent performance during continuous operation while preventing data inconsistencies, duplicate records, and operational failures.

3.9.4 Black-Box Testing

Black-Box Testing was employed to evaluate the external functionality of the application without considering its internal program code. The researchers focused on verifying whether the system produced the expected outputs based on various user inputs while ensuring that every feature behaved according to the specified functional requirements.

Different test cases were prepared using both valid and invalid input data. These included login attempts, reservation submissions, transaction processing, inventory updates, report generation, and user account management. The researchers verified whether appropriate validation messages were displayed whenever incorrect or incomplete information was entered into the system.

The successful completion of Black-Box Testing demonstrated that the application functioned correctly from the perspective of the end users while maintaining proper input validation and operational consistency.

3.9.5 User Acceptance Testing (UAT)

User Acceptance Testing (UAT) was conducted to determine whether the developed system satisfied the operational requirements of Crates N' Plates. Unlike technical testing procedures that focused primarily on software functionality, UAT emphasized the practical usability of the application during actual restaurant operations.

The owner of Crates N' Plates served as the primary evaluator during this phase. The client interacted with the completed system by performing actual operational tasks, including customer reservation verification, sales transaction processing, inventory monitoring, report generation, and administrative management. The client evaluated whether the developed functionalities aligned with the restaurant's operational requirements and whether the application effectively addressed the challenges identified during the initial requirements gathering phase.

Feedback obtained during User Acceptance Testing was used to implement minor refinements before the final deployment of the application. This collaborative validation process ensured that the completed system met the expectations and operational needs of Crates N' Plates.

Summary

The combination of Unit Testing, Integration Testing, System Testing, Black-Box Testing, and User Acceptance Testing enabled the researchers to evaluate the functionality, reliability, and operational readiness of the Online Restaurant Management System. These testing procedures ensured that all software modules performed correctly, communicated effectively, and satisfied the business requirements identified during the data gathering phase. Through continuous testing and refinement, the researchers successfully developed a reliable and efficient restaurant management system capable of supporting the daily operations of Crates N' Plates.

3.10 Evaluation Criteria

The quality and acceptability of the developed **Online Restaurant Management System with Sales Analytics and Reservation Scheduling** were evaluated using the **ISO/IEC 25010 Software Quality Model**, an internationally recognized standard for assessing software quality. The ISO/IEC 25010 model provides a comprehensive framework for evaluating software products based on defined quality characteristics that determine the effectiveness, efficiency, reliability, security, and overall usability of a software application (ISO/IEC, 2023).

The researchers adopted the ISO/IEC 25010 Software Quality Model because it provides standardized evaluation criteria applicable to web-based information systems. The model enables software developers and researchers to assess whether a developed application satisfies user requirements while maintaining acceptable levels of software quality. Through this evaluation, the researchers determined whether the Online Restaurant Management System effectively addressed the operational needs of Crates N' Plates and met acceptable software engineering standards.

The developed application was evaluated based on seven software quality characteristics specified by ISO/IEC 25010, namely **Functional Suitability,**

Performance Efficiency, Usability, Reliability, Security, Maintainability, and Portability. Each characteristic measured a specific aspect of software quality that contributed to the overall effectiveness of the system.

3.10.1 Functional Suitability

Functional Suitability refers to the degree to which the developed system provides appropriate functions that satisfy the operational requirements of Crates N' Plates. This criterion evaluated whether the implemented features performed their intended functions accurately, completely, and appropriately.

The evaluation determined whether the system successfully automated sales transaction processing, reservation scheduling, inventory monitoring, kitchen order management, report generation, sales analytics, and user management according to the operational requirements identified during the data gathering phase.

3.10.2 Performance Efficiency

Performance Efficiency measured the capability of the system to provide acceptable response times while efficiently utilizing available computing resources. This criterion evaluated the speed of transaction processing, reservation handling, report generation, database retrieval, and overall system responsiveness during normal operation.

The researchers also assessed whether the application maintained stable performance while multiple users simultaneously accessed different system modules.

3.10.3 Usability

Usability evaluated the ease with which intended users could learn, understand, operate, and interact with the Online Restaurant Management System. The assessment focused on interface organization, navigation consistency, readability, accessibility, ease of learning, and overall user satisfaction.

Particular attention was given to determining whether customers, cashiers, kitchen personnel, and administrators could perform their assigned tasks efficiently without requiring extensive technical knowledge or prolonged training.

3.10.4 Reliability

Reliability measured the ability of the developed system to perform its intended functions consistently without failure during continuous operation. This criterion evaluated whether the application maintained operational stability while processing reservations, sales transactions, inventory updates, report generation, and user authentication.

The researchers also observed whether the system successfully prevented data loss, software crashes, and unexpected interruptions that could affect restaurant operations.

3.10.5 Security

Security evaluated the ability of the system to protect confidential business information from unauthorized access while maintaining data confidentiality, integrity, and accountability. This criterion assessed the effectiveness of user authentication, Role-Based Access Control (RBAC), user authorization, and database security mechanisms implemented within the application.

The evaluation also determined whether each user could only access the functionalities appropriate to his or her assigned organizational role, thereby protecting sensitive operational information.

3.10.6 Maintainability

Maintainability assessed the ease with which the developed application could be modified, corrected, enhanced, or updated in response to changing business requirements. This criterion evaluated the organization of the program code, modular system architecture, database structure, and documentation prepared throughout the development process.

The modular design of the application allowed future developers to improve individual system components without significantly affecting the remaining modules.

3.10.7 Portability

Portability measured the capability of the application to operate successfully across different computing environments and hardware platforms. Since the Online Restaurant Management System was developed as a web-based application, the evaluation determined whether authorized users could access the system through various web browsers and internet-connected devices, including desktop computers, laptops, tablets, and smartphones.

The portability assessment also verified that the application maintained consistent functionality regardless of the operating system or device utilized by the users.

Evaluation Scale

To interpret the results of the software evaluation, the researchers utilized a **five-point Likert scale**. Each criterion was rated according to the respondents' level of agreement regarding the quality and performance of the developed system.

Table 3.1

Five-Point Likert Scale Used in the Evaluation

SCALE	RANGE	INTERPRETATION
5	4.21 - 5.00	EXCELLENT
4	341 - 4.20	VERY GOOD
3	2.61 - 340	GOOD
2	1.81 - 2.60	FAIR
1	1.00 -1.80	POOR

The weighted mean was computed for each ISO/IEC 25010 quality characteristic to determine the overall level of acceptability of the developed system. The interpretation of the computed mean followed the verbal interpretations presented in Table 3.1.

Summary

The ISO/IEC 25010 Software Quality Model provided a comprehensive framework for evaluating the developed Online Restaurant Management System in terms of its functionality, efficiency, usability, reliability, security, maintainability, and portability. By adopting internationally recognized software quality standards, the researchers were able to objectively assess whether the developed application effectively satisfied the operational requirements of Crates N' Plates while maintaining acceptable software engineering quality. The results of this evaluation are presented and discussed in **Chapter IV**.